DEPARTMENT OF
MATHEMATICS

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BSc in Mathematics

MICROLOCAL EQUIVALENCE OF QUASI-ORDINARY SURFACES

MASTER IN MATHEMATICS AND APPLICATIONS
SPECIALIZATION IN PURE MATHEMATICS

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Microlocal Equivalence of Quasi-Ordinary Surfaces

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Se ter um bom orientador é um privilégio, o que se há de dizer de ter dois!

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"To live without hope is to cease to live."

— **Fyodor Dostoevsky**, *The Brothers Karamazov*

Abstract

Resolution of singularities has been an area of active research in mathematics for the past three hundred years, going back to I. Newton [24]. Afterwards, significant results and generalizations have derived from this work, for instance, by O. Zariski [31], S. Abhyankar [1], H. Hironaka [14], and J. Lipman [17].

The focus throughout this dissertation lies on one particular generalization of plane curves: quasi-ordinary surfaces. Because handling fractional parametrizations in multiple variables inherently complicates the local structure of the singularity, achieving a resolution in this setting remains highly non-trivial. More recently, authors like O. Neto [23], M. Hernandes [13], and M. Villa [9] have shown that the relevance of this topic remains till this day.

Given a quasi-ordinary surface Y with one characteristic exponent λ, the goal will be to find a quasi-ordinary surface Y' with the same characteristic exponent λ that admits a Puiseux parametrization whose support is finite up to composition of (possibly infinitely many) analytic transformations.

The construction of Y' will be done via the composition of finitely many contact and analytic transformations applied to $\mathbb{P}^*_Y\mathbb{C}^3$. The latter may also be seen as contact transformations. Given that the set of all contact transformations forms a group under composition, it is possible to establish an equivalence relation between quasi-ordinary surfaces that we call *microlocal equivalence*.

Keywords: Quasi-ordinary surfaces · Puiseux parametrization · Contact geometry · Microlocal equivalence

Resumo

Resolução de singularidades é uma área de investigação ativa em matemática desde há três séculos, tempos que remotam a I. Newton [24]. Desde então, generalizações e resultados significantes derivaram deste trabalho, por exemplo, por O. Zariski [31], S. Abhyankar [1], H. Hironaka [14], e J. Lipman [17].

O foco desta dissertação resume-se essencialmente a uma generalização particular de curvas planas: superfícies quasi-ordinárias. Dado que trabalhar com parametrizações fracionárias em várias variáveis dificulta inerentemente a estrutura local da singularidade, obter uma resolução neste panorama mantém-se um problema altamente não trivial. Mais recentemente, autores como O. Neto [23], M. Hernandes [13], e M. Villa [9] têm demonstrado nas suas publicações a relevância que o estudo destes objetos ainda possui.

Dada uma superfície quasi-ordinária Y com um expoente característico λ, o objetivo será determinar uma superfície quasi-ordinária Y' com o mesmo expoente característico λ que admite uma parametrização de Puiseux com suporte finito, a menos de uma composição (possivelmente em número infinito) de transformações analíticas.

A construção de Y' será realizada através da composição de um número finito de transformações de contacto e analíticas aplicadas a $\mathbb{P}^*_Y\mathbb{C}^3$. Estas últimas também podem ser interpretadas como transformações de contacto. Dado que o conjunto de todas as transformações de contacto munido da composição forma um grupo, é possível estabelecer uma relação de equivalência entre superfícies quasi-ordinárias a que denominamos de *equivalência microlocal*.

Palavras-chave: Superfícies Quasi-ordinárias · Parametrização de Puiseux · Geometria de Contacto · Equivalência Microlocal

Disclosure of Usage of Generative Artificial Intelligence

The author, Miguel Ventura Garcia, acknowledges the use of the following generative artificial intelligence tools in the development of this thesis. These technologies were utilized under strict supervision, and all AI-generated outputs were critically reviewed and verified for accuracy. The author accepts full responsibility for the validity and originality of the content, confirming that the use of these tools aligns with all institutional policies and standards of academic integrity.*

Generative AI tools used: ChatGPT-5.2 and Gemini 3 Pro.

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- [x] **Hypothesis viability assessment**
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Literature Review

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- [x] **Literature summarization and synthesis**
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- [ ] Refactoring and Optimization
- [ ] Process automation
- [ ] Code documentation and comment generation

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- [ ] Data collection
- [ ] Validation
- [ ] Data cleaning
- [ ] Data curation and organization
- [ ] Data analysis
- [ ] Quantitative Plotting and Charting
- [ ] Reproducibility and rerun checks
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- [x] **Polishing and Editing**
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- [ ] Tone Adjustment
- [ ] Translation
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- [ ] Monitoring compliance with ethical standards
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Additional comment #2: All mathematical content and verifications of logical derivations were performed manually by the author.

CONTENTS

List of Figures

Acronyms

q.o. quasi-ordinary *(p. 8)*

Symbols

λ	Characteristic exponent of a quasi-ordinary hypersurface *(p. 9)*
Y	Analytic germ at the origin of a quasi-ordinary hypersurface *(p. 8)*
υ	Puiseux parametrization *(p. 9)*
ζ	Quasi-ordinary branch *(p. 8)*
$\mathbb{P}^*_X\mathbb{C}^{n+1}$	Conormal of an analytic subset X of $\mathbb{C}^{n+1}$ *(p. 14)*
$\mathbb{P}^*\mathbb{C}^{n+1}$	Projective cotangent bundle *(p. 14)*
π	Canonical projection : $\mathbb{P}^*\mathbb{C}^{n+1} \longrightarrow \mathbb{C}^{n+1}$ *(p. 14)*
π_i	i-th fake projection *(p. 14)*
$\Sigma_o(X)$	Limit of tangents of X at o *(p. 14)*
$\mathrm{Fr}(\lambda)$	Frobenius vector of $\Gamma(\lambda)$ *(p. 23)*
$\Gamma(\tilde{\lambda}_1, \ldots, \tilde{\lambda}_s)$	Semigroup associated to a quasi-ordinary hypersurface with characteristic exponents $\tilde{\lambda}_1, \ldots, \tilde{\lambda}_s$ *(p. 10)*
$\Lambda(\lambda)$	Legendrian semigroup *(p. 24)*
$\Lambda_i(\lambda)$	i-th fake Legendrian semigroup *(p. 24)*
$L_{\tau,i}$	*(p. 48)*
$M_{\tau,i}$	*(p. 48)*
$N_{\tau,i}$	*(p. 48)*
$\mathbb{1}_i$	i-th vector of the canonical basis for $\mathbb{R}^n$ as a real vector space *(p. 10)*
$\mathbb{1}$	$\sum_{j=1}^{n} \mathbb{1}_j$ *(p. 15)*
$\mathcal{R}_i(\sigma)$	*(p. 33)*
R	*(p. 48)*

1

Introduction

A primary objective in mathematical research is the reduction of complex problems into simpler, more manageable components. From a technical standpoint, this often involves transforming a given problem into one or more "equivalent" counterparts that are easier to analyze.

One significant generalization of plane curves is the concept of quasi-ordinary hypersurfaces. Let f be a Weierstrass polynomial defining such a hypersurface. According to the Jung-Abhyankar Theorem (cf. [2]), all branches of f are fractional power series. Furthermore, J. Lipman demonstrated in [19] the existence of a specific set of exponents, now termed *characteristic exponents*, which possess fundamental structural properties.

Inspired by Proposition 1.2 of [30], our goal is to eliminate terms of a branch ζ in an equivalent manner, a framework we define as *microlocal equivalence*, reducing the current problem to an analytic one.

Before diving into the realm of the theory of quasi-ordinary hypersurfaces, Chapter 2 establishes the needed common ground in order to fully grasp the concepts reviewed and developed in this thesis.

Chapter 3 establishes the necessary framework for understanding quasi-ordinary hypersurfaces, building upon the plane curve case. Specifically, for a plane curve branch ζ with a Puiseux parametrization v (cf. [6]), O. Zariski investigated analytically equivalent branches in [30]. He succeeded in deriving a "short form" for these branches, which utilizes an analytical isomorphism to eliminate a substantial portion of terms from the Puiseux parametrization.

This thesis focuses on the specific case of specific quasi-ordinary surface branches with a single characteristic exponent interlinked with the fact that the conormal of these quasi-ordinary surface branches will have an isolated singularity. In this setting, the associated semigroup has certain properties that rise from geometric ones, exhibiting a simpler structure in comparison with the general case, which loses the property of the conormal having an isolated singularity. In Chapter 5, we explore properties of said semigroup that facilitate the construction of contact transformations. The underlying principles of contact geometry, which allow us to compute these transformations, are detailed in Chapter 4. The classification of the irreducible quasi-ordinary surfaces whose conormal has an isolated singularity at the origin is also dealt in Chapter 4.

We begin Chapter 6 with technical reparametrization results that serve as a foundation for the study of the obtained parametrizations via the contact transformations constructed. In this chapter we also study specific contact transformations, allowing us to transform quasi-ordinary hypersurfaces into quasi-ordinary hypersurfaces.

The concluding sections of Chapter 6 contain the main results and provide illustrative examples where the aforementioned transformations exhibit suboptimal behavior. These cases highlight the limitations of the current framework and identify specific configurations where the contact transformations utilized lose some interest.

Finally, Chapter 7 outlines several promising directions for future research.

2

Preliminaries

Before outlining the contextual framework of this study, it is essential to establish a common conceptual foundation. This section delineates the fundamental definitions and notation requisite for a comprehensive understanding of the subsequent analysis. These terms are presented sequentially, following the order of their introduction within the text.

Notation 2.0.1. *We denote the ring of polynomials, formal power series, convergent power series, and Laurent power series in the variables $x_1, \ldots, x_n$ with coefficients in $\mathbb{C}$, respectively, by $\mathbb{C}[x_1, \ldots, x_n]$, $\mathbb{C}[[x_1, \ldots, x_n]]$, $\mathbb{C}\{x_1, \ldots, x_n\}$, and $\mathbb{C}((x_1, \ldots, x_n))$. If $\underline{x} = (x_1, \ldots, x_n)$, we often write $\mathbb{C}[\underline{x}]$, $\mathbb{C}[[\underline{x}]]$, $\mathbb{C}\{\underline{x}\}$, and $\mathbb{C}((\underline{x}))$ to denote, respectively, $\mathbb{C}[x_1, \ldots, x_n]$, $\mathbb{C}[[x_1, \ldots, x_n]]$, $\mathbb{C}\{x_1, \ldots, x_n\}$, and $\mathbb{C}((x_1, \ldots, x_n))$.*

Definition 2.0.2. *A function $h : \mathbb{C}^n \to \mathbb{C}$ is said to be **holomorphic** if for each $z \in \mathbb{C}^n$, there exists a neighborhood of z such that h is complex differentiable.*

Definition 2.0.3. *A subset $A \subseteq \mathbb{C}^n$ is called a **complex analytic subset** if for each $z \in A$, there exists an open neighborhood $U \subseteq \mathbb{C}^n$ of z and finitely many holomorphic functions $f_1, \ldots, f_k : U \to \mathbb{C}$ such that*

$$A \cap U = \{\tilde{z} \in U \mid f_1(\tilde{z}) = \ldots = f_k(\tilde{z}) = 0\}.$$

Definition 2.0.4. *An element f of $\mathbb{C}\{x_1, \ldots, x_{n-1}\}[x_n]$ is called a **Weierstrass polynomial** if f is of the form*

$$f = x_n^m + \sum_{k=0}^{m-1} a_k(x_1, \ldots, x_{n-1}) x_n^k,$$

where the a_k are holomorphic functions in a neighborhood of the origin, which vanish at the origin.

Definition 2.0.5. *Let C be a totally ordered finite set. A finite sequence of symbols of C is called a **word**. The **lexicographical order**, denoted by $\geq_{lex}$, is defined on a set of words by*

$$a_1 \ldots a_n \geq_{lex} b_1 \ldots b_n$$

if and only if either $a_i = b_i$, for each $i \in \{1, \ldots, n\}$, or the first index i where $a_i \neq b_i$ satisfies $a_i > b_i$.

Definition 2.0.6. *A partially ordered set (poset) $(L, \leq)$ is called a **lattice** if each two-element subset $\{a, b\} \subseteq L$ has a least upper bound, i.e., an element $x \in L$ (denoted as $a \vee b$ or the **join**) such that $a \leq x$ and $b \leq x$, and for any other element y where $a \leq y$ and $b \leq y$, then $x \leq y$ and a greatest lower bound, i.e., an element $z \in L$ (denoted as $a \wedge b$ or the **meet**) such that $z \leq a$ and $z \leq b$, and for any other element w where $w \leq a$ and $w \leq b$, then $w \leq z$.*

Example 2.0.7. *Let $\tau = (\tau_1, \ldots, \tau_n), \sigma = (\sigma_1, \ldots, \sigma_n) \in \mathbb{Z}^n$. We define the partial order $\leq$ on $\mathbb{Z}^n$ as*

$$\tau \leq \sigma \iff \tau_i \leq \sigma_i, \text{ for all } i \in \{1, \ldots, n\}.$$

Then $\tau \vee \sigma = (\max\{\tau_1, \sigma_1\}, \ldots, \max\{\tau_n, \sigma_n\})$ and $\tau \wedge \sigma = (\min\{\tau_1, \sigma_1\}, \ldots, \min\{\tau_n, \sigma_n\})$. Thus $(\mathbb{Z}^n, \leq)$ is a lattice.

Definition 2.0.8. *A **semigroup** is an ordered pair $(S, \cdot)$ where S is a set and $\cdot$ is an associative binary operation on S. An **affine semigroup** is any semigroup that is isomorphic to a finitely generated sub-semigroup of $\mathbb{Z}^n$ for some $n \in \mathbb{N}$.*

Definition 2.0.9. *Let $S \subseteq \mathbb{N}_0$ be a semigroup such that $0 \in S$. We say that S is a **numerical semigroup** if $\mathbb{N}_0 \setminus S$ is finite.*

Definition 2.0.10. *Let S be an affine semigroup. The **cone** generated by S, denoted as cone(S), is the set of all finite linear combinations of elements of S with non-negative real coefficients, i.e.,*

$$cone(S) = \left\{ \sum_{k=1}^{\ell} c_k s_k : c_k \in \mathbb{R}_{\geq 0}, s_k \in S, \ell \in \mathbb{N} \right\}.$$

Definition 2.0.11. *An **affine simplicial semigroup** is an affine semigroup S such that cone(S) is generated by a set of linearly independent vectors over $\mathbb{R}$.*

Definition 2.0.12. *Let K be a field. A map $\nu : K \to \mathbb{Z} \cup \{\infty\}$ is said to be a **valuation** if it satisfies the following properties for all $a, b \in K$:*

$\nu(a) = \infty$ if and only if $a = 0$,

$\nu(ab) = \nu(a) + \nu(b)$,

$\nu(a + b) \geq \min(\nu(a), \nu(b))$, with equality if $\nu(a) \neq \nu(b)$.

Definition 2.0.13. *Let $f \in \mathbb{C}[[\underline{x}]]$ be a non-zero formal power series. We can write*

$$f(\underline{x}) = \sum_{k \geq i} f_k(\underline{x}),$$

*where each f_k is a homogeneous polynomial of degree k and $f_i \neq 0$. We call i the **order** of f, and denote it by $\operatorname{ord}(f)$. It is standard convention to state that $ord(0) = \infty$.*

Definition 2.0.14. *Let $f \in \mathbb{C}\{\underline{x}\}$ and X be the hypersurface defined by the equation $f = 0$. We define the **singular locus** of X, $\operatorname{Sing}(X)$, as the analytic set defined by the equations $f = 0$ and $\frac{\partial f}{\partial x_i} = 0$, for $i = 1, \ldots, n$.*

Definition 2.0.15. *Let $f \in \mathbb{C}\{\underline{x}\}$ be a convergent power series on an open set U, and X be the hypersurface defined by the equation $f = 0$. Let $o \in X$ and V be a sufficiently small open subset of U that contains o. After a translation, we can assume that o is the origin and that $X \cap V$ is defined by $h = 0$, where $h \in \mathbb{C}\{\underline{x}\}$. We define the* ***multiplicity*** *of X at o, $\mathrm{m}_o(X)$, as $\mathrm{ord}(h)$.*

Definition 2.0.16. *Let X be a topological space and A be a subset of X. We define the* ***interior*** *of A, $int(A)$, to be the union of all open subsets of X that are contained in A.*

Definition 2.0.17. *[15] The* ***n-dimensional complex projective space****, denoted by $\mathbb{P}^n\mathbb{C}$ or simply $\mathbb{P}^n$, is defined to be the set of 1-dimensional linear subspaces of $\mathbb{C}^{n+1}$, with the quotient topology determined by the natural map $\pi : \mathbb{C}^{n+1} \setminus \{0\} \longrightarrow \mathbb{P}^n\mathbb{C}$ sending each point $\underline{x} \in \mathbb{C}^{n+1} \setminus \{0\}$ to the subspace spanned by $\underline{x}$.*

When working with formal power series, we sometimes arrive at the question if the composition of formal power series is well-defined. In order to guarantee this, we review how $\mathbb{C}[[\underline{x}]]$ can be equipped with a topology, making it a topological space.

Definition 2.0.18 (cf. [11] pg.357)**.** *Let $\mathfrak{m} = (x_1, \ldots, x_n)\mathbb{C}[[\underline{x}]]$ denote the maximal ideal of $\mathbb{C}[[\underline{x}]]$.*

i) Let $\mathcal{T} = \left\{\mathfrak{m}^j : j \in \mathbb{N}\right\}$. Consider $(\mathbb{C}[[\underline{x}]], \mathcal{T})$ as a topological space, where $\mathcal{T}$ is a fundamental system of neighborhoods of 0. This topology is called the $\mathfrak{m}$***-adic topology****.*

ii) A sequence $\{\varepsilon_\ell\}_{\ell \in \mathbb{N}}$, $\varepsilon_\ell \in \mathbb{C}[[\underline{x}]]$, is called a ***Cauchy sequence*** *if, for every $j \in \mathbb{N}$, there exists $k \in \mathbb{N}$ such that*

$$\varepsilon_\ell - \varepsilon_s \in \mathfrak{m}^j, \text{ for all } \ell, s \geq k.$$

iii) A sequence $\{\varepsilon_\ell\}_{\ell \in \mathbb{N}}$, $\varepsilon_\ell \in \mathbb{C}[[\underline{x}]]$, is called ***convergent*** *if there exists a power series $\varepsilon \in \mathbb{C}[[\underline{x}]]$ such that for every $j \in \mathbb{N}$ there exists some $k \in \mathbb{N}$ satisfying $\varepsilon - \varepsilon_\ell \in \mathfrak{m}^j$ for all $\ell \geq k$. Then ε is uniquely determined, and we write, as usual, $\varepsilon =: \lim_{\ell \to \infty} \varepsilon_\ell$.*

Notation 2.0.19. *Let $(a_1, \ldots, a_n) \in \mathbb{R}^n$. We write*

$$|(a_1, \ldots, a_n)| = |a_1| + \ldots + |a_n|.$$

3

Fundamentals of Quasi-Ordinary Hypersurfaces

Puiseux series (cf. Chapter III Section 8.3 of [6]), introduced by Victor Puiseux (and historically attributed also to Isaac Newton), allow us to overcome the difficulties that arise when analyzing germs of plane curves at the origin. In this chapter, we review the extension of these concepts to the study of quasi-ordinary hypersurfaces.

Let $\mathbb{N}_0$ (resp. $\mathbb{N}$) denote the set of non-negative integers (resp. positive integers), $\mathbb{Q}_{\geq 0}$ (resp. $\mathbb{Q}_{>0}$) the set of non-negative rationals (resp. positive rationals), and $\mathbb{R}_{\geq 0}$ the set of non-negative real numbers. We denote the set of non-zero complex numbers by $\mathbb{C}^*$.

Given two vectors $\sigma = (\sigma_1, \ldots, \sigma_n)$ and $\tau = (\tau_1, \ldots, \tau_n)$ in $\mathbb{R}^n$, we define the partial order $\leq$ by:

$$\sigma \leq \tau \iff \sigma_j \leq \tau_j, \quad \forall j \in \{1, \ldots, n\}.$$

Moreover, we define the **corner** of σ, denoted by $\llcorner(\sigma)$, to be

$$\llcorner(\sigma) := \{\tau \in \mathbb{R}^n : \sigma \leq \tau\}.$$

Let $\underline{x} = (x_1, \ldots, x_n)$ and $\sigma = (\sigma_1, \ldots, \sigma_n) \in \mathbb{R}^n_{\geq 0}$. We define the monomial $\underline{x}^\sigma$ as

$$\underline{x}^\sigma := x_1^{\sigma_1} \cdots x_n^{\sigma_n}.$$

Definition 3.0.1. *Let $f(\underline{x}) = \sum_{\tau \in \mathbb{N}_0^n} c_\tau \underline{x}^\tau \in \mathbb{C}[[\underline{x}]]$. The **support** of f is the set*

$$Supp(f) = \{\tau \in \mathbb{N}_0^n : c_\tau \neq 0\}.$$

3.1 Quasi-Ordinary Singularities

Let A be an analytic subset of $\mathbb{C}^n$. We say that A has *normal crossings* at a point $o \in A$ if there exists an open neighborhood U of o, a system of local coordinates $(x_1, \ldots, x_n)$ centered at o, and an integer $v \in \{1, \ldots, n\}$ such that

$$A \cap U = \{\underline{x} \in U : x_1 \cdots x_v = 0\}.$$

We say that A is a *divisor with normal crossings* in $\mathbb{C}^n$ if it has normal crossings at every point $o \in A$.

The analytic germ at the origin, Y, of a hypersurface in $\mathbb{C}^{n+1}$ is called **quasi-ordinary (q.o.)** if there exists a finite projection $q : (\mathbb{C}^{n+1}, 0) \to (\mathbb{C}^n, 0)$, whose restriction to Y is finite and is a local isomorphism outside a divisor with normal crossings.

When working with q.o. singularities, we can assume without loss of generality (cf. pg.51 of [19]) that we have coordinates $(\underline{x}, y)$ such that $q(\underline{x}, y) = \underline{x}$ and the hypersurface is defined by a Weierstrass polynomial in y.

Given a Weierstrass polynomial $f \in \mathbb{C}\{\underline{x}\}[y]$, the *apparent contour* of f, denoted by Z, is the set defined by the equations $f = \frac{\partial f}{\partial y} = 0$. The set $\Delta_y f = q(Z)$ is called the **discriminant** of f relative to the projection q.

We call $f \in \mathbb{C}\{\underline{x}\}[y]$ a **q.o. polynomial** if f is a Weierstrass polynomial in y with a discriminant $\Delta_y f$ of the form:

$$\Delta_y f = \underline{x}^{\sigma} u(\underline{x}),$$

where $u \in \mathbb{C}\{\underline{x}\}$ is a unit and $\sigma \in \mathbb{N}_0^n$. If f is a q.o. polynomial, the equation $f = 0$ defines a quasi-ordinary hypersurface.

Let Y be a singular irreducible q.o. hypersurface defined by the q.o. polynomial f. The roots of f are fractional power series, i.e., elements of the ring $\mathbb{C}\{x_1^{1/m}, \dots, x_n^{1/m}\}$, that we shall denote by $\mathbb{C}\{\underline{x}^{1/m}\}$ (cf. Section 2.1 of [9]). These roots are called the **branches** of Y. The notions of order and support can be naturally extended to elements of $\mathbb{C}\{\underline{x}^{1/m}\}$ and the notion of degree to elements of $\mathbb{C}[\underline{x}^{1/m}]$.

Lemma 3.1.1 (cf. Lemma 2.2 of [9])**.** *Let ζ be a branch of Y. There exist vectors $\tilde{\lambda}_1, \dots, \tilde{\lambda}_s \in$ Supp(ζ), called the **characteristic exponents** of Y, satisfying*

1. $\tilde{\lambda}_1 < \tilde{\lambda}_2 < \cdots < \tilde{\lambda}_s$.

2. *For all* $i \in \{1, \dots, s\}$, $\tilde{\lambda}_i \notin \mathbb{Z}^n + \sum_{j=1}^{i-1} \mathbb{Z}\tilde{\lambda}_j$.

3. *If* $\tau \in$ *Supp*(ζ), *then* $\tau \in \mathbb{Z}^n + \sum_{\tilde{\lambda}_j \le \tau} \mathbb{Z}\tilde{\lambda}_j$.

Notation. We denote by $(\lambda_{j,1}, \dots, \lambda_{j,n})$ the coordinates of the characteristic exponent $\tilde{\lambda}_j$ with respect to the canonical basis of $\mathbb{Q}^n$.

Definition 3.1.2. *A q.o. branch ζ as in Lemma 3.1.1 is said to be **strongly normalized** if the following conditions hold.*

(i) $(\lambda_{1,1}, \dots, \lambda_{s,1}) \ge_{lex} \cdots \ge_{lex} (\lambda_{1,n}, \dots, \lambda_{s,n})$.

(ii) $\tilde{\lambda}_1$ *is not of the form* $(\lambda_{1,1}, 0, \dots, 0)$ *with* $\lambda_{1,1} < 1$.

(iii) *There exists* $H \in \mathbb{C}\{\underline{x}\}$ *such that* $H(0) \neq 0$ *and:*

$$\zeta = \underline{x}^{\tilde{\lambda}_1} H(\underline{x}^{1/m}).$$

The relevance of the characteristic exponents comes from the fact that they characterize the topological type of irreducible q.o. hypersurfaces (see, for instance, [19]).

Given a branch ζ of Y, the set of conjugates of ζ over $\mathbb{C}[[\underline{x}]]$ is the set of the fractional power series $\{\zeta(\varrho_1 x_1^{1/m}, \ldots, \varrho_n x_n^{1/m})\}$, where ϱ_i, for $i = 1, \ldots, n$, runs through all m-th roots of unity.

Let ζ be a branch of Y and let $\zeta_1, \ldots, \zeta_m$ be the distinct conjugates of ζ over $\mathbb{C}[[\underline{x}]]$. Our irreducible q.o. polynomial f equals

$$\prod_{j=1}^{m} \left(y - \zeta_j(\underline{x}^{1/m}) \right). \tag{3.1}$$

Assuming the conditions of Lemma 3.1.1 hold, we introduce the lattices $M_0 := \mathbb{Z}^n$ and $M_i := M_{i-1} + \mathbb{Z}\tilde{\lambda}_i$ for $i = 1, \ldots, s$. For convenience, we denote the lattice M_s by M. Additionally, define m_i to be the smallest positive integer such that $m_i \tilde{\lambda}_i \in M_{i-1}$ for each $i = 1, \ldots, s$. The degree of the q.o. polynomial f is given by $m = m_1 \cdots m_s$ (cf. Definition 2.5 of [9]). For $s = 1$, $m = m_1$ and we call m the **ramification number** of $\lambda = \tilde{\lambda}_1$. Set $k_i = \prod_{j=i}^{s} m_j$ and $k_{s+1} = 1$.

Given $h_1, \ldots, h_k \in \mathbb{C}\{\underline{x}\}$, we shall denote by $V(h_1, \ldots, h_k)$ the set defined by the equations $h_1 = \ldots = h_k = 0$. In Chapter 2 of [18], J. Lipman studies the singular locus and the multiplicity of a q.o. surface. The following lemma is based on Theorem 2.7 of [18] and its respective proof.

Lemma 3.1.3. *Let Y be an irreducible q.o. surface, defined by a Weierstrass polynomial f, with a normalized branch and characteristic exponents $\tilde{\lambda}_1, \ldots, \tilde{\lambda}_s$. The following statements hold:*

(a) The three conditions below are equivalent.
   (i) $\Delta_y f$ equals $V(x_1)$.
   (ii) $\lambda_{i,2} = 0$, $i = 1, \ldots, s$.
   (iii) Y is equisingular along $V(x_1, y)$.

(b) If $\lambda_{1,2} = 0$ and $\lambda_{s,2} \neq 0$, $\mathrm{Sing}(Y)$ contains $Y \cap V(x_1) = V(x_1, y)$, and is contained in the union of $V(x_1, y)$ and $Y \cap V(x_2)$. Let r be the smallest integer such that $1 \leq r \leq s$ and $\lambda_{r,2} \neq 0$. Then $\mathrm{m}_0(Y) = m$, $\mathrm{m}_{(0,a,0)}(Y) = \min(m, m\lambda_{1,1})$, for $0 < |a| \ll 1$, and $\mathrm{m}_{o'}(Y) = \min(k_r, k_r \lambda_{r,2})$, for $o' \in Y \cap V(x_2)$ and $0 < |o'| \ll 1$.

(c) If $\lambda_{1,1}, \lambda_{1,2} > 0$, $\mathrm{Sing}(Y)$ is a subset of $V(x_1 x_2, y)$, $\mathrm{m}_0(Y) = \min(m, m(\lambda_{1,1} + \lambda_{1,2}))$, $\mathrm{m}_{(0,a,0)}(Y) = \min(m, m\lambda_{1,1})$ and $\mathrm{m}_{(a,0,0)}(Y) = \min(m, m\lambda_{1,2})$, for $0 < |a| \ll 1$.

Remark 3.1.4. *Statement $(a.iii)$ of Lemma 3.1.3 means that for any point $(0, a, 0)$ with $0 < |a| \ll 1$, we can find a sufficiently small open neighborhood U_a containing $(0, a, 0)$ such that $Y \cap U_a$ is an irreducible q.o. surface. Its characteristic exponents are contained in $\{(\lambda_{1,1}, 0), \ldots, (\lambda_{s,1}, 0)\}$ and are independent of the choice of a.*

Definition 3.1.5. *Let $\underline{t} = (t_1, \ldots, t_n)$ and ζ be a branch of Y defined by a q.o. polynomial f of degree m. This branch induces a parametrization v of Y (i.e., $v^* f = 0$) by setting $x_i = t_i^m$ for $i \in \{1, \ldots, n\}$, and consequently $y = \zeta(\underline{t}^m)$. We call v the* ***Puiseux parametrization*** *of Y.*

Example 3.1.6. *To illustrate these concepts, let*

$$\zeta = x_1^{7/4}x_2^{3/2} + x_1^{29/4}x_2^{11/2} + \sum_{k=2}^{\infty} x_1^{(21/4)+2k}x_2^{(9/2)+k}.$$

be a branch of Y. We parametrize Y as follows:

$$C : \begin{cases} x_1 = t_1^4, \\ x_2 = t_2^4, \\ y = t_1^7 t_2^6 + t_1^{29} t_2^{22} + \sum_{k=2}^{\infty} t_1^{21+8k} t_2^{18+4k}. \end{cases}$$

Let $\lambda = (7/4, 3/2)$. The smallest positive integer m such that $m\lambda \in \mathbb{Z}^2$ is 4. Observing the support of ζ, we note that for any $\tau \in Supp(\zeta)$, we have $\tau \in M = \mathbb{Z}^2 + \mathbb{Z}\lambda$. Thus, λ is the only characteristic exponent of ζ.

Let $\{\mathbb{1}_i\}_{i=1,\dots,n}$ be the canonical basis of $\mathbb{R}^n$.

Definition 3.1.7. *Consider a q.o. hypersurface with strongly normalized branches and characteristic exponents $\tilde{\lambda}_1, \dots, \tilde{\lambda}_s$. Let $\Gamma(\tilde{\lambda}_1, \dots, \tilde{\lambda}_s)$ be the semigroup generated by:*

$$m\mathbb{1}_1, \dots, m\mathbb{1}_n, \quad \psi_1 = m\tilde{\lambda}_1, \quad \text{and} \quad \psi_{j+1} = m_j\psi_j + m\tilde{\lambda}_{j+1} - m\tilde{\lambda}_j \quad \text{for } j = 1, \dots, s-1.$$

We call $\Gamma(\tilde{\lambda}_1, \dots, \tilde{\lambda}_s)$ the semigroup associated to an irreducible q.o. hypersurface with a strongly normalized branch and characteristic exponents $\tilde{\lambda}_1, \dots, \tilde{\lambda}_s$.

For the remainder of this work, unless stated otherwise, we consider only singular irreducible q.o. hypersurfaces with strongly normalized branches defined by q.o. polynomials in $\mathbb{C}\{\underline{x}\}[y]$.

3.2 The Case of Plane Curves ($n = 1$)

When $n = 1$, our q.o. singularities correspond to germs of plane curves. In this case, as we shall see, the associated semigroup Γ is a numerical semigroup.

Let $f \in \mathbb{C}\{x_1\}[y]$ be a q.o. polynomial of degree m and Y be defined by the equation $f = 0$. Set $\mathcal{O} = \mathbb{C}[[x_1, y]]/(f)$. Given $g \in \mathbb{C}[[x_1, y]]$, there are uniquely determined $h \in \mathbb{C}[[x_1, y]]$ and $r_0, \dots, r_{m-1} \in \mathbb{C}[[x_1]]$ such that $g = hf + r$, where $r = \sum_{i=0}^{m-1} r_i y^i$. See, for example, Section 6.2 of [12]. Since v^*f is the zero series, we have a map $\vartheta : \mathcal{O} \to \mathbb{N}_0 \cup \{\infty\}$, defined by $\vartheta(g) = \operatorname{ord}(v^*g) = \operatorname{ord}(v^*r)$. This map is, in fact, a valuation with the following property:

Proposition 3.2.1 (Proposition 1.1 of [30])**.** *The semigroup $\Gamma = \vartheta(\mathcal{O})$ contains all sufficiently large integers. More precisely, there exists a conductor $c \geq 0$ such that for all integers $i \geq c$, we have $i \in \Gamma$.*

O. Zariski also proved in [30] that $\vartheta(\mathcal{O}) = \Gamma(\tilde{\lambda}_1, \dots, \tilde{\lambda}_s) \cup \{+\infty\}$.

Definition 3.2.2. *Let C and D be two branches of plane curves. We say that C and D are* ***analytically equivalent****, written $C \cong D$, if there exists an analytic isomorphism $T : U \to U'$, where U and U' are neighborhoods of the origin in $\mathbb{C}^2$, such that:*

(a) C is defined in U and D is defined in U'.

(b) $T(C \cap U) = D \cap U'$.

A fundamental result by O. Zariski [30] demonstrates that we can obtain a branch analytically equivalent to Y (in the sense of Definition 3.2.2) that eliminates exponents belonging to the semigroup Γ.

Theorem 3.2.3 (Proposition 1.2 of [30]). *Let C be a branch of a singular irreducible plane curve germ Y at the origin. Let $\nu_1 < \nu_2 < \cdots < \nu_q$ be the integers in the set $\{\lambda + 1, \ldots, c\}$ which do not belong to Γ. Then there exists a branch C', with $C \cong C'$, parametrized as follows:*

$$\begin{cases} x' = t^m, \\ y' = t^\lambda + \sum_{i=1}^q a'_{\nu_i} t^{\nu_i}. \end{cases}$$

This form is hereafter referred to as a ***short parametrization*** *of Y.*

3.3 Generalization to Surfaces

We now outline a possible approach to generalizing Theorem 3.2.3 for the case of surfaces ($n = 2$). Let Y be a q.o. surface with a single characteristic exponent, denoted by λ. Let ζ be a strongly normalized branch of Y which admits a parametrization of the type

$$x_1 = t_1^m, \quad x_2 = t_2^m, \quad y = \underline{t}^{m\lambda} + \sum_{\sigma \in mM_1} c_\sigma \underline{t}^\sigma.$$

For each $\sigma \in \mathrm{Supp}(\upsilon^* y)$, it follows that $\sigma \geq m\lambda$ and $\sigma \in m\mathbb{Z}^2 + m\lambda\mathbb{Z}$. To Y we will associate an affine semigroup $\Lambda(\lambda)$ that we shall not explicitly define yet.

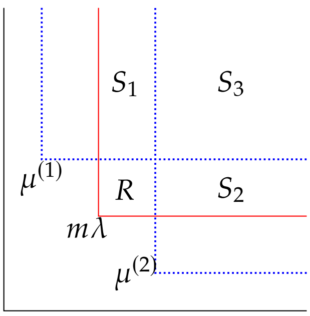

Figure 3.1: Sectioning of the support of a q.o. surface

We construct two affine semigroups, $\Lambda_1(\lambda)$ and $\Lambda_2(\lambda)$, such that

$$\Lambda_1(\lambda)_{\mathcal{G}} + \Lambda_2(\lambda)_{\mathcal{G}} = \Lambda(\lambda)_{\mathcal{G}},$$

where the subscript $\mathcal{G}$ denotes the group generated by the respective semigroup.

This construction allows us to compute the **Frobenius vectors** (cf. [5]) of $\Lambda_1(\lambda)$ and $\Lambda_2(\lambda)$. The Frobenius vector $\mu^{(i)}$ of the affine semigroup $\Lambda_i(\lambda)$ behaves analogously to a Frobenius number of a numerical semigroup: if $\sigma \in \Lambda_i(\lambda)_{\mathcal{G}}$ and $\sigma \in \operatorname{int}(\llcorner(\mu^{(i)}))$, then $\sigma \in \Lambda_i(\lambda)$.

As illustrated in Figure 3.1, the support of v^*y lies in $\llcorner(m\lambda)$. More specifically,

$$R = \llcorner(m\lambda) \setminus (\operatorname{int}(\llcorner(\mu^{(1)})) \cup \operatorname{int}(\llcorner(\mu^{(2)}))),$$

$$S_3 = \operatorname{int}(\llcorner(\mu^{(1)})) \cap \operatorname{int}(\llcorner(\mu^{(2)})),$$

$$S_1 = \operatorname{int}(\llcorner(\mu^{(1)})) \cap \llcorner(m\lambda) \setminus S_3,$$

and

$$S_2 = \operatorname{int}(\llcorner(\mu^{(2)})) \cap \llcorner(m\lambda) \setminus S_3.$$

Aiming to obtain a similar result to Theorem 3.2.3, our first objective is to identify the terms with exponents $\sigma \in (S_1 \cup S_2) \cap (\Gamma(\lambda)_{\mathcal{G}} \setminus \Gamma(\lambda))$ and then eliminate them via contact transformations. Moreover, by means of analytic and contact transformations, we ideally eliminate some terms lying in $R \cap (\Lambda_1(\lambda) + \Lambda_2(\lambda))$. This would leave us with a parametrization whose support is contained in $\llcorner(m\lambda) \cap S_3$ up to a finite number of terms that lie in R.

4

Contact Geometry

While contemporary contact geometry is a prominent research area within modern topology and differential geometry, its origins lie in the 19th-century formalization of classical mechanics and geometric optics. The physical intuition for the field was first captured by Christiaan Huygens, whose eponymous *Huygens' Principle* describes the evolution of wavefronts (see pgs. 272–279 [22]). It was then in 1871 that Sophus Lie introduced the concept of contact transformations in the context of differential equations and the geometric foundations of classical mechanics and thermodynamics in his doctoral dissertation (see [16]).

4.1 Fundamental Geometric Structures

Fix coordinates $(\underline{x}, y)$ in $\mathbb{C}^{n+1}$. At any point $o \in \mathbb{C}^{n+1}$, the cotangent space at o, $T_o^*\mathbb{C}^{n+1}$, is defined as the dual space of the tangent space $T_o\mathbb{C}^{n+1}$. Therefore, any covector can be uniquely written as

$$\sum_{i=1}^{n} \xi_i \, dx_i + \eta \, dy,$$

for some $\xi_1, \ldots, \xi_n, \eta \in \mathbb{C}$. We consider the disjoint union

$$T^*\mathbb{C}^{n+1} = \bigsqcup_{o \in \mathbb{C}^{n+1}} T_o^*\mathbb{C}^{n+1} = \mathbb{C}^{n+1} \times \mathbb{C}^{n+1},$$

called the **cotangent bundle** of $\mathbb{C}^{n+1}$. This bundle is naturally endowed with the canonical 1-form

$$\theta = \sum_{i=1}^{n} \xi_i \, dx_i + \eta \, dy.$$

Set $\underline{\xi} = (\xi_1, \ldots, \xi_n)$ and $(\underline{x}, y, \underline{\xi}, \eta) = (x_1, \ldots, x_n, y, \xi_1, \ldots, \xi_n, \eta)$. Let $\mathbb{P}^n$ denote the complex projective space of dimension n. We identify the zero section $\mathbb{C}^{n+1} \times \{0\}$ of $T^*\mathbb{C}^{n+1}$ with $\mathbb{C}^{n+1}$.

4.1.1 Projective Cotangent Bundle

Definition 4.1.1. *We define the* ***projective cotangent bundle***, $\mathbb{P}^*\mathbb{C}^{n+1}$, *as*

$$\left\{(\underline{x}, y; \xi_1 : \ldots : \xi_n : \eta) : (\underline{x}, y, \underline{\xi}, \eta) \in T^*\mathbb{C}^{n+1} \setminus \mathbb{C}^{n+1}\right\} \cong \mathbb{C}^{n+1} \times \mathbb{P}^n.$$

We shall write $(\underline{x}, y; \xi_1 : \ldots : \xi_n : \eta)$ as $(\underline{x}, y; \widehat{\xi} : \eta)$. Define

$$\pi : \mathbb{P}^*\mathbb{C}^{n+1} \to \mathbb{C}^{n+1}$$

as the canonical projection of $\mathbb{P}^*\mathbb{C}^{n+1}$, that is,

$$\pi(\underline{x}, y; \widehat{\xi} : \eta) = (\underline{x}, y).$$

Let U be the open set

$$\left\{(\widehat{\xi} : \eta) = (\xi_1 : \ldots : \xi_n : \eta) \in \mathbb{P}^n : \eta \neq 0\right\}.$$

For $i = 1, \ldots, n$, set $p_i = -\xi_i/\eta$ and $\underline{p} = (p_1, \ldots, p_n)$. The chart $\mathbb{C}^{n+1} \times U$ of $\mathbb{P}^*\mathbb{C}^{n+1}$ is isomorphic to $\mathbb{C}^{2n+1}$ endowed with coordinates $(\underline{x}, y, \underline{p}) = (x_1, \ldots, x_n, y, p_1, \ldots, p_n)$. Furthermore,

$$\left.\frac{\theta}{\eta}\right|_V = dy - \sum_{i=1}^{n} p_i\, dx_i = \omega,$$

where V is the open set

$$\left\{(\underline{x}, y, \underline{\xi}, \eta) \in T^*\mathbb{C}^{n+1} : \eta \neq 0\right\}.$$

The restriction of the canonical projection of $\mathbb{P}^*\mathbb{C}^{n+1}$ to $\mathbb{C}^{n+1} \times U$ will also be denoted by π.

Definition 4.1.2. *For $i = 1, \ldots, n$, define $\pi_i : \mathbb{C}^{2n+1} \to \mathbb{C}^{n+1}$, the* ***$i$-th fake projection*** *as $\pi_i(\underline{x}, y, \underline{p}) = (\underline{x}, p_i) = (x_1, \ldots, x_n, p_i)$.*

4.1.2 Legendrian Sets and Conormal

Definition 4.1.3. *Let $L \subset \mathbb{C}^{2n+1}$ be an analytic set of dimension n, with singular locus W. We say that L is* ***Legendrian*** *if*

$$\omega|_{L \setminus W} = 0.$$

A Legendrian set L is in ***generic position*** *at $o \in L$ if $L \cap \pi^{-1}(o)$ is a finite set.*

Definition 4.1.4. *Given an irreducible analytic subset X of $\mathbb{C}^{n+1}$, we define the* ***conormal*** *of X, and denote it by $\mathbb{P}^*_X\mathbb{C}^{n+1}$, as the smallest Legendrian set L such that $\pi(L) = X$. If X has irreducible components X_i, $i \in I$, we set $\mathbb{P}^*_X\mathbb{C}^{n+1} = \bigcup_{i \in I} \mathbb{P}^*_{X_i}\mathbb{C}^{n+1}$.*

Let $o \in X$. We define the ***limit of tangents*** *of X at o to be the set*

$$\Sigma_o(X) = \mathbb{P}^*_X\mathbb{C}^{n+1} \cap \pi^{-1}(o).$$

4.2 Application to Quasi-Ordinary Singularities

Consider an irreducible q.o. hypersurface Y with characteristic exponents $\tilde{\lambda}_1, \ldots, \tilde{\lambda}_s$. Assume further that $\mathbb{1} < \tilde{\lambda}_1$, where $\mathbb{1} = \sum_{j=1}^{n} \mathbb{1}_j$. Theorem 2.10 of [3] states that $\mathbb{P}^*_Y\mathbb{C}^{n+1}$ is in generic position at 0 and $\Sigma_0(Y)$ is the origin of $\mathbb{C}^{2n+1}$. We shall denote $\mathbb{C}^{2n+1} \cap U$, where U is an arbitrary small polydisc centered at the origin, endowed with ω, by $(\mathbb{C}^{2n+1}, \omega)$. The condition $\mathbb{1} < \tilde{\lambda}_1$ implies that $\mathbb{P}^*_Y\mathbb{C}^{n+1}$ is a subset of $(\mathbb{C}^{2n+1}, \omega)$.

4.2.1 Parametrization of the Conormal

For the following proposition, we refer to Definition 3.1.5.

Proposition 4.2.1. *Let Y be a q.o. hypersurface with characteristic exponents $\tilde{\lambda}_1, \ldots, \tilde{\lambda}_s$. Assume that $\mathbb{1} < \tilde{\lambda}_1$. The Puiseux parametrization v of Y induces a parametrization of $\mathbb{P}^*_Y\mathbb{C}^{n+1}$.*

Proof. Let $\tilde{\lambda}_1 = (\lambda_{1,1}, \ldots, \lambda_{1,n})$ and v be the Puiseux parametrization of Y defined, on a polydisc centered at zero, by

$$x_i = t_i^m,\ i = 1, \ldots, n,\ y = \underline{t}^{m\tilde{\lambda}_1} + \epsilon(\underline{t}), \tag{4.1}$$

where $\epsilon \in \mathbb{C}\{\underline{t}\}$ and $\mathrm{Supp}(\epsilon) \subseteq \llcorner(m\tilde{\lambda}_1) \setminus \{m\tilde{\lambda}_1\}$. For $t_1, \ldots, t_n \neq 0$,

$$v^*\omega = \sum_{i=1}^{n} \left(m\lambda_{1,i}\underline{t}^{m\tilde{\lambda}_1 - \mathbb{1}_i} + \frac{\partial \epsilon}{\partial t_i}(\underline{t}) - m t_i^{m-1} p_i \right) dt_i.$$

Therefore, $v^*\omega$ is zero if and only if, for all $i \in \{1, \ldots, n\}$

$$p_i = \lambda_{1,i}\underline{t}^{m\tilde{\lambda}_1 - m\mathbb{1}_i} + \frac{1}{m t_i^{m-1}} \frac{\partial \epsilon}{\partial t_i}(\underline{t}). \tag{4.2}$$

Since $\mathbb{1} < \tilde{\lambda}_1$, then $(0, \ldots, 0) < m(\tilde{\lambda}_1 - \mathbb{1}_i)$. Let $\sigma \in \mathrm{Supp}(\epsilon)$. Then $m\tilde{\lambda}_1 < \sigma$, which implies $m(\tilde{\lambda}_1 - \mathbb{1}_i) < \sigma - m\mathbb{1}_i$. Hence,

$$\frac{1}{m t_i^{m-1}} \frac{\partial \epsilon}{\partial t_i} \in \mathbb{C}\{\underline{t}\}.$$

We obtain a parametrization, for a small enough punctured polydisc centered at zero, of $\mathbb{P}^*_Y\mathbb{C}^{n+1}$ defined by

$$x_i = t_i^m,\ y = \underline{t}^{m\tilde{\lambda}_1} + \epsilon(\underline{t}),\ p_i = \lambda_{1,i}\underline{t}^{m\tilde{\lambda}_1 - m\mathbb{1}_i} + \frac{1}{m t_i^{m-1}} \frac{\partial \epsilon}{\partial t_i}(\underline{t}),\ i = 1, \ldots, n. \tag{4.3}$$

This parametrization extends, by closure, to $t_i = 0$, for each $i \in \{1, \ldots, n\}$. □

Definition 4.2.2. *Let Y be an irreducible q.o. hypersurface with characteristic exponents $\tilde{\lambda}_1, \ldots, \tilde{\lambda}_s$ such that $\mathbb{1} < \tilde{\lambda}_1$. Let Υ be the parametrization defined by (4.3). We call Υ the **Puiseux parametrization** of $\mathbb{P}^*_Y\mathbb{C}^{n+1}$.*

Lemma 4.2.3. *Let Y be an irreducible q.o. hypersurface with characteristic exponent λ. Assume that $\mathbb{1} < \lambda$. Then, for all $i \in \{1, \dots, n\}$, $\pi_i(\mathbb{P}^*_Y\mathbb{C}^{n+1})$ is a q.o. hypersurface with characteristic exponent $\lambda - \mathbb{1}_i$, with branches, up to a change of coordinates, strongly normalized and that belong to the ring $\mathbb{C}\left\{\underline{x}^{1/m}\right\}$.*

Proof. We start the proof from (4.3). Suppose Y only has $\lambda = (\lambda_1, \dots, \lambda_n)$ as its characteristic exponent. Notice that $m\mathbb{Z}^n + m\mathbb{Z}\lambda = m\mathbb{Z}^n + m\mathbb{Z}(\lambda - \mathbb{1}_i)$. If $\sigma \in \operatorname{Supp}(\frac{1}{mt_i^{m-1}}\frac{\partial \epsilon}{\partial t_i})$, then $\sigma = \tau - m\mathbb{1}_i$, for some $\tau \in \operatorname{Supp}(\epsilon) \subseteq m\mathbb{Z}^n + m\mathbb{Z}\lambda$. Hence $\sigma \in m\mathbb{Z}^n + m\mathbb{Z}(\lambda - \mathbb{1}_i)$. Thus, $\pi_i(\mathbb{P}^*_Y\mathbb{C}^{n+1})$ is a q.o. hypersurface with characteristic exponent $\lambda - \mathbb{1}_i$. On the other hand, since $\tau \geq m\lambda$, then $\sigma = \tau - m\mathbb{1}_i \geq m(\lambda - \mathbb{1}_i)$. Therefore, after possibly a change of coordinates, we may assume that the branch is strongly normalized.

As for the last part, let m' be the ramification number of $\lambda - \mathbb{1}_i$. By definition,

$$m'(\lambda - \mathbb{1}_i) = (a_1, \dots, a_n),$$

for some $a_1, \dots, a_n \in \mathbb{Z}$. Therefore,

$$m'\lambda = (a_1, \dots, a_n) + m'\mathbb{1} \in \mathbb{Z}^n,$$

and we conclude that $m \leq m'$. Since

$$m(\lambda - \mathbb{1}_i) = m\lambda - m\mathbb{1}_i \in \mathbb{Z}^n,$$

we also conclude $m \geq m'$. Hence, $m = m'$. $\square$

4.3 Singular Locus of the Conormal

In the deformation theory of singularities, the assumption that an analytic set has an isolated singularity is often essential to obtain a deformation with a finite base (see, for example, Theorem 1.16 of [10]). This requirement motivates our study of when the conormal, in generic position, of a q.o. surface has an isolated singularity at the origin. Definition 2.0.14 describes the singular locus for the hypersurface case. For the conormal of a surface, however, characterizing the singular locus is more delicate. In this context, without going into too much detail, we will define a singular point as one where Theorem 4.3.15 of [8] fails to hold. We begin by relating the singular locus of a surface to that of its conormal.

Proposition 4.3.1. *Let $h \in \mathbb{C}\{x_1, x_2, y\}$ and X be the surface defined by the equation $h = 0$. Assume that the origin does not belong to the singular locus of X. Then $\Sigma_0(X)$ is just a point that does not belong to the singular locus of $\mathbb{P}^*_X\mathbb{C}^3$.*

Proof. Since the origin does not belong to the singular locus of X, we can apply the Implicit Function Theorem. There is an open set U and, possibly after swapping the order of the variables, $\varphi \in \mathbb{C}\{x_1, x_2\}$ such that the origin belongs to U and $X \cap U$ is defined by the equation $y - \varphi = 0$. Hence, $\mathbb{P}^*_X\mathbb{C}^3 \cap \pi^{-1}(U)$ is defined by the equations

$$y - \varphi = 0,\ p_1 - \frac{\partial \varphi}{\partial x_1} = 0,\ p_2 - \frac{\partial \varphi}{\partial x_2} = 0.$$

Therefore

$$\Sigma_0(X) = \left\{\left(0,0,0,\frac{\partial \varphi}{\partial x_1}(0,0),\frac{\partial \varphi}{\partial x_2}(0,0)\right)\right\}$$

and the Jacobian matrix of the defining functions of the conormal is

$$\begin{pmatrix} -\frac{\partial \varphi}{\partial x_1} & -\frac{\partial \varphi}{\partial x_2} & 1 & 0 & 0 \\ -\frac{\partial^2 \varphi}{\partial x_1^2} & -\frac{\partial^2 \varphi}{\partial x_1 \partial x_2} & 0 & 1 & 0 \\ -\frac{\partial^2 \varphi}{\partial x_2 \partial x_1} & -\frac{\partial^2 \varphi}{\partial x_2^2} & 0 & 0 & 1 \end{pmatrix}.$$

Since the Jacobian matrix has maximal rank at the point $\Sigma_0(X)$, the proposition holds. □

Remark 4.3.2. *An immediate consequence of Proposition 4.3.1 is that*

$$\pi\left(\mathrm{Sing}\left(\mathbb{P}^*_X\mathbb{C}^3\right)\right) \subseteq \mathrm{Sing}(X).$$

For Theorem 4.3.3, we will need to study the rank of the Jacobian matrix of a Puiseux parametrization of the conormal of a q.o. surface. The problem is that this type of parametrization might not be "good". For example, consider Y defined by the equation $y^6 - x_1^9 x_2^8 = 0$. Then Y is a q.o. surface with characteristic exponent $(3/2, 4/3)$ that admits the "good" parametrization

$$x_1 = t_1^2,\ x_2 = t_2^3,\ y = t_1^3 t_2^4.$$

So, for our purpose, we need to refine the parametrization (4.3). Let $i \in \{1,2\}$ and $h \in \mathbb{C}\{t_1, t_2\}$. Define $\mathrm{Supp}_i(h)$ as the projection of $\mathrm{Supp}(h)$ onto the i-th coordinate. Set ϖ_i as the greatest common divisor of the set $\{m, m\lambda_{1,i}\} \cup \mathrm{Supp}_i(\epsilon)$. In (4.3), we perform the reparametrization defined by $s_i = t_i^{\varpi_i}$, and obtain

$$\begin{aligned} x_1 &= s_1^{k_1}, \\ x_2 &= s_2^{k_2}, \\ y &= s_1^{k_1\lambda_{1,1}} s_2^{k_2\lambda_{1,2}} + \varepsilon(s_1, s_2), \\ p_1 &= \lambda_{1,1} s_1^{k_1(\lambda_{1,1}-1)} s_2^{k_2\lambda_{1,2}} + \frac{1}{k_1 s_1^{k_1-1}} \frac{\partial \varepsilon}{\partial s_1}(s_1, s_2), \\ p_2 &= \lambda_{1,2} s_1^{k_1\lambda_{1,1}} s_2^{k_2(\lambda_{1,2}-1)} + \frac{1}{k_2 s_2^{k_2-1}} \frac{\partial \varepsilon}{\partial s_2}(s_1, s_2), \end{aligned} \tag{4.4}$$

where $\varepsilon \in \mathbb{C}\{s_1, s_2\}$, $\mathrm{Supp}(\varepsilon) \subseteq \llcorner(k_1\lambda_{1,1}, k_2\lambda_{1,2}) \setminus \{(k_1\lambda_{1,1}, k_2\lambda_{1,2})\}$ and the greatest common divisor of the set $\{k_i, k_i\lambda_{1,i}\} \cup \mathrm{Supp}_i(\varepsilon)$ is one for $i = 1, 2$.

Theorem 4.3.3. *Let Y be an irreducible q.o. surface with characteristic exponents $\tilde{\lambda}_1, \ldots, \tilde{\lambda}_s$. Assume that, for $i = 1, 2$, $\lambda_{1,i} > 1$. Then $\mathbb{P}^*_Y\mathbb{C}^3$ has an isolated singularity at the origin if and only if $s = 1$ and*

$$\lambda = \left(\frac{m+1}{m}, \frac{m+1}{m}\right).$$

Proof. From Lemma 3.1.3, we know that $\mathrm{Sing}(Y) = V(y, x_1x_2)$. Remark 4.3.2 implies

$$\mathrm{Sing}\left(\mathbb{P}^*_Y\mathbb{C}^3\right) \subseteq \mathbb{P}^*_Y\mathbb{C}^3 \cap \pi^{-1}(V(y, x_1x_2)) = V(y, x_1x_2, p_1, p_2).$$

Assume that $\mathbb{P}^*_Y\mathbb{C}^3$ has an isolated singularity at the origin. Let $a \in \mathbb{C} \setminus \{0\}$ with $|a| \ll 1$ and let V be a sufficiently small open set such that $(0, a, 0) \in V$. Then $Y \cap V$ is equisingular, that is, all the irreducible components of $Y \cap V$ have the same characteristic exponent and verify statement $(a)(ii)$ of Lemma 3.1.3. If $Y \cap V$ is not irreducible, then the conormal of $Y \cap V$ is the union of the conormal of each irreducible component, which implies that $(0, a, 0, 0, 0)$ is a singular point of $\mathbb{P}^*_Y\mathbb{C}^3$. Therefore, $Y \cap V$ must be irreducible.

Let (4.4) be a parametrization of $\mathbb{P}^*_Y\mathbb{C}^3$. Set

$$\varepsilon_i(s_1, s_2) = \frac{1}{k_i s_i^{k_i-1}} \frac{\partial \varepsilon}{\partial s_i}(s_1, s_2), \ i = 1, 2.$$

The Jacobian matrix $J(s_1, s_2)$ of (4.4) equals

$$\begin{bmatrix} k_1\, s_1^{k_1-1} & 0 \\ 0 & k_2 s_2^{k_2-1} \\ k_1\lambda_{1,1}\, s_1^{k_1\lambda_{1,1}-1} s_2^{k_2\lambda_{1,2}} + \frac{\partial \varepsilon}{\partial s_1} & k_2\lambda_{1,2}\, s_1^{k_1\lambda_{1,1}} s_2^{k_2\lambda_{1,2}-1} + \frac{\partial \varepsilon}{\partial s_2} \\ k_1\lambda_{1,1}(\lambda_{1,1}-1)\, s_1^{k_1(\lambda_{1,1}-1)-1} s_2^{k_2\lambda_{1,2}} + \frac{\partial \varepsilon_1}{\partial s_1} & k_2\lambda_{1,1}\lambda_{1,2}\, s_1^{k_1(\lambda_{1,1}-1)} s_2^{k_2\lambda_{1,2}-1} + \frac{\partial \varepsilon_1}{\partial s_2} \\ k_1\lambda_{1,1}\lambda_{1,2}\, s_1^{k_1\lambda_{1,1}-1} s_2^{k_2(\lambda_{1,2}-1)} + \frac{\partial \varepsilon_2}{\partial s_1} & k_2\lambda_{1,2}(\lambda_{1,2}-1)\, s_1^{k_1\lambda_{1,1}} s_2^{k_2(\lambda_{1,2}-1)-1} + \frac{\partial \varepsilon_2}{\partial s_2} \end{bmatrix}$$

Since $\lambda_{1,1} > 1$, we cannot have $k_1\lambda_{1,1} - 1 = 0$ and the rank of $J(0, a^{1/k_2})$ is two if and only if $k_1(\lambda_{1,1} - 1) - 1 = 0$, that is, if and only if

$$\lambda_{1,1} = \frac{k_1 + 1}{k_1}$$

Therefore, by symmetry, for $\mathbb{P}^*_Y\mathbb{C}^3$ to have an isolated singularity, we must have

$$\lambda_{1,i} = \frac{k_i + 1}{k_i}, \ i = 1, 2.$$

Note that $Y \cap V$ has first characteristic exponent

$$\left(\frac{k_1+1}{k_1}, 0\right).$$

Assume that $k_1 \neq k_2$ and let $\widetilde{k}$ be the greatest common divisor between k_1 and k_2. Then, by definition,

$$m_1 = \frac{k_1 k_2}{\widetilde{k}}.$$

This implies that $Y \cap V$ has at most multiplicity $k_1 m_2 \cdots m_s$ at $(0, a, 0)$ and at least $k_2/\widetilde{k}$ irreducible components. Hence, we cannot have $k_1 \neq k_2$.

Assume that $k_1 = k_2 < m$. Since $m_1 = k_1 < m$, we must have $s \geq 2$. Since, for $i = 1, 2$, $m\lambda_{2,i} = \ell_i \varpi_i$, for some positive integer ℓ_i, and $m = k_1 \varpi_i$, we obtain $\widetilde{\lambda}_2 = (\ell_1/k_1, \ell_2/k_2)$. But

$$\left(\frac{\ell_1}{k_1}, 0\right) = (-\ell_1, 0) + \ell_1 \left(\frac{k_1+1}{k_1}, 0\right) \in \mathbb{Z}^2 + \mathbb{Z}\left(\frac{k_1+1}{k_1}, 0\right),$$

which implies that $Y \cap V$ has at most multiplicity $k_1 m_3 \ldots m_s$ at $(0, a, 0)$ and at least m_2 irreducible components. Hence, this case also cannot happen.

Assume that $k_1 = k_2 = m$. Then $m = m_1$ and we have $s = 1$. Then $Y \cap V$ is irreducible, with characteristic exponent

$$\left(\frac{m+1}{m}, 0\right)$$

and both matrices $J(0, a^{1/m})$ and $J(a^{1/m}, 0)$ have rank 2. We conclude that in this case, $\mathbb{P}^*_Y\mathbb{C}^3$ has an isolated singularity at the origin. $\square$

Remark 4.3.4. *The techniques used in the proof of Theorem 4.3.3 show us that if Y has one characteristic exponent equal to $((m+1)/m, 0)$, then the singular locus of $\mathbb{P}^*_Y\mathbb{C}^3$ is the empty set.*

Example 4.3.5. *This example shows that Theorem 4.3.3 does not hold for $n \geq 3$. Consider the q.o. hypersurface Y with branch*

$$\zeta = x_1^{5/4} x_2^{5/4} x_3^{5/4}.$$

The singular locus of $\mathbb{P}^*_Y\mathbb{C}^4$ is defined by the equations

$$p_1 = p_2 = p_3 = y = x_1 x_2 = x_1 x_3 = x_2 x_3 = 0.$$

This computation was made using the following prompts in a Singular.jl *session:*

```
using Latexify
using Singular
var_names = vcat(["t_$i" for i in 1:3], ["x_$i" for i in 1:4],
                 ["p_$i" for i in 1:3])
R, all_vars = polynomial_ring(QQ, var_names; ordering=ordering_ds())
t = all_vars[1:3]
x = all_vars[4:7]
p = all_vars[8:10]
conorpar = Ideal(R, x[1]-t[1]^4, x[2]-t[2]^4,
                 x[3]-t[3]^4, x[4]-t[1]^5*t[2]^5*t[3]^5,
                 p[1]-5//4*t[1]*t[2]^5*t[3]^5,
                 p[2]-5//4*t[1]^5*t[2]*t[3]^5,
                 p[3]-5//4*t[1]^5*t[2]^5*t[3])
conor = eliminate(eliminate(eliminate(conorpar, t[1]), t[2]), t[3])
for g in gens(conor)
```

```
    display(latexify(string(g)))
end
singl = Singular.LibPrimdec.radical(std(Singular.LibSing.slocus(conor)))
for g in gens(singl)
    display(latexify(string(g)))
end
```

Example 4.3.6. *This example helps clarify that Theorem 4.3.3 also does not hold when we have a q.o. surface with $\lambda_{1,2} = 0$ and $\lambda_{s,2} \neq 0$. Consider the q.o. surface Y with branch*

$$\zeta = x_1^{5/4} - x_1^{5/4} x_2^{5/4}$$

Then Y has characteristic exponents $(5/4, 0), (5/4, 5/4)$ and its singular locus equals

$$V(x_1, y) \cup V(y^4 - x_1^5, x_2).$$

Using Singular.jl*, one obtains*

$$\operatorname{Sing}\left(\mathbb{P}_Y^* \mathbb{C}^3\right) = V(x_1, y, p_1, p_2).$$

Theorem 4.3.3 motivates the following definition.

Definition 4.3.7. *We say that an irreducible q.o. surface Y is **plain** if it is a surface with one characteristic exponent equal to*

$$\left(\frac{m+1}{m}, \frac{m+1}{m}\right).$$

4.4 Infinitesimal Contact Transformations

Definition 4.4.1. *The germ of a holomorphic map $\Phi : (\mathbb{C}^{2n+1}, \omega) \to (\mathbb{C}^{2n+1}, \omega)$ is said to be a **contact transformation** if there is $\varphi \in \mathbb{C}\{\underline{x}, y, \underline{p}\}$ such that $\varphi(\underline{0}, 0, \underline{0}) \neq 0$ and $\Phi^* \omega = \varphi \omega$.*

Example 4.4.2. *Consider the plane curve C with Puiseux parametrization*

$$x = t^2,\ y = t^3.$$

Then $\mathbb{P}_C^* \mathbb{C}^2$ admits the parametrization

$$x = t^2,\ y = t^3,\ p = \frac{3}{2} t.$$

The map $\Phi(x, y, p) = (-p, y - xp, x)$ is a contact transformation, where $\Phi^* \omega = \varphi \omega$ with $\varphi(x, y, p) = 1$, and $D = \pi(\Phi(\mathbb{P}_C^* \mathbb{C}^2))$ is a plane curve with parametrization

$$x = -\frac{3}{2} t,\ y = -\frac{1}{2} t^3.$$

The curve C is singular at the origin with characteristic exponent $3/2$, while D is smooth.

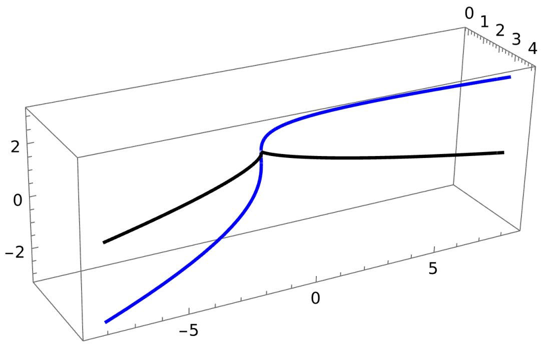

Figure 4.1: Parametrization of C and of $\mathbb{P}^*_C\mathbb{C}^2$

The set of all contact transformations is a group. Let $\mathcal{M}$ denote the maximal ideal of $\mathbb{C}\{\underline{x}, y, \underline{p}\}$. We are especially interested in the subgroup of the **infinitesimal** contact transformations, that is, contact transformations of the type

$$(\underline{x}, y, \underline{p}) \mapsto (x_1 + \alpha_1, \ldots, x_n + \alpha_n, y + \beta, p_1 + \gamma_1, \ldots, p_n + \gamma_n), \tag{4.5}$$

where $\alpha_i, \beta, \gamma_i \in \mathbb{C}\{\underline{x}, y, \underline{p}\}$ and $\partial\alpha_i/\partial x_j, \beta, \partial\beta/\partial y, \partial\gamma_i/\partial p_j \in \mathcal{M}$, for all $i, j \in \{1, \ldots, n\}$. We shall denote $(x_1 + \alpha_1, \ldots, x_n + \alpha_n, y + \beta, p_1 + \gamma_1, \ldots, p_n + \gamma_n)$ by $(\underline{x} + \underline{\alpha}, y + \beta, \underline{p} + \underline{\gamma})$ and the map (4.5) by $\Phi_{\underline{\alpha},\beta}$.

The following theorem is a valuable tool to build infinitesimal contact transformations.

Theorem 4.4.3 (Theorem 4.1 of [21])**.** *Let $\alpha_1, \ldots, \alpha_n \in \mathbb{C}\{\underline{x}, y, \underline{p}\}$, $\beta_0 \in \mathbb{C}\left\{\underline{x}, y\right\}$ be such that*

$$\frac{\partial\alpha_i}{\partial x_j}, \frac{\partial\beta_0}{\partial y} \in \mathcal{M},\ i, j = 1, \ldots, n.$$

There are uniquely determined $\beta, \gamma_1, \ldots, \gamma_n \in \mathbb{C}\{\underline{x}, y, \underline{p}\}$ such that

$$\beta - \beta_0 \in (\underline{p})\mathbb{C}\{\underline{x}, y, \underline{p}\}$$

and $\alpha_1, \ldots, \alpha_n, \beta, \gamma_1, \ldots, \gamma_n$ define an infinitesimal contact transformation $\Phi_{\underline{\alpha},\beta}$.

Definition 4.4.4 (Microlocal equivalence)**.** *We say that two germs of hypersurfaces at the origin, Y_1 and Y_2, are **microlocally equivalent** if there is a contact transformation Ψ such that $\Psi(\mathbb{P}^*_{Y_1}\mathbb{C}^{n+1}) = \mathbb{P}^*_{Y_2}\mathbb{C}^{n+1}$.*

Example 4.4.2 shows that if Ψ is a contact transformation and Y is a q.o. hypersurface, even if $\pi(\Psi(\mathbb{P}^*_Y\mathbb{C}^{n+1}))$ is a q.o. hypersurface, the two hypersurfaces might not have the same characteristic exponents. The main goal of this thesis is to study infinitesimal contact transformations that transform the conormal of a q.o. hypersurface with one characteristic exponent into the conormal of a q.o. hypersurface with the same characteristic exponent. The goals of this study are inspired by Theorem 3.2.3.

5

LEGENDRIAN SEMIGROUP

Having established the necessary foundational concepts, we are now positioned to examine the case of a single characteristic exponent. The following discussion explores the algebraic framework underlying this geometric problem. We denote the group generated by a semigroup $\mathcal{S}$ by $\mathcal{S}_{\mathcal{G}}$.

Let $\mathcal{S}$ be a subsemigroup of the additive semigroup $\mathbb{N}_0^n$. We say that $\sigma \in \mathbb{Z}^n$ is the **Frobenius vector** of $\mathcal{S}$ (see Theorem 1 of [5]) if $\sigma \notin \mathcal{S}$ and

$$\mathcal{S}_{\mathcal{G}} \cap \operatorname{int}(\llcorner(\sigma)) \subset \mathcal{S}.$$

For this section, we consider an irreducible q.o. hypersurface Y to have one characteristic exponent λ. Recall Definition 3.1.7. The Frobenius vector of $\Gamma(\lambda)$ is denoted by $\mathrm{Fr}(\lambda)$.

We now proceed with the exposition of some results and notions that will be necessary later on.

Lemma 5.0.1. *The Frobenius vector of* $\Gamma(\lambda)$ *is*

$$(m-1)m\lambda - m\mathbb{1}.$$

Proof. Let

$$D_1 = \det(mI_n) = m^n,$$

where I_n denotes the $n \times n$ identity matrix, and

$$D_2 = \gcd\left(\det(mI_n), \det(mI_n^{(i)}\ (m\lambda)^T) : i = 1, \dots, n\right),$$

where $I_n^{(i)}$ denotes the matrix I_n with the i-th column removed.

In order to compute D_2, notice that

$$\left|\det\begin{bmatrix} & m\lambda_1 \\ mI_n^{(i)} & \vdots \\ & m\lambda_n \end{bmatrix}\right| = m^n \left|\det\begin{bmatrix} & \lambda_1 \\ I_n^{(i)} & \vdots \\ & \lambda_n \end{bmatrix}\right| = m^n\lambda_i.$$

Hence,

$$D_2 = \gcd(m^n, m^n\lambda_i : i = 1, \dots, n) = m^{n-1}\gcd(m, m\lambda_i : i = 1, \dots, n).$$

We claim that $\gcd(m, m\lambda_i : i = 1, \ldots, n) = 1$. In fact, let $k \in \mathbb{N}$ be such that k divides m and $m\lambda_i$, for each $i = 1, \ldots, n$. Then there exist $d, r_1, \ldots, r_n \in \mathbb{N}$ such that

$$m = dk, m\lambda_i = r_i k, \ i = 1, \ldots, n.$$

Therefore,

$$\lambda_i = \frac{r_i k}{m} = \frac{r_i}{d}, \ i = 1, \ldots, n.$$

Thus $d\lambda \in \mathbb{Z}^n$, which implies that $d = m$ and $k = 1$, as desired.
Considering

$$e_1 = \frac{D_1}{D_2} = \frac{m^n}{m^{n-1}} = m,$$

by Theorem 1.1 of [5], one has that

$$Fr(\lambda) = (m-1)m\lambda - m\mathbb{1}.$$

□

Definition 5.0.2. *Let $\lambda > \mathbb{1}$. We define the **i-th fake Legendrian semigroup of** $\Gamma(\lambda)$, $\Lambda_i(\lambda)$, as the semigroup generated by*

$$m\mathbb{1}_1, \ldots, m\mathbb{1}_n, \text{ and } m\lambda - m\mathbb{1}_i,$$

for each $i = 1, \ldots, n$. We call the semigroup

$$\Lambda(\lambda) = \Lambda_1(\lambda) + \ldots + \Lambda_n(\lambda)$$

*the **Legendrian semigroup** of $\Gamma(\lambda)$.*

Notice that, by Lemma 4.2.3, the i-th fake Legendrian semigroup of $\Gamma(\lambda)$ is precisely the semigroup associated to the i-th fake projection of $\mathbb{P}^*_Y\mathbb{C}^{n+1}$.

Proposition 5.0.3. *Let $\lambda > \mathbb{1}$. Then, for each $i = 1, \ldots, n$,*

$$\Lambda_i(\lambda) \subseteq \Lambda(\lambda) \tag{5.1}$$

and

$$\Gamma(\lambda)_{\mathcal{G}} = \Lambda_i(\lambda)_{\mathcal{G}} = \Lambda(\lambda)_{\mathcal{G}}. \tag{5.2}$$

Proof. Relation (5.1) follows immediately from the definition of $\Lambda(\lambda)$. We now prove (5.2) for each $i = 1, \ldots, n$. Since $m\lambda - m\mathbb{1}_i \in \Gamma(\lambda)_{\mathcal{G}}$ and $m\lambda = (m\lambda - m\mathbb{1}_i) + m\mathbb{1}_i \in \Lambda_i(\lambda)_{\mathcal{G}}$, we conclude that $\Gamma(\lambda)_{\mathcal{G}} = \Lambda_i(\lambda)_{\mathcal{G}}$.

Also, given (5.1), we have that $\Lambda_i(\lambda)_{\mathcal{G}} \subseteq \Lambda(\lambda)_{\mathcal{G}}$. On the other hand, given $x \in \Lambda(\lambda)_{\mathcal{G}}$, there exist $a_1, \ldots, a_n, b_1, \ldots, b_n \in \mathbb{Z}$ such that

$$\begin{aligned} x &= a_1 m\mathbb{1}_1 + \ldots + a_n m\mathbb{1}_n + b_1(m\lambda - m\mathbb{1}_1) + \ldots + b_n(m\lambda - m\mathbb{1}_n) \\ &= (a_1 - b_1)m\mathbb{1}_1 + \ldots + (a_n - b_n)m\mathbb{1}_n + (b_1 + \ldots + b_n)m\lambda \in \Gamma(\lambda)_{\mathcal{G}} = \Lambda_i(\lambda)_{\mathcal{G}}. \end{aligned}$$

Hence, $\Lambda_i(\lambda)_{\mathcal{G}} = \Lambda(\lambda)_{\mathcal{G}}$. □

Proposition 5.0.4. *Let $\lambda > \mathbb{1}$. An element τ of $\mathbb{N}_0^n$ belongs to $\Lambda(\lambda)_{\mathcal{G}}$ if and only if there are integers $a_1, \ldots, a_n, b_1, \ldots, b_n \in \mathbb{Z}$ such that $(0, 0, \ldots, 0) \leq (a_1, a_2, \ldots, a_n) \leq (m-1, m-1, \ldots, m-1)$ and*

$$\tau = \sum_{k=1}^{n} b_k m \mathbb{1}_k + \sum_{k=1}^{n} a_k (m\lambda - m\mathbb{1}_k).$$

Furthermore, τ belongs to $\Lambda(\lambda)$ if and only if $b_1, \ldots, b_n$ are non-negative integers.

Proof. Assume that τ belongs to $\Lambda(\lambda)_{\mathcal{G}}$. There are integers $\widetilde{a}_1, \ldots, \widetilde{a}_n, \widetilde{b}_1, \ldots, \widetilde{b}_n$ such that

$$\tau = \sum_{k=1}^{n} \widetilde{b}_k m \mathbb{1}_k + \sum_{k=1}^{n} \widetilde{a}_k (m\lambda - m\mathbb{1}_k). \tag{5.3}$$

Let a_k, d_k, for each $k = 1, \ldots, n$, be the unique integers such that $a_k \in \{0, \ldots, m-1\}$ and $\widetilde{a}_k = d_k m + a_k$. Now,

$$\begin{aligned}
\tau &= \sum_{k=1}^{n} \widetilde{b}_k m \mathbb{1}_k + \sum_{k=1}^{n} \widetilde{a}_k (m\lambda - m\mathbb{1}_k) \\
&= \sum_{k=1}^{n} \widetilde{b}_k m \mathbb{1}_k + \sum_{k=1}^{n} (d_k m + a_k)(m\lambda - m\mathbb{1}_k) \\
&= \sum_{k=1}^{n} \widetilde{b}_k m \mathbb{1}_k + \sum_{k=1}^{n} d_k m (m\lambda - m\mathbb{1}_k) + \sum_{k=1}^{n} a_k (m\lambda - m\mathbb{1}_k).
\end{aligned}$$

Writing $\lambda = \sum_{k=1}^{n} \lambda_k \mathbb{1}_k$, one gets

$$\begin{aligned}
\tau &= \sum_{k=1}^{n} (\widetilde{b}_k - d_k m) m \mathbb{1}_k + \sum_{k=1}^{n} \left(d_k m \sum_{j=1}^{n} m \lambda_j \mathbb{1}_j \right) + \sum_{k=1}^{n} a_k (m\lambda - m\mathbb{1}_k) \\
&= \sum_{k=1}^{n} (\widetilde{b}_k - d_k m) m \mathbb{1}_k + \sum_{k=1}^{n} \left(\sum_{j=1}^{n} m d_j m \lambda_k \mathbb{1}_k \right) + \sum_{k=1}^{n} a_k (m\lambda - m\mathbb{1}_k) \\
&= \sum_{k=1}^{n} \left(\widetilde{b}_k - d_k m + m\lambda_k \sum_{j=1}^{n} d_j \right) m \mathbb{1}_k + \sum_{k=1}^{n} a_k (m\lambda - m\mathbb{1}_k).
\end{aligned}$$

Set, for each $k = 1, \ldots, n$,

$$b_k := \widetilde{b}_k - d_k m + m\lambda_k \sum_{j=1}^{n} d_j.$$

If $\widetilde{a}_1, \ldots, \widetilde{a}_n, \widetilde{b}_1, \ldots, \widetilde{b}_n$ are non-negative integers then, since $\lambda > \mathbb{1}$, $b_1, \ldots, b_n$ are non-negative integers. The other implications are trivial. □

Remark 5.0.5. *i) Let $b_1, \ldots, b_n$ be integers, $a_1, \ldots, a_n \in \{0, \ldots, m-1\}$, and*

$$\tau = \sum_{k=1}^{n} b_k m \mathbb{1}_k + \sum_{k=1}^{n} a_k (m\lambda - m\mathbb{1}_k).$$

We can write τ as

$$\tau = \sum_{\substack{k=1 \\ k\notin\{i,j\}}}^{n} b_k m\mathbb{1}_k + \sum_{\substack{k=1 \\ k\notin\{i,j\}}}^{n} a_k(m\lambda - m\mathbb{1}_k) + (b_i - d_i)m\mathbb{1}_i + (b_j + d_i)m\mathbb{1}_j + \\ + (a_i - d_i)(m\lambda - m\mathbb{1}_i) + (a_j + d_i)(m\lambda - m\mathbb{1}_j),$$

where $d_i \in \{0, \dots, a_j\}$. If $b_1, \dots, b_n$ are non-negative integers, we cannot guarantee that each $b_i - d_i$ is a non-negative integer, even when $a_i + a_j \leq m-1$. An example of this is, for $\lambda = (7/4, 3/2)$,

$$(7,6) - (0,4) = (4,0) - (0,4) + ((7,6) - (4,0)).$$

ii) Since cone$(\Gamma(\lambda)) = \mathbb{R}^n_{\geq 0}$, $\Gamma(\lambda)$ is an affine simplicial semigroup. Lemma 1.3 of [25] then tells us that an element τ of $\Gamma(\lambda)_{\mathcal{G}}$ can be uniquely written as

$$\sum_{k=1}^{n} b_k m\mathbb{1}_k + am\lambda,$$

where $b_1, \dots, b_n$ are integers and $a \in \{0, 1, \dots, m-1\}$. Furthermore, using the same reasoning as in the proof of Proposition 5.0.4, one can verify that τ belongs to $\Gamma(\lambda)$ if and only if $b_1, \dots, b_n$ are non-negative. Moreover, for each $i \in \{1, \dots, n\}$, cone$(\Lambda_i(\lambda)) = \mathbb{R}^n_{\geq 0}$, so that $\Lambda_i(\lambda)$ is also an affine simplicial semigroup. Similarly, every $\tau \in \Lambda_i(\lambda)_{\mathcal{G}}$ can be written uniquely as

$$\sum_{k=1}^{n} b_k m\mathbb{1}_k + a(m\lambda - m\mathbb{1}_i),$$

where $b_1, \dots, b_n$ are integers and $a \in \{0, 1, \dots, m-1\}$.

Lemma 5.0.6. *Let $\lambda > \mathbb{1}$. The following statements hold:*

i) If $m = 2$ then

$$(\Gamma(\lambda)_{\mathcal{G}} \setminus \Gamma(\lambda)) \cap \llcorner(2\lambda) = \emptyset.$$

ii) If $m \geq 3$ then

$$(\Gamma(\lambda)_{\mathcal{G}} \setminus \Gamma(\lambda)) \cap \llcorner(m\lambda)$$

is not finite.

iii) If $n = 2$, $m \geq 3$, and $\lambda_1, \lambda_2 < m/(m-2)$, then

$$(\Gamma(\lambda)_{\mathcal{G}} \setminus \Lambda(\lambda)) \cap \llcorner(m\lambda) \subseteq [m\lambda_1, (m-1)m\lambda_1 - m] \times [m\lambda_2, (m-1)m\lambda_2 - m].$$

iv) If $n = 2$, $m \geq 3$, and $\lambda_i \geq m/(m-2)$, for some $i \in \{1, 2\}$, then

$$(\Gamma(\lambda)_{\mathcal{G}} \setminus \Lambda(\lambda)) \cap \llcorner(m\lambda)$$

is not finite.

Proof. We start by proving i). Let $\tau \in \Gamma(\lambda)_{\mathcal{G}} \cap \llcorner(2\lambda)$, then $\tau \in \Gamma(\lambda)_{\mathcal{G}}$ and $\tau \geq 2\lambda$. By Lemma 5.0.1, $Fr(\lambda) = 2\lambda - 2\mathbb{1}$. Therefore $\tau \in \llcorner(\mathrm{int}(Fr(\lambda)))$ and, by the definition of the Frobenius vector, $\tau \in \Gamma(\lambda)$. Thus, $\Gamma(\lambda)_{\mathcal{G}} \cap \llcorner(2\lambda) \subseteq \Gamma(\lambda)$, and the result follows.

We now prove ii). We begin by checking that, for $m \geq 3$,

$$m\lambda < Fr(\lambda) = (m-1)m\lambda - m\mathbb{1}.$$

This statement is equivalent to

$$\mathbb{1} < (m-2)\lambda,$$

which holds, since $m \geq 3$ and $\lambda > (1, 1, \ldots, 1)$. For all $\ell \in \mathbb{N}_0$,

$$Fr(\lambda) + \ell m\mathbb{1}_1 = (\ell - 1)m\mathbb{1}_1 - \sum_{k=2}^{n} m\mathbb{1}_k + (m-1)m\lambda$$

is written uniquely as an element of $\Gamma(\lambda)_{\mathcal{G}}$. Therefore, each $Fr(\lambda) + \ell m\lambda_1$ does not belong to $\Gamma(\lambda)$ (see ii) of Remark 5.0.5). Also, $m\lambda < Fr(\lambda) \leq Fr(\lambda) + \ell m\mathbb{1}_1$ which means

$$\{Fr(\lambda) + \ell m\mathbb{1}_1 : \ell \in \mathbb{N}_0\} \subset (\Gamma(\lambda)_{\mathcal{G}} \setminus \Gamma(\lambda)) \cap \llcorner(m\lambda).$$

To prove iii), let $\tau = (\tau_1, \tau_2) \in (\Gamma(\lambda)_{\mathcal{G}} \setminus \Gamma(\lambda)) \cap \llcorner(m\lambda)$. In particular $\tau \in \llcorner(m\lambda)$, which implies that $m\lambda_i \leq \tau_i$, for $i = 1, 2$. On the other hand, by Lemma 5.0.1, we can apply Theorem 1.1 of [5] to the semigroups $\Lambda_1(\lambda), \ldots, \Lambda_n(\lambda)$. Let $\mu^{(i)}$ denote the Frobenius vector of $\Lambda_i(\lambda)$, for $i = 1, 2$. Then,

$$\mu^{(i)} = (m-1)m(\lambda - \mathbb{1}_i) - m\mathbb{1}.$$

Consider the following computations:

$$(m-1)(m\lambda_i - m) - m < m\lambda_i \Leftrightarrow (m-1)\lambda_i - (m-1) - 1 < \lambda_i \Leftrightarrow (m-2)\lambda_i < m,$$

for each $i = 1, 2$. Our hypothesis enables us to conclude that each $\mathrm{int}(\llcorner(\mu^{(i)}))$ is not contained in $\llcorner(m\lambda)$.

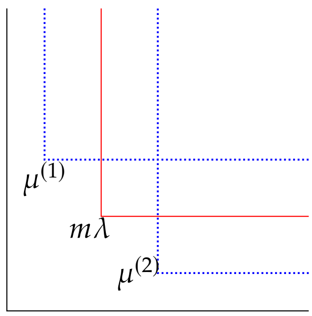

Figure 5.1: Frobenius vectors outside $\llcorner(m\lambda)$

Let $i \in \{1, 2\}$ and $\tau \in \Lambda(\lambda)_{\mathcal{G}} \cap \mathrm{int}(\llcorner(\mu^{(i)}))$. By the definition of $\mu^{(i)}$, we have that $\tau \in \Lambda_i(\lambda)$. Using (5.1) of Proposition 5.0.3, we conclude that $\tau \in \Lambda(\lambda)$. Thus,

$$\Lambda(\lambda)_{\mathcal{G}} \cap (\mathrm{int}(\llcorner(\mu^{(1)})) \cup \mathrm{int}(\llcorner(\mu^{(2)}))) \subseteq \Lambda(\lambda).$$

Now, notice that

$$\llcorner(m\lambda) = \bigcup_{i=1}^{2} (\llcorner(m\lambda) \cap \operatorname{int}(\llcorner(\mu^{(i)}))) \cup R,$$

where

$$R = [m\lambda_1, (m-1)m\lambda_1 - m] \times [m\lambda_2, (m-1)m\lambda_2 - m].$$

Therefore, by (5.2) of Proposition 5.0.3,

$$(\Gamma(\lambda)_{\mathcal{G}} \backslash \Lambda(\lambda)) \cap \llcorner(m\lambda) = (\Lambda(\lambda)_{\mathcal{G}} \backslash \Lambda(\lambda)) \cap \llcorner(m\lambda) \subseteq R.$$

We now proceed to prove iv). Assume that $\lambda_i \geq m/(m-2)$, for some $i \in \{1, 2\}$ and, for each $i \in \{1, 2\}$, let $\mu^{(i)}$ be the same as in the proof of iii). This means $\mu^{(i)} > m\lambda$, as seen in the proof of statement ii).

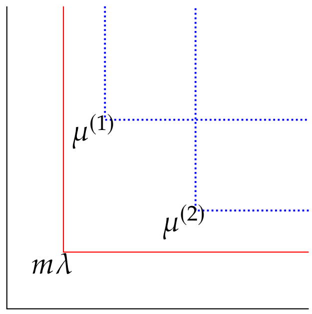

Figure 5.2: Frobenius vectors inside $\llcorner(m\lambda)$

We know that, for all $r \in \{0, \dots, m-1\}$, $\ell \in \mathbb{N}_0$, and $j \in \{1, 2\} \setminus \{i\}$,

$$\begin{aligned}
\mu^{(i)} + \ell m \mathbb{1}_j &= (m-1)(m\lambda - m\mathbb{1}_i) - m\mathbb{1} + \ell m \mathbb{1}_j \\
&= -m\mathbb{1}_1 - m\mathbb{1}_2 + \ell m \mathbb{1}_j + (m-1)(m\lambda - m\mathbb{1}_i) \\
&= -(r+1)m\mathbb{1}_i + (\ell + r - 1)m\mathbb{1}_j + (m-1-r)(m\lambda - m\mathbb{1}_i) + r(m\lambda - m\mathbb{1}_j).
\end{aligned}$$

Proposition 5.0.4 allows us to conclude that $\mu^{(i)} + \ell m \mathbb{1}_j$ belongs to $\Lambda(\lambda)_{\mathcal{G}}$, which equals $\Gamma(\lambda)_{\mathcal{G}}$ by Proposition 5.0.3. Also, by Proposition 5.0.4, we know that $\mu^{(i)} + \ell m \mathbb{1}_j$ does not belong to $\Lambda(\lambda)$. Hence, $\{\mu^{(i)} + \ell m \mathbb{1}_j : \ell \in \mathbb{N}_0\} \subset (\Gamma(\lambda)_{\mathcal{G}} \setminus \Lambda(\lambda)) \cap \llcorner(m\lambda)$. □

Lemma 5.0.6 yields significant insights that merit further discussion. Specifically, part i) seems to permit a construction analogous to Zariski's Theorem 3.2.3 for the case $m = 2$ by substituting the conductor c with the Frobenius vector $Fr(\lambda)$, which exhibits the requisite conductor-like behavior. However, part ii) demonstrates that this reasoning cannot be generalized for $m \geq 3$. Conversely, part iii) establishes that by decomposing the semigroup into a sum of two semigroups, one recovers a specific case for $n = 2$ where the principles of Theorem 3.2.3 seem to remain applicable. Finally, part iv) confirms that the hypotheses in iii) for $n = 2$ represent the optimal conditions achievable.

The following example illustrates the restriction imposed on n in iii) and iv) of Lemma 5.0.6.

Example 5.0.7. *Let $\lambda = (\frac{5}{4}, \frac{5}{4}, \frac{5}{4})$. Then*

$$\Gamma(\lambda)_{\mathcal{G}} = \langle (4,0,0), (0,4,0), (0,0,4), (5,5,5) \rangle_{\mathcal{G}}.$$

Consider

$$(7,7,7) = -2 \cdot (4,0,0) - 2 \cdot (0,4,0) - 2 \cdot (0,0,4) + 3 \cdot (5,5,5) \in \Gamma(\lambda)_{\mathcal{G}} \cap \llcorner(m\lambda).$$

Note that

$$(7,7) \notin \langle (4,0), (0,4), (1,5), (5,1) \rangle.$$

Therefore,

$$(7,7,7) + \ell m \mathbb{1}_3 \notin \Lambda(\lambda),$$

for all $\ell \geq 0$; otherwise, we could write

$$(7,7,7) + \ell \mathbb{1}_3 = 4 \cdot (b_1, b_2, b_3) + a_1 \cdot (1,5,5) + a_2 \cdot (5,1,5) + a_3 \cdot (5,5,1),$$

for non-negative integers $b_1, b_2, b_3, a_1, a_2, a_3$. This in turn implies that

$$(7,7) = 4 \cdot (b_1, b_2) + a_1 \cdot (1,5) + a_2 \cdot (5,1) + a_3 \cdot (5,5),$$

and therefore

$$(7,7) \in \langle (4,0), (0,4), (1,5), (5,1) \rangle.$$

which is impossible. Thus,

$$(\Gamma(\lambda)_{\mathcal{G}} \setminus \Lambda(\lambda)) \cap \llcorner(m\lambda)$$

is not finite.

6

Microlocal Equivalence

This chapter formulates the process of eliminating terms in a Puiseux parametrization of a quasi-ordinary (q.o.) surface. To provide a comprehensive discussion, we restrict our focus to $n = 2$. Recall that we use contact transformations in order to eliminate terms in a Puiseux parametrization of a q.o. surface Y with one characteristic exponent. We associate to Y a Puiseux parametrization v. Recall that $\pi : \mathbb{P}^*\mathbb{C}^3 \to \mathbb{C}^3$ denotes the canonical projection (see Section 4.2.1). Given an infinitesimal contact transformation $\Psi : \mathbb{P}^*\mathbb{C}^3 \to \mathbb{P}^*\mathbb{C}^3$ defined by

$$(\underline{x}, y, \underline{p}) \mapsto (x_1 + \alpha_1, x_2 + \alpha_2, y + \beta, p_1 + \gamma_1, p_2 + \gamma_2), \tag{6.1}$$

one would ideally aim for $Y' = (\pi \circ \Psi)(\mathbb{P}^*_Y\mathbb{C}^3)$ to be a q.o. surface with the same characteristic exponent as Y. To verify this, we must construct a Puiseux parametrization for Y'. Notice that v induces a parametrization for $\mathbb{P}^*_Y\mathbb{C}^3$, $\Psi(\mathbb{P}^*_Y\mathbb{C}^3)$, and, in particular, for Y'. Moreover, Y' is parametrized by

$$x_1 = t_1^m + v^*\alpha_1, \quad x_2 = t_2^m + v^*\alpha_2, \quad y' = v^*y + v^*\beta$$

(cf. Proposition 4.2.1). Consequently, achieving this elimination requires addressing the following questions:

Q_1. How can we apply the reparametrizations

$$u_1^m = t_1^m + v^*\alpha_1 \quad \text{and} \quad u_2^m = t_2^m + v^*\alpha_2(u_1, t_2)$$

to obtain a Puiseux parametrization of the type

$$x_1 = u_1^m, \quad x_2 = u_2^m, \quad y' = \underline{u}^{m\lambda} + \sum_{\tau \in \Gamma(\lambda)_{\mathcal{G}}} c_\tau \underline{u}^\tau$$

for Y', where $\underline{u} = (u_1, u_2)$?
Answer: ii) of Lemma 6.1.7, which is constructed upon Proposition 6.1.4, Lemma 6.1.5, and i) of Lemma 6.1.7.

Q_2. How is the support of v^*y' affected by the previous reparametrizations?
Answer: Proposition 6.1.6.

Q_3. How are the coefficients of $v^*y'(u_1, u_2)$ related to their counterparts in v^*y?
Answer: Most results in the following section describe how the reparametrizations alter the coefficients.

Q_4. Is Y' a q.o. surface with characteristic exponent λ?
Answer: After successful reparametrization, this answer will depend solely on α_1, α_2, and β, motivating Definitions 6.1.1 and 6.1.2. In Theorem 6.2.4, we present sufficient conditions for this to be the case.

Because the elimination process is inherently technical, we provide an informal geometric overview of the strategy before proceeding to the formal definitions. The support of v^*y can be represented geometrically, as illustrated in Figure 6.1.

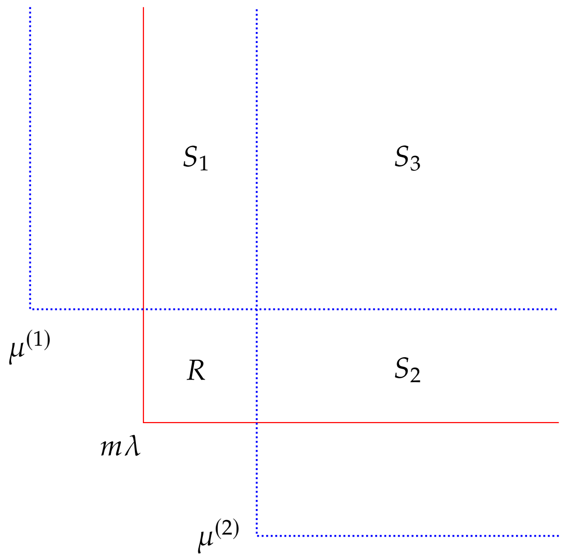

Figure 6.1: Sectioning of the support of a q.o. surface

For each $i \in \{1, 2\}$, $\mu^{(i)}$ denotes the Frobenius vector of the i-th fake Legendrian semigroup of λ (to recall these notions we refer to Chapter 5), $S_3 = \mathrm{int}(\llcorner(Fr(\lambda)))$, and $S_i = \llcorner(m\lambda) \cap \mathrm{int}(\llcorner(\mu^{(i)})) \setminus S_3$. First, we eliminate the half-lines $m\lambda + \ell m \mathbb{1}_i$, where $\ell \geq 1$ and $i \in \{1, 2\}$. Next we aim to eliminate every term in S_1. Let $\sigma = (\sigma_1, \sigma_2) \in S_1$. By the definition of $\mu^{(1)}$, one may write $\sigma = b_{\sigma,1} m \mathbb{1}_1 + b_{\sigma,2} m \mathbb{1}_2 + a_\sigma(m\lambda - m\mathbb{1}_1)$, where $b_{\sigma,1}, b_{\sigma,2}, a_\sigma \in \mathbb{N}_0$. Proposition 5.0.4 then plays a crucial role by stating that we may assume that $a_\sigma \in \{0, 1, \ldots, m-1\}$ and that $b_{\sigma,1}, b_{\sigma,2}$, and a_σ are unique. We will then eliminate *at once* all the terms of the form $\tau_\ell = \sigma + \ell m \mathbb{1}_2 \in S_1$, for $\ell \geq 0$. These transformations generate terms only in $\llcorner(\sigma) \setminus \{\tau = (\tau_1, \tau_2) \in \mathbb{Z}^n : \tau_1 = \sigma_1\}$. Hence, we can apply this process for all (finite) possible values of a_σ to eliminate all terms in the half line $\sigma + \ell \mathbb{1}_2$, $\ell \geq 0$. Thus, we can iteratively eliminate every half line in S_1 in finitely many iterations. By symmetry, we can apply this exact reasoning to eliminate every term in S_2.

The construction of the briefly described contact transformations is developed in the second section of this chapter. The formalization of the illustrated elimination process takes place in the third section. Finally, some examples are introduced in the fourth and final section of this chapter in order to illustrate the subtleties of the elimination process and demonstrate the boundaries of this approach.

6.1 Reparametrization Results

Let $\underline{t} = (t_1, \ldots, t_n)$. Let $i \in \{1, \ldots, n\}$, $\sigma = (\sigma_1, \ldots, \sigma_n) \in \mathbb{Z}^n$, and $f = \sum_{\tau \geq 0} c_\tau \underline{t}^\tau \in \mathbb{C}[[\underline{t}]]$. We establish the following notation

$$\mathcal{R}_i(\sigma) = \llcorner(\sigma) \cap \{\tau = (\tau_1, \ldots, \tau_n) \in \mathbb{Z}^n : \tau_i = \sigma_i\} = \left\{ \sigma + \sum_{\substack{j=1 \\ j \neq i}}^{n} \ell_j \mathbb{1}_j : \ell_j \in \mathbb{N}_0^n, j \in \{1, \ldots, n\} \right\}.$$

A keen observation† allows us to see that, if we relax the conditions of Chapter 5 of [27] to allow zero vectors, $\mathcal{R}_i(\sigma)$ equals the fiber of the n-set $\{0, \ldots, 0, \mathbb{1}_i, 0, \ldots, 0\}$ over $\sigma_i \mathbb{1}_i$.

Definition 6.1.1. *Let G be a subgroup of the additive group $\mathbb{Z}^n$ and $A \subseteq \mathbb{Z}^n$. Let $\sigma \in A$. We say that A is* ***dominated by*** *(G, σ) if $A \subseteq G \cap \llcorner(\sigma)$.*

Definition 6.1.2. *Let $f \in \mathbb{C}[[\underline{t}]]$ and $\sigma \in Supp(f)$. Assume that $Supp(f)$ is dominated by (G, σ). We call σ the* ***dominating exponent*** *of f and the coefficient of $\underline{t}^\sigma$ the* ***dominating coefficient****, which we denote by $DC(f)$.*

Example 6.1.3. *Let Y be a q.o. hypersurface with characteristic exponent $\lambda = (\lambda_1, \ldots, \lambda_n)$ that admits a Puiseux parametrization of the type*

$$x_1 = t_1^m, \quad x_2 = t_2^m, \quad y = \sum_{\tau \in \Gamma(\lambda)_{\mathcal{G}}} c_\tau \underline{t}^\tau.$$

Since we are only working with strongly normalized branches (cf. Definition 3.1.2), $Supp(\upsilon^* y)$ is dominated by $(\Gamma(\lambda)_{\mathcal{G}}, m\lambda)$.

Moreover, by Lemma 4.2.3, $\pi_i(\mathbb{P}^*_Y \mathbb{C}^{n+1})$ is a q.o. hypersurface with characteristic exponent $\lambda - \mathbb{1}_i$, for each $i \in \{1, \ldots, n\}$. Hence, $Supp(\upsilon^* p_i)$ is dominated by $(\Gamma(\lambda)_{\mathcal{G}}, m\lambda - m\mathbb{1}_i)$. In particular, by (4.2), $DC(\upsilon^* p_i) = \lambda_i$.

Throughout this section, we let G denote a subgroup of the additive group $\mathbb{Z}^n$.

Proposition 6.1.4. *Let $\varepsilon \in \mathbb{C}[[\underline{t}]]$, $\delta \in \mathbb{C}[[\underline{t}]]$, $\sigma \in Supp(\varepsilon)$, and $\tau \in Supp(\delta)$. Suppose that $Supp(\varepsilon)$ is dominated by (G, σ).*

i) If $Supp(\delta) \subseteq G$, then $Supp(\varepsilon\delta) \subseteq G \cap \llcorner(\sigma)$.

ii) If $Supp(\delta)$ is dominated by (G, τ), then $Supp(\varepsilon\delta)$ is dominated by $(G, \sigma + \tau)$ and $DC(\varepsilon\delta) = DC(\varepsilon)DC(\delta)$.

† This perspective stems from an astute observation made by the Examiner, Professor Marcelo Hernandes, during the public defense of this dissertation.

Proof. First, we prove i). Write $\varepsilon = \sum_{\rho \geq 0} \varepsilon_\rho \underline{t}^\rho$ and $\delta = \sum_{\rho \geq 0} \delta_\rho \underline{t}^\rho$. Then

$$\varepsilon\delta = \sum_{\rho \geq 0} \left(\sum_{0 \leq \varpi \leq \rho} \varepsilon_{\rho-\varpi} \delta_\varpi \right) \underline{t}^\rho. \tag{6.2}$$

If $\rho \in \mathrm{Supp}(\varepsilon\delta)$, then

$$\sum_{0 \leq \varpi \leq \rho} \varepsilon_{\rho-\varpi}\delta_\varpi \neq 0.$$

Then, there exists ϖ such that $\varepsilon_{\rho-\varpi}\delta_\varpi \neq 0$. Hence, $\rho - \varpi \in \mathrm{Supp}(\varepsilon)$ and $\varpi \in \mathrm{Supp}(\delta)$, so that

$$\rho = (\rho - \varpi) + \varpi \in \mathrm{Supp}(\varepsilon) + \mathrm{Supp}(\delta) \subseteq G + G = G.$$

Therefore,

$$\mathrm{Supp}(\varepsilon\delta) \subseteq \mathrm{Supp}(\varepsilon) + \mathrm{Supp}(\delta) \subseteq G.$$

Moreover, since $\mathrm{Supp}(\varepsilon)$ is dominated by (G, σ), then $\rho - \varpi \geq \sigma$, so that $\rho \geq \rho - \varpi \geq \sigma$. Hence, $\rho \in \llcorner(\sigma)$ and $\mathrm{Supp}(\varepsilon\delta) \subseteq \llcorner(\sigma)$. Thus, $\mathrm{Supp}(\varepsilon\delta) \subseteq G \cap \llcorner(\sigma)$, as desired.

We now prove ii). By i), we conclude that $\mathrm{Supp}(\varepsilon\delta) \subseteq G$. Let $\rho \in \mathrm{Supp}(\varepsilon\delta) \subseteq \mathrm{Supp}(\varepsilon) + \mathrm{Supp}(\delta)$. There exist $\varpi_1 \in \mathrm{Supp}(\varepsilon)$ and $\varpi_2 \in \mathrm{Supp}(\delta)$ such that $\rho = \varpi_1 + \varpi_2$. Therefore, $\rho = \varpi_1 + \varpi_2 \geq \sigma + \tau$. Hence, $\mathrm{Supp}(\varepsilon\delta) \subseteq G \cap \llcorner(\sigma + \tau)$. In order to show that $\mathrm{Supp}(\varepsilon\delta)$ is dominated by $(G, \sigma + \tau)$, we need to prove that $\sigma + \tau \in \mathrm{Supp}(\varepsilon\delta)$. As in (6.2), if $\rho = \sigma + \tau$, then

$$\begin{aligned}
\sum_{0 \leq \varpi \leq \sigma+\tau} \varepsilon_{\sigma+\tau-\varpi}\delta_\varpi &= \sum_{\substack{0 \leq \varpi \leq \sigma+\tau \\ \varpi \geq \tau \\ \sigma+\tau-\varpi \geq \sigma}} \varepsilon_{\sigma+\tau-\varpi}\delta_\varpi \\
&= \sum_{\substack{0 \leq \varpi \leq \sigma+\tau \\ \varpi = \tau}} \varepsilon_{\sigma+\tau-\varpi}\delta_\varpi \\
&= \varepsilon_\sigma \delta_\tau = \mathrm{DC}(\varepsilon)\,\mathrm{DC}(\delta) \neq 0.
\end{aligned}$$

Thus, $\sigma + \tau \in \mathrm{Supp}(\varepsilon\delta)$. Consequently, $\mathrm{Supp}(\varepsilon\delta)$ is dominated by $(G, \sigma + \tau)$ and, in particular,

$$\mathrm{DC}(\varepsilon\delta) = \sum_{0 \leq \varpi \leq \sigma+\tau} \varepsilon_{\sigma+\tau-\varpi}\delta_\varpi = \mathrm{DC}(\varepsilon)\,\mathrm{DC}(\delta).$$

□

Lemma 6.1.5. *Let $m \in \mathbb{N} \setminus \{1\}$, $\kappa_1 \in \mathbb{C}^*$ and $\varepsilon \in (t_1, \ldots, t_n)\mathbb{C}[[\underline{t}]]$ such that Supp(ε) is dominated by (G, σ). Then there exist $\kappa_2 \in \mathbb{C}$ and $\delta \in (t_1, \ldots, t_n)\mathbb{C}[[\underline{t}]]$ such that*

$$(\kappa_2 + \delta(\underline{t}))^m = \kappa_1 + \varepsilon(\underline{t}).$$

Moreover, $\kappa_2 \in \mathbb{C}^*$, Supp$(\delta)$ is dominated by (G, σ), and DC$(\delta) = \frac{1}{m\kappa_2^{m-1}}DC(\varepsilon)$. In particular, if $\varepsilon \in (t_i)\mathbb{C}[[\underline{t}]]$, for some $i \in \{1, \ldots, n\}$, then $\delta \in (t_i)\mathbb{C}[[\underline{t}]]$. Under this assumption, for each $\tau \in \mathcal{R}_i(\sigma)$,

$$\delta_\tau = \frac{\varepsilon_\tau}{m\kappa_2^{m-1}},$$

where δ_τ and ε_τ denote the coefficients of $\underline{t}^\tau$, respectively, in δ and in ε.

Proof. Using the well-known binomial expansion for $(1+z)^{\frac{1}{m}}$, we may write

$$(1+z)^{\frac{1}{m}} = \sum_{\ell \geq 0} d_\ell z^\ell,$$

for some $d_\ell \in \mathbb{C}$. In particular, $d_0 = 1$, $d_1 = \frac{1}{m}$, and $d_\ell \neq 0$, for each $\ell \geq 2$.

Note that $\varepsilon \in (t_1, \ldots, t_n)\mathbb{C}[[\underline{t}]]$, so that $\varepsilon(\underline{t})$ has no constant term and, consequently, $\kappa_1 + \varepsilon(\underline{t})$ has a nonzero constant term.

We claim that the binomial expansion of

$$(\kappa_1 + \varepsilon(\underline{t}))^{1/m} = \kappa_1^{\frac{1}{m}}(1 + \kappa_1^{-1}\varepsilon(\underline{t}))^{1/m} = \kappa_1^{\frac{1}{m}} \sum_{\ell \geq 0} d_\ell (\kappa_1^{-1}\varepsilon(\underline{t}))^\ell$$

is well-defined. Let $\mathfrak{m}$ be the maximal ideal in $\mathbb{C}[[\underline{t}]]$ and let

$$\chi(\underline{t}) = \sum_{\ell \geq 0} d_\ell (\kappa_1^{-1}\varepsilon(\underline{t}))^\ell.$$

Consider the sequence $\{\chi_j\}_{j\in\mathbb{N}}$ in $\mathbb{C}[[\underline{t}]]$ defined as

$$\chi_j(\underline{t}) = \sum_{\ell=0}^{j} d_\ell (\kappa_1^{-1}\varepsilon(\underline{t}))^\ell,$$

for each $j \in \mathbb{N}$. We claim that $\{\chi_j\}_{j\in\mathbb{N}}$ is convergent and that

$$\lim_{j\to\infty} \chi_j = \chi$$

in the $\mathfrak{m}$-adic topology (see Definition 2.0.18). To prove the former, we will show that $\{\chi_j\}_{j\in\mathbb{N}}$ is a Cauchy sequence, which converges due to the completeness of $\mathbb{C}[[\underline{t}]]$ guaranteed by Theorem 6.1.8 of [11]. Let $s_1 \in \mathbb{N}$ and choose $s_2 = s_1 - 1$. Then, for all $j, k \geq s_2$, where $j \geq k$,

$$\begin{aligned}
\chi_j - \chi_k &= \sum_{\ell=0}^{j} d_\ell (\kappa_1^{-1}\varepsilon(\underline{t}))^\ell - \sum_{\ell=0}^{k} d_\ell (\kappa_1^{-1}\varepsilon(\underline{t}))^\ell \\
&= \sum_{\ell=k+1}^{j} d_\ell (\kappa_1^{-1}\varepsilon(\underline{t}))^\ell \in \mathfrak{m}^{k+1} \subseteq \mathfrak{m}^{s_1},
\end{aligned}$$

since $\varepsilon(\underline{t}) \in \mathfrak{m}$ and $k+1 \geq s_2 + 1 = s_1$. We now prove the latter. Fix $s \in \mathbb{N}$. Therefore, by choosing $k = s-1$, we then have that, for all $j \geq k$,

$$\begin{aligned}
\chi_j - \chi &= \sum_{\ell=0}^{j} d_\ell (\kappa_1^{-1}\varepsilon(\underline{t}))^\ell - \sum_{\ell\geq 0} d_\ell (\kappa_1^{-1}\varepsilon(\underline{t}))^\ell \\
&= -\sum_{\ell=j+1}^{\infty} d_\ell (\kappa_1^{-1}\varepsilon(\underline{t}))^\ell.
\end{aligned}$$

But then, given that $\varepsilon(\underline{t}) \in \mathfrak{m}$, it follows that

$$\chi_j - \chi \in \mathfrak{m}^{j+1} \subseteq \mathfrak{m}^s,$$

since $j+1 \geq k+1 = s$. This shows that χ is well-defined.

This allows us to guarantee the existence of such $\kappa_2 \in \mathbb{C}$ and $\delta \in (t_1, \ldots, t_n)\mathbb{C}[[\underline{t}]]$. In particular, it follows that $\kappa_2^m = \kappa_1 \neq 0$, so that $\kappa_2 \neq 0$. On the other hand,

$$\delta(\underline{t}) = \kappa_1^{\frac{1}{m}} \sum_{\ell \geq 1} d_\ell (\kappa_1^{-1} \varepsilon(\underline{t}))^\ell, \tag{6.3}$$

so that

$$\mathrm{Supp}(\delta) \subseteq \cup_{\ell \geq 1} \mathrm{Supp}(\varepsilon(\underline{t})^\ell) \subseteq G \cap \llcorner(\sigma).$$

Moreover, since $\mathrm{Supp}(\varepsilon)$ is dominated by (G, σ) and $\varepsilon \in \mathfrak{m}$, then $\sigma > 0$. This shows that $\sigma \notin \mathrm{Supp}(\varepsilon(\underline{t})^\ell)$, whenever $\ell > 1$, for $\mathrm{Supp}(\varepsilon(\underline{t})^\ell)$ is dominated by $(G, \ell\sigma)$, by part $ii)$ of Proposition 6.1.4, and $\sigma < \ell\sigma$. Thus, $\sigma \in \mathrm{Supp}(\delta)$ and $\mathrm{Supp}(\delta)$ is dominated by (G, σ).

In particular, by (6.3),

$$\mathrm{DC}(\delta) = \kappa_1^{\frac{1}{m}-1} d_1 \, \mathrm{DC}(\varepsilon) = \frac{\mathrm{DC}(\varepsilon)}{m\kappa_2^{m-1}},$$

for $d_1 = \frac{1}{m}$.

To prove the second part of the lemma, write $\sigma = (\sigma_1, \ldots, \sigma_n)$. Note that $\varepsilon \in (t_i)\mathbb{C}[[\underline{t}]]$, for some $i \in \{1, \ldots, n\}$, and $\mathrm{Supp}(\varepsilon)$ being dominated by (G, σ) together imply that $\sigma_i \geq 1$. But then, $\mathrm{Supp}(\delta)$ is also dominated by (G, σ). Therefore, for all $\tau = (\tau_1, \ldots, \tau_n) \in \mathrm{Supp}(\delta)$, $\tau_i \geq \sigma_i \geq 1$. Thus, $\delta \in (t_i)\mathbb{C}[[\underline{t}]]$.

More generally, let $\tau = (\tau_1, \ldots, \tau_n) \in \mathcal{R}_i(\sigma)$. Since $\mathrm{Supp}(\delta)$ is dominated by (G, σ) and $\delta \in (t_i)\mathbb{C}[[\underline{t}]]$, it follows that $\sigma_i \geq 1$. Since $\mathrm{Supp}(\delta^\ell) \subseteq \llcorner(\sigma + (\ell-1)\mathbb{1}_i)$, for each $\ell \in \{1, \ldots, m\}$, it follows that $\tau \notin \mathrm{Supp}(\delta^\ell)$ whenever $\ell > 1$. Therefore, δ_τ is fully determined by the equation

$$\delta_\tau = \kappa_1^{\frac{1}{m}-1} d_1 \varepsilon_\tau = \frac{\varepsilon_\tau}{m\kappa_2^{m-1}}.$$

□

Proposition 6.1.6. *Let $\varepsilon = \sum_{\rho \geq \sigma} \varepsilon_\rho \underline{t}^\rho \in \mathbb{C}[[\underline{t}]]$ and $\sigma \in Supp(\varepsilon)$ be such that $Supp(\varepsilon)$ is dominated by (G, σ). Let $\delta \in (t_i)\mathbb{C}[[\underline{t}]]$, $\kappa \in \mathbb{C}$, and*

$$h(\underline{t}) = \varepsilon(t_1, \ldots, t_{i-1}, t_i(\kappa + \delta(\underline{t})), t_{i+1}, \ldots, t_n).$$

i) If $\kappa \neq 0$ and $Supp(\delta) \subseteq G$, then $Supp(h)$ is dominated by (G, σ).

ii) For each $\tau = (\tau_1, \ldots, \tau_n)$,

$$\varepsilon_\tau \underline{t}^\tau (\kappa + \delta(\underline{t}))^{\tau_i} = \kappa^{\tau_i} \varepsilon_\tau \underline{t}^\tau + \chi(\underline{t}),$$

where $\chi \in \mathbb{C}[[\underline{t}]]$ and $Supp(\chi) \subseteq \llcorner(\tau) \setminus \mathcal{R}_i(\tau)$.

iii) Suppose that Supp(δ) *is dominated by* (G,ϖ), *for some* $\varpi\in$ *Supp*(δ). *Then, for each* $\tau=(\tau_1,\ldots,\tau_n)$,

$$\varepsilon_\tau \underline{t}^\tau(\kappa+\delta(\underline{t}))^{\tau_i}=\varepsilon_\tau\kappa^{\tau_i}\underline{t}^\tau+\varepsilon_\tau\tau_i\kappa^{\tau_i-1}\,\mathrm{DC}(\delta)\underline{t}^{\tau+\varpi}+\chi(\underline{t}),$$

where $\chi\in\mathbb{C}[[\underline{t}]]$ *and Supp*$(\chi)\subseteq\llcorner(\varpi+\tau)\setminus\{\varpi+\tau\}$.

Proof. We first prove i). Suppose that $\kappa\neq 0$. If $\delta=0$, there is nothing to prove. Suppose otherwise. Letting $\rho=(\rho_1,\ldots,\rho_n)$, by the definition of h,

$$h(\underline{t})=\sum_{\rho\geq\sigma}\varepsilon_\rho\underline{t}^\rho(\kappa+\delta(\underline{t}))^{\rho_i}=\sum_{\rho\geq\sigma}\varepsilon_\rho\kappa^{\rho_i}\underline{t}^\rho+\sum_{\rho\geq\sigma}\left(\sum_{j=1}^{\rho_i}\binom{\rho_i}{j}\varepsilon_\rho\kappa^{\rho_i-j}\delta(\underline{t})^j\right)\underline{t}^\rho.$$

Let $\tilde{\chi}(\underline{t})=\sum_{\rho\geq\sigma}\left(\sum_{j=1}^{\rho_i}\binom{\rho_i}{j}\varepsilon_\rho\kappa^{\rho_i-j}\delta(\underline{t})^j\right)\underline{t}^\rho$. Given that $\mathrm{Supp}(\delta)\subseteq G$, it follows by i) of Proposition 6.1.4 that $\mathrm{Supp}(\tilde{\chi})\subseteq G\cap\llcorner(\sigma)$. Hence, $\mathrm{Supp}(h)\subseteq\mathrm{Supp}(\varepsilon)\cup\mathrm{Supp}(\tilde{\chi})\subseteq G\cap\llcorner(\sigma)$. On the other hand, let $\tau\in\mathrm{Supp}(\tilde{\chi})$. Then

$$\tau=\rho+\varpi^{(1)}+\ldots+\varpi^{(j)},$$

for some $\rho\in\mathrm{Supp}(\varepsilon)$, $j\in\{1,\ldots,\rho_i\}$, and $\varpi^{(1)},\ldots,\varpi^{(j)}\in\mathrm{Supp}(\delta)$. If we let the subscript i denote the i-th coordinate of the respective n-tuple, we note that

$$\tau_i=\rho_i+\varpi_i^{(1)}+\ldots+\varpi_i^{(j)}>\rho_i\geq\sigma_i,$$

for $\varpi^{(1)},\ldots,\varpi^{(j)}\in\mathrm{Supp}(\delta)$, and $\delta\in(t_i)\mathbb{C}[[\underline{t}]]$. Hence $\tau>\sigma$ and $\sigma\notin\mathrm{Supp}(\tilde{\chi})$. Therefore, $\mathrm{Supp}(h)$ is dominated by (G,σ), for $\sigma\in\mathrm{Supp}(\varepsilon)$ and $\kappa\neq 0$.

To prove part ii) of the proposition, we follow a similar reasoning. Indeed, for each $\tau=(\tau_1,\ldots,\tau_n)$,

$$\varepsilon_\tau\underline{t}^\tau(\kappa+\delta(\underline{t}))^{\tau_i}=\kappa^{\tau_i}\varepsilon_\tau\underline{t}^\tau+\chi(\underline{t}),$$

where

$$\chi(\underline{t})=\left(\sum_{j=1}^{\tau_i}\binom{\tau_i}{j}\varepsilon_\tau\kappa^{\tau_i-j}\delta(\underline{t})^j\right)\underline{t}^\tau. \tag{6.4}$$

If $\tau_i=0$ or $\varepsilon_\tau=0$, then $\chi(\underline{t})=0$. Its support is empty, and the condition $\mathrm{Supp}(\chi)\subseteq\llcorner(\tau)\setminus\mathcal{R}_i(\tau)$ is vacuously satisfied. Assume $\tau_i\geq 1$ and $\varepsilon_\tau\neq 0$. As before, we conclude that if $\rho=(\rho_1,\ldots,\rho_n)\in\mathrm{Supp}(\chi)$, then

$$\rho=\tau+\varpi^{(1)}+\ldots+\varpi^{(j)},$$

for some $j\in\{1,\ldots,\tau_i\}$, and $\varpi^{(1)},\ldots,\varpi^{(j)}\in\mathrm{Supp}(\delta)$, where $\delta\in(t_i)\mathbb{C}[[\underline{t}]]$. Notice that $\mathrm{Supp}(\delta)\neq\emptyset$, for $\delta\neq 0$. Additionally, since $\delta\in(t_i)\mathbb{C}[[\underline{t}]]$, the i-th coordinate of any element in $\mathrm{Supp}(\delta)$ is at least 1. Therefore, $\rho_i>\tau_i$ and $\rho\geq\tau$, so that $\rho\in\llcorner(\tau)\setminus\mathcal{R}_i(\tau)$.

In order to prove iii), notice that by extracting the first term in (6.4),

$$\chi(\underline{t})=\varepsilon_\tau\tau_i\kappa^{\tau_i-1}\underline{t}^\tau\delta(\underline{t})+\chi_1(\underline{t}),$$

where

$$\chi_1(\underline{t}) = \left(\sum_{j=2}^{\tau_i} \binom{\tau_i}{j} \varepsilon_\tau \kappa^{\tau_i - j} \delta(\underline{t})^j \right) \underline{t}^\tau,$$

so that $\mathrm{Supp}(\chi_1) \subseteq \llcorner(2\varpi + \tau)$, and $\varpi + \tau \notin \llcorner(2\varpi + \tau)$. In particular,

$$\varepsilon_\tau \tau_i \kappa^{\tau_i - 1} \underline{t}^\tau \delta(\underline{t}) = \varepsilon_\tau \tau_i \kappa^{\tau_i - 1} DC(\delta) \underline{t}^{\varpi + \tau} + \chi_2(\underline{t}),$$

where

$$\chi_2(\underline{t}) = \varepsilon_\tau \tau_i \kappa^{\tau_i - 1} \underline{t}^\tau (\delta(\underline{t}) - \mathrm{DC}(\delta) \underline{t}^\varpi),$$

so that $\mathrm{Supp}(\chi_2) \subseteq \llcorner(\varpi + \tau) \setminus \{\varpi + \tau\}$. Letting the remainder for part iii) be $\tilde{\chi} = \chi_1 + \chi_2$, since $\mathrm{Supp}(\chi_1) \subseteq \llcorner(2\varpi + \tau) \subseteq \llcorner(\varpi + \tau) \setminus \{\varpi + \tau\}$ and $\mathrm{Supp}(\chi_2) \subseteq \llcorner(\varpi + \tau) \setminus \{\varpi + \tau\}$, it follows that their sum satisfies the required support condition, which proves the claim. □

Lemma 6.1.7. *Let $\underline{w} = (t_1, \ldots, t_{i-1}, u, t_{i+1}, \ldots, t_n)$. Let $\varepsilon \in (t_i)\mathbb{C}[[\underline{t}]]$ and $\sigma = (\sigma_1, \ldots, \sigma_n) \in$* $\mathrm{Supp}(\varepsilon)$ *be such that Supp(ε) is dominated by (G, σ).*

i) Let $\kappa_1 \in \mathbb{C}^*$. There exist $\kappa_2 \in \mathbb{C}^*$ and $\delta \in (u)\mathbb{C}[[\underline{w}]]$ such that $t_i = u(\kappa_2 + \delta(\underline{w}))$ solves $u = t_i(\kappa_1 + \varepsilon(\underline{t}))$. Moreover, δ is unique, Supp(δ) is dominated by (G, σ), and

$$\mathrm{DC}(\delta) = -\frac{\kappa_2^{\sigma_i + 1}}{\kappa_1} \mathrm{DC}(\varepsilon).$$

More generally, for each $\tau \in \mathcal{R}_i(\sigma)$,

$$c_\tau = -\frac{\kappa_2^{\sigma_i + 1}}{\kappa_1} d_\tau,$$

where c_τ is the coefficient of $\underline{w}^\tau$ in δ, and d_τ is the coefficient of $\underline{t}^\tau$ in ε.

ii) If $\varepsilon \in (t_i)\mathbb{C}\{\underline{t}\}$ and

$$u^m = t_i^m (1 + \varepsilon(\underline{t})), \tag{6.5}$$

then there exists a unique $\delta \in (u)\mathbb{C}\{\underline{w}\}$ such that $t_i = u(1 + \delta(\underline{w}))$ solves (6.5), and $\mathrm{Supp}(\delta)$ *is dominated by (G, σ). In particular,*

$$\mathrm{DC}(\delta) = -\frac{\mathrm{DC}(\varepsilon)}{m}.$$

More generally, for each $\tau \in \mathcal{R}_i(\sigma)$,

$$c_\tau = -\frac{d_\tau}{m},$$

where c_τ is the coefficient of $\underline{w}^\tau$ in δ, and d_τ is the coefficient of $\underline{t}^\tau$ in ε.

Proof. We shall prove i). To see that such κ_2 and δ in fact exist, let

$$F(\underline{t}, u) = u - t_i(\kappa_1 + \varepsilon(\underline{t})).$$

Then $\frac{\partial F}{\partial t_i} = -\kappa_1 - \varepsilon(\underline{t}) - t_i \frac{\partial \varepsilon}{\partial t_i}(\underline{t})$. Since $\varepsilon \in (t_i)\mathbb{C}[[\underline{t}]]$, then $\frac{\partial F}{\partial t_i}(0) = -\kappa_1 \neq 0$. By the Implicit Function Theorem for formal power series, $t_i = \varphi(\underline{w})$, for some unique $\varphi \in \mathbb{C}[[\underline{w}]]$.

We now determine the explicit form of this solution. Note that

$$t_i = \kappa_1^{-1} u - \kappa_1^{-1} t_i \varepsilon(\underline{t}).$$

By repeatedly applying this substitution to the RHS of the above equation, one obtains a formal power series $\chi \in \mathbb{C}[[\underline{w}]]$ such that

$$t_i = \kappa_1^{-1} u - \kappa_1^{-1} \chi(\underline{w}).$$

Using a similar argument as in the proof of Lemma 6.1.5, this iteration is well-defined. We claim that $\chi \in (u^2)\mathbb{C}[[\underline{w}]]$. Note that $\varepsilon \in (t_i)\mathbb{C}[[\underline{t}]]$ and $\mathrm{Supp}(\varepsilon)$ being dominated by (G, σ) imply that $\sigma_i \geq 1$ and that $\mathrm{Supp}(t_i \varepsilon(\underline{t}))$ is dominated by $(\mathbb{Z}^n, \sigma + \mathbb{1}_i)$. Moreover, $\sigma_i + 1 \geq 2$. By applying at each substitution part ii) of Proposition 6.1.6, one concludes that $\mathrm{Supp}(\chi) \subseteq \llcorner(\sigma + \mathbb{1}_i)$, so that $\chi \in (u^2)\mathbb{C}[[\underline{w}]]$. We can then write $\chi(\underline{w}) = -\kappa_1 u \delta(\underline{w})$, where $\delta \in (u)\mathbb{C}[[\underline{w}]]$, so that

$$t_i = u(\kappa_1^{-1} + \delta(\underline{w})).$$

By setting $\kappa_2 = \kappa_1^{-1}$, this proves the existence of κ_2 and δ under the desired conditions. Their uniqueness follows directly from the uniqueness of φ.

Next, we prove that $\mathrm{Supp}(\delta) \subseteq \llcorner(\sigma)$. Let $\underline{v} = (t_1, \ldots, t_{i-1}, u(\kappa_2 + \delta(\underline{w})), t_{i+1}, \ldots, t_n)$. We have that

$$\begin{aligned} u &= t_i(\kappa_1 + \varepsilon(\underline{t})) = u(\kappa_2 + \delta(\underline{w}))(\kappa_1 + \varepsilon(\underline{v})) \\ &= \kappa_2 \kappa_1 u + u \kappa_1 \delta(\underline{w}) + u(\kappa_2 + \delta(\underline{w}))\varepsilon(\underline{v}) \\ &= u + u\kappa_1 \delta(\underline{w}) + u(\kappa_2 + \delta(\underline{w}))\varepsilon(\underline{v}). \end{aligned}$$

Therefore,

$$-\kappa_1 u \delta(\underline{w}) = u(\kappa_2 + \delta(\underline{w}))\varepsilon(\underline{v}).$$

We then observe that

$$-\kappa_1 \delta(\underline{w}) = \kappa_2 \varepsilon(\underline{v}) + \delta(\underline{w})\varepsilon(\underline{v}). \tag{6.6}$$

By ii) of Proposition 6.1.6, $\mathrm{Supp}(\varepsilon(\underline{v})) \subseteq \llcorner(\sigma)$ so that

$$\begin{aligned} \mathrm{Supp}(\delta(\underline{w})) &\subseteq \mathrm{Supp}(\varepsilon(\underline{v})) \cup \mathrm{Supp}(\delta(\underline{w})\varepsilon(\underline{v})) \\ &\subseteq \llcorner(\sigma) \cup (\llcorner(\sigma) + \mathrm{Supp}(\delta(\underline{w}))) \\ &\subseteq \llcorner(\sigma). \end{aligned}$$

We now want to show that $\mathrm{Supp}(\delta(\underline{w})) \subseteq G$. Since $\varepsilon \in (t_i)\mathbb{C}[[\underline{t}]]$, then by ii) of Proposition 6.1.6, $\varepsilon(\underline{v}) \in (u)\mathbb{C}[[\underline{v}]]$. Hence, $1 + \kappa_2 \varepsilon(\underline{v})$ is a unit. From (6.6), we write

$$\delta(\underline{w}) = -\kappa_2^2 \varepsilon(\underline{v})(1 + \kappa_2 \varepsilon(\underline{v}))^{-1} = -\kappa_2^2 \varepsilon(\underline{v}) \sum_{j=0}^{\infty} (-\kappa_2)^j \varepsilon(\underline{v})^j = -\sum_{j=1}^{\infty} (-\kappa_2)^{j+1} \varepsilon(\underline{v})^j. \tag{6.7}$$

We claim that $\operatorname{Supp}(\varepsilon(\underline{v})) \subseteq G$. Let $\tau \in \operatorname{Supp}(\varepsilon(\underline{v}))$. Note that

$$\begin{aligned}\varepsilon(\underline{v}) &= \sum_{\rho \geq \sigma} \varepsilon_\rho \underline{w}^\rho (\kappa_2 + \delta(\underline{w}))^{\rho_i} \\ &= \sum_{\rho \geq \sigma} \varepsilon_\rho \kappa_2^{\rho_i} \underline{w}^\rho + \sum_{\rho \geq \sigma} \left(\sum_{\ell=1}^{\rho_i} \binom{\rho_i}{\ell} \varepsilon_\rho \kappa_2^{\rho_i - \ell} \delta(\underline{w})^\ell \right) \underline{w}^\rho.\end{aligned}$$

If $\tau \in \operatorname{Supp}(\varepsilon)$, then $\tau \in G$. Suppose otherwise. Then

$$\tau = \varpi_1 + \ldots + \varpi_\ell + \rho$$

for some $\rho = (\rho_1, \ldots, \rho_n) \in \operatorname{Supp}(\varepsilon)$, $\ell \in \{1, \ldots, \rho_i\}$, and $\varpi_1, \ldots, \varpi_\ell \in \operatorname{Supp}(\delta)$. Since $\delta \in (u)\mathbb{C}[[\underline{w}]]$, then $\varpi_j < \tau$, for all $j \in \{1, \ldots, \ell\}$. We claim that $\varpi_1, \ldots, \varpi_\ell \in G$. Suppose otherwise and, without loss of generality, say that $\varpi_1 \notin G$.

From (6.7), ϖ_1 is a sum of elements in $\operatorname{Supp}(\varepsilon(\underline{v}))$. Therefore, there exists at least one element $\tau^{(1)} \in \operatorname{Supp}(\varepsilon(\underline{v}))$ such that $\tau^{(1)} \notin G$ and $\varpi_1 \geq \tau^{(1)}$. Because $\tau^{(1)} \notin G$, it must arise from the second sum in the expansion of $\varepsilon(\underline{v})$. Thus, we may write

$$\tau^{(1)} = \varpi_1^{(1)} + \ldots + \varpi_k^{(1)} + \rho^{(1)}$$

for some $\rho^{(1)} = (\rho_1^{(1)}, \ldots, \rho_n^{(1)}) \in \operatorname{Supp}(\varepsilon)$, $k \in \left\{1, \ldots, \rho_i^{(1)}\right\}$, and $\varpi_1^{(1)}, \ldots, \varpi_k^{(1)} \in \operatorname{Supp}(\delta)$. Since $\rho^{(1)} \geq \sigma > 0$, we have $\tau^{(1)} > \varpi_j^{(1)}$ for all $j \in \{1, \ldots, k\}$. Since $\tau^{(1)} \notin G$ and $\rho^{(1)} \in G$, we may assume without loss of generality that $\varpi_1^{(1)} \notin G$.

Combining these inequalities yields $\varpi_1 \geq \tau^{(1)} > \varpi_1^{(1)}$. If one continues this process indefinitely, we obtain an infinite strictly descending chain

$$\varpi_1 > \varpi_1^{(1)} > \cdots > \varpi_1^{(s)} > \cdots.$$

Since the coordinate sum strictly decreases at each step, this strictly descending chain in $\mathbb{N}_0^n$ must be finite, which is impossible. Therefore, $\varpi_1, \ldots, \varpi_\ell \in G$. Thus,

$$\tau = \varpi_1 + \ldots + \varpi_\ell + \rho \in G.$$

The claim then follows. Therefore, $\operatorname{Supp}(\varepsilon(\underline{v})^j) \subseteq G$, for each $j \in \mathbb{N}$. From (6.7), we conclude that $\operatorname{Supp}(\delta) \subseteq G$.

We now aim to prove that $\sigma \in \operatorname{Supp}(\delta)$. In order to do this, note that, for each $s \in \mathbb{N}$, $\operatorname{Supp}(\varepsilon(\underline{v})^s)$ is dominated by $(G, s\sigma)$, by $ii)$ of Proposition 6.1.4. Given that $\sigma > 0$, then $\sigma \notin \operatorname{Supp}(\varepsilon(\underline{v})^s)$, for any $s > 1$, because $s\sigma > \sigma$. Therefore, $\kappa_2 \neq 0$ implies that $\sigma \in \operatorname{Supp}(\delta)$. Thus, $\operatorname{Supp}(\delta)$ is dominated by (G, σ).

Finally, we compute $\operatorname{DC}(\delta)$. From the expansion in (6.7), the dominating term of $\delta(\underline{w})$ is given by the $j = 1$ term, namely $-\kappa_2^2 \varepsilon(\underline{v})$. By $ii)$ of Proposition 6.1.6, the dominating coefficient of $\varepsilon(\underline{v})$ is $\kappa_2^{\sigma_i} \operatorname{DC}(\varepsilon)$. Consequently, the dominating coefficient of δ is $-\kappa_2^{\sigma_i+2} \operatorname{DC}(\varepsilon)$. Since $\kappa_2 = \kappa_1^{-1}$, we have $-\kappa_2^{\sigma_i+2} = -\frac{\kappa_2^{\sigma_i+1}}{\kappa_1}$, yielding $\operatorname{DC}(\delta) = -\frac{\kappa_2^{\sigma_i+1}}{\kappa_1} \operatorname{DC}(\varepsilon)$, as required.

More generally, this last argument can be made for every $\tau = (\tau_1, \ldots, \tau_n) \in \mathcal{R}_i(\sigma)$. In particular, $\tau_i = \sigma_i \geq 1$, since $\varepsilon \in (t_i)\mathbb{C}[[\underline{t}]]$ and $\operatorname{Supp}(\varepsilon) \subseteq \llcorner(\sigma)$. Thus, $\tau_i < s\sigma_i$, for each

$s > 1$. Hence $\tau \notin \operatorname{Supp}(\varepsilon(\underline{v})^s)$, for any $s > 1$. Moreover, if one denotes by c_τ the coefficient of $\underline{w}^\tau$ in δ, then c_τ is given by the $j = 1$ term, namely $-\kappa_2^2 d_\tau$, where d_τ is the coefficient of $\underline{t}^\tau$ in ε. By ii) of Proposition 6.1.6, $c_\tau = -\kappa_2^{\sigma_i+2} d_\tau$. Since $\kappa_2 = \kappa_1^{-1}$, we have $-\kappa_2^{\sigma_i+2} = -\frac{\kappa_2^{\sigma_i+1}}{\kappa_1}$, yielding $c_\tau = -\frac{\kappa_2^{\sigma_i+1}}{\kappa_1} d_\tau$, as required.

We now prove ii). By taking the principal m-th root in Lemma 6.1.5,

$$u = t_i(1 + \varepsilon_1(\underline{t})), \text{ where } \varepsilon_1 \in (t_i)\mathbb{C}\{\underline{t}\}, \tag{6.8}$$

$\operatorname{Supp}(\varepsilon_1)$ is dominated by (G, σ), $\operatorname{DC}(\varepsilon_1) = \frac{\operatorname{DC}(\varepsilon)}{m}$, and the coefficients of $\underline{t}^\tau$ in ε_1 and ε also verify the analogous equality, for each $\tau \in \mathcal{R}_i(\sigma)$. Moreover by part i),

$$t_i = u(1 + \delta(\underline{w})),$$

where $\delta \in (u)\mathbb{C}[[\underline{w}]]$, and $\operatorname{Supp}(\delta)$ is dominated by (G, σ). Additionally,

$$\operatorname{DC}(\delta) = -\operatorname{DC}(\varepsilon_1) = -\frac{\operatorname{DC}(\varepsilon)}{m}$$

and, for each $\tau \in \mathcal{R}_i(\sigma)$, $c_\tau = -\frac{d_\tau}{m}$. Given that $\frac{\partial u}{\partial t_i}(0) = 1$ in (6.8), by the Implicit Function Theorem and the uniqueness of δ, then $\delta \in (u)\mathbb{C}\{\underline{w}\}$. □

6.2 Contact Transformations

Let $\underline{p} = (p_1, \ldots, p_n)$ and $\pi : \mathbb{C}^{2n+1} \to \mathbb{C}^{n+1}$ be defined by $\pi(\underline{x}, y, \underline{p}) = (\underline{x}, y)$, for all $(\underline{x}, y, \underline{p}) \in \mathbb{C}^{2n+1}$. If $A = \emptyset$, we define $\sum_{i \in A} i = 0$.

Lemma 6.2.1. *Let $\alpha \in \mathbb{C}\{\underline{x}, p_i\}$, for some $i \in \{1, \ldots, n\}$, be such that*

$$\frac{\partial \alpha}{\partial x_\ell} \in (p_i)\mathbb{C}\{\underline{x}, p_i\}, \text{ for each } \ell \in \{1, \ldots, n\}.$$

There are unique $\beta, \gamma_1, \ldots, \gamma_n \in \mathbb{C}\{\underline{x}, p_i\}$ such that $\beta \in (p_i)^2\mathbb{C}\left\{\underline{x}, p_i\right\}$ and

$$(\underline{x}, y, \underline{p}) \mapsto (\underline{x} + \alpha \mathbb{1}_i, y + \beta, \underline{p} + \underline{\gamma}) \tag{6.9}$$

is an infinitesimal contact transformation. Furthermore, if $\beta = \sum_{\ell \geq 1} \beta_\ell(\underline{x}) p_i^\ell$ and $\alpha = \sum_{\ell \geq 1} \alpha_\ell(\underline{x}) p_i^\ell$, where $\beta_\ell, \alpha_\ell \in \mathbb{C}\{\underline{x}\}$, then $\beta_1 = 0$ and

$$\beta_{\ell+1} = \frac{\ell}{\ell+1}\alpha_\ell + \frac{1}{\ell+1}\sum_{k=0}^{\ell}\left((k+1)\alpha_{k+1}\frac{\partial \beta_{\ell-k}}{\partial x_i} - (k+1)\beta_{k+1}\frac{\partial \alpha_{\ell-k}}{\partial x_i}\right), \quad \ell \geq 1.$$

Proof. Set $\beta_0 = 0$. The existence and uniqueness of $\beta, \gamma_1, \ldots, \gamma_n \in \mathbb{C}\{\underline{x}, y, \underline{p}\}$ are assured by Theorem 4.4.3. Let $\varphi \in \mathbb{C}\{\underline{x}, y, \underline{p}\}$ be such that $\varphi(0) \neq 0$. Then

$$\begin{aligned} d(y+\beta) - \sum_{j=1}^{n}(p_j+\gamma_j)dx_j - (p_i+\gamma_i)d\alpha = {} & \varphi\left(dy - \sum_{j=1}^{n} p_j dx_j\right) \\ & + (1-\varphi)dy + (\varphi-1)\sum_{j=1}^{n} p_j dx_j \\ & + \sum_{j=1}^{n} \frac{\partial\beta}{\partial x_j}dx_j + \frac{\partial\beta}{\partial y}dy + \sum_{j=1}^{n}\frac{\partial\beta}{\partial p_j}dp_j - \sum_{j=1}^{n}\gamma_j dx_j \\ & - \sum_{j=1}^{n}(p_i+\gamma_i)\frac{\partial\alpha}{\partial x_j}dx_j - (p_i+\gamma_i)\frac{\partial\alpha}{\partial y}dy - \sum_{j=1}^{n}(p_i+\gamma_i)\frac{\partial\alpha}{\partial p_j}dp_j. \end{aligned}$$

Collecting the coefficients of each differential form, one gets

$$\begin{aligned} d(y+\beta) - \sum_{j=1}^{n}(p_j+\gamma_j)dx_j - (p_i+\gamma_i)d\alpha = {} & \varphi\left(dy - \sum_{j=1}^{n} p_j dx_j\right) \\ & + \left(1-\varphi+\frac{\partial\beta}{\partial y} - (p_i+\gamma_i)\frac{\partial\alpha}{\partial y}\right)dy \\ & + \sum_{j=1}^{n}\left((\varphi-1)p_j + \frac{\partial\beta}{\partial x_j} - \gamma_j - (p_i+\gamma_i)\frac{\partial\alpha}{\partial x_j}\right)dx_j \\ & + \sum_{j=1}^{n}\left(\frac{\partial\beta}{\partial p_j} - (p_i+\gamma_i)\frac{\partial\alpha}{\partial p_j}\right)dp_j. \end{aligned}$$

Therefore (6.9) is a contact transformation if and only if

$$\varphi = 1 + \frac{\partial\beta}{\partial y} - (p_i+\gamma_i)\frac{\partial\alpha}{\partial y},$$

$$(\varphi-1)p_j + \frac{\partial\beta}{\partial x_j} - \gamma_j - (p_i+\gamma_i)\frac{\partial\alpha}{\partial x_j} = 0, \quad j = 1, \ldots, n,$$

$$\frac{\partial\beta}{\partial p_j} = (p_i+\gamma_i)\frac{\partial\alpha}{\partial p_j}, \quad j = 1, \ldots, n.$$

Our hypotheses allow us to simplify the above equations to

$$\varphi = 1+\frac{\partial\beta}{\partial y}, \quad \frac{\partial\beta}{\partial p_i} = (p_i+\gamma_i)\frac{\partial\alpha}{\partial p_i}, \quad \frac{\partial\beta}{\partial p_j} = 0, j \neq i, \quad \frac{\partial\beta}{\partial y}p_j + \frac{\partial\beta}{\partial x_j} - \gamma_j - (p_i+\gamma_i)\frac{\partial\alpha}{\partial x_j} = 0, j = 1, \ldots, n.$$

We write the last equations as

$$\gamma_j + \gamma_i\frac{\partial\alpha}{\partial x_j} = \frac{\partial\beta}{\partial y}p_j + \frac{\partial\beta}{\partial x_j} - p_i\frac{\partial\alpha}{\partial x_j}, \quad j = 1, \ldots, n,$$

or in matrix form $AX = B$, where

$$A = \begin{bmatrix} 1 & & & & \frac{\partial\alpha}{\partial x_1} & & & & \\ & 1 & & & \frac{\partial\alpha}{\partial x_2} & & & & \\ & & \ddots & & & & & & \\ & & & 1 & \frac{\partial\alpha}{\partial x_{i-1}} & & & & \\ & & & & 1+\frac{\partial\alpha}{\partial x_i} & & & & \\ & & & & \frac{\partial\alpha}{\partial x_{i+1}} & 1 & & & \\ & & & & & & \ddots & & \\ & & & & \frac{\partial\alpha}{\partial x_{n-1}} & & & 1 & \\ & & & & \frac{\partial\alpha}{\partial x_n} & & & & 1 \end{bmatrix}, \quad B = \begin{bmatrix} \frac{\partial\beta}{\partial y}p_1 + \frac{\partial\beta}{\partial x_1} - p_i\frac{\partial\alpha}{\partial x_1} \\ \vdots \\ \frac{\partial\beta}{\partial y}p_i + \frac{\partial\beta}{\partial x_i} - p_i\frac{\partial\alpha}{\partial x_i} \\ \vdots \\ \frac{\partial\beta}{\partial y}p_n + \frac{\partial\beta}{\partial x_n} - p_i\frac{\partial\alpha}{\partial x_n} \end{bmatrix}, \quad \text{and} \quad X = \begin{bmatrix} \gamma_1 \\ \vdots \\ \gamma_n \end{bmatrix}.$$

Given that $\frac{\partial\alpha}{\partial x_i} \in (p_i)\mathbb{C}\left\{\underline{x}, p_i\right\}$, then $|A| = 1 + \frac{\partial\alpha}{\partial x_i}$ is a unit in $\mathbb{C}\left\{\underline{x}, y, \underline{p}\right\}$. Therefore, the linear system has a unique solution. Explicitly, $|A|\gamma_j = |A_j|$, where A_j is equal to A up to the j-th column, which is equal to B. Now, for $j \neq i$,

$$|A_j| = \pm\left(\left(1+\frac{\partial\alpha}{\partial x_i}\right)\left(\frac{\partial\beta}{\partial y}p_j + \frac{\partial\beta}{\partial x_j} - p_i\frac{\partial\alpha}{\partial x_j}\right) - \left(\frac{\partial\alpha}{\partial x_j}\right)\left(\frac{\partial\beta}{\partial y}p_i + \frac{\partial\beta}{\partial x_i} - p_i\frac{\partial\alpha}{\partial x_i}\right)\right), \tag{6.10}$$

meanwhile for $j = i$,

$$|A_i| = \frac{\partial\beta}{\partial y}p_i + \frac{\partial\beta}{\partial x_i} - p_i\frac{\partial\alpha}{\partial x_i}. \tag{6.11}$$

Multiplying both sides of $\frac{\partial\beta}{\partial p_i} = (p_i + \gamma_i)\frac{\partial\alpha}{\partial p_i}$ by $|A|$, one obtains

$$\left(1+\frac{\partial\alpha}{\partial x_i}\right)\frac{\partial\beta}{\partial p_i} = \left(1+\frac{\partial\alpha}{\partial x_i}\right)p_i\frac{\partial\alpha}{\partial p_i} + \left(\frac{\partial\beta}{\partial y}p_i + \frac{\partial\beta}{\partial x_i} - p_i\frac{\partial\alpha}{\partial x_i}\right)\frac{\partial\alpha}{\partial p_i},$$

so that

$$\left(1+\frac{\partial\alpha}{\partial x_i}\right)\frac{\partial\beta}{\partial p_i} = \left(p_i + p_i\frac{\partial\beta}{\partial y} + \frac{\partial\beta}{\partial x_i}\right)\frac{\partial\alpha}{\partial p_i}. \tag{6.12}$$

Write

$$\beta = \sum_{\ell\geq 0}\beta_\ell p_i^\ell, \quad \alpha = \sum_{\ell\geq 0}\alpha_\ell p_i^\ell,$$

where $\beta_\ell \in \mathbb{C}\{\underline{x}, y\}$ and $\alpha_\ell \in \mathbb{C}\{\underline{x}\}$, for each $\ell \geq 0$. Therefore, from (6.12) one gets

$$\begin{aligned}\sum_{\ell\geq 0}(\ell+1)\beta_{\ell+1}p_i^\ell + \sum_{\ell\geq 0}\left(\sum_{k=0}^{\ell}(k+1)\beta_{k+1}\frac{\partial\alpha_{\ell-k}}{\partial x_i}\right)p_i^\ell &= \sum_{\ell\geq 0}\ell\alpha_\ell p_i^\ell + \sum_{\ell\geq 0}\left(\sum_{k=0}^{\ell}k\alpha_k\frac{\partial\beta_{\ell-k}}{\partial y}\right)p_i^\ell \\ &\quad + \sum_{\ell\geq 0}\left(\sum_{k=0}^{\ell}(k+1)\alpha_{k+1}\frac{\partial\beta_{\ell-k}}{\partial x_i}\right)p_i^\ell.\end{aligned}$$

Consequently, the following recursive formula for β holds

$$\begin{cases}\left(1+\frac{\partial\alpha_0}{\partial x_i}\right)\beta_1 = \alpha_1\frac{\partial\beta_0}{\partial x_i}, \\ \beta_{\ell+1} = \frac{1}{\ell+1}\left(\ell\alpha_\ell + \sum_{k=0}^{\ell}\left(k\alpha_k\frac{\partial\beta_{\ell-k}}{\partial y} + (k+1)\alpha_{k+1}\frac{\partial\beta_{\ell-k}}{\partial x_i} - (k+1)\beta_{k+1}\frac{\partial\alpha_{\ell-k}}{\partial x_i}\right)\right), \quad \ell \geq 1.\end{cases}$$

Moreover, our initial hypothesis tells us that $\frac{\partial \alpha_0}{\partial x_i} \in (p_i)\mathbb{C}\left\{\underline{x}, p_i\right\}$, so that $1 + \frac{\partial \alpha_0}{\partial x_i}$ is a unit in $\mathbb{C}\left\{\underline{x}, y, \underline{p}\right\}$. Since $\beta_0 = 0$, then $\beta_1 = 0$. Let $\ell \geq 1$ and suppose that for all $k \leq \ell$ one has $\frac{\partial \beta_k}{\partial y} = 0$. Since $\alpha \in \mathbb{C}\left\{\underline{x}, p_i\right\}$, then $\frac{\partial \alpha_k}{\partial y} = 0$, for all k. Therefore,

$$\frac{\partial \beta_{\ell+1}}{\partial y} = \frac{\ell}{\ell+1}\frac{\partial \alpha_\ell}{\partial y} - \frac{1}{\ell+1}\sum_{k=0}^{\ell}\left((k+1)\frac{\partial \beta_{k+1}}{\partial y}\frac{\partial \alpha_{\ell-k}}{\partial x_i}\right) = 0.$$

Thus, $\frac{\partial \beta_\ell}{\partial y} = 0$ for all ℓ, that is, $\frac{\partial \beta}{\partial y} = 0$ and $\beta \in \mathbb{C}\{\underline{x}, p_i\}$, given that

$$\frac{\partial \beta}{\partial p_j} = 0, \quad \text{for all } j \neq i.$$

Finally, from (6.10) and (6.11), one concludes that

$$\gamma_1, \ldots, \gamma_n \in \mathbb{C}\{\underline{x}, p_i\}.$$

□

Corollary 6.2.2. *Suppose α, β as in Lemma 6.2.1. If $\alpha(\underline{x}, p_i) = \alpha_\ell(\underline{x})p_i^\ell$, for some $\ell \geq 1$, then*

$$\beta(\underline{x}, p_i) = \frac{\ell}{\ell+1}\alpha_\ell(\underline{x})p_i^{\ell+1}\varepsilon(\underline{x}, p_i),$$

where $\varepsilon \in \mathbb{C}\{\underline{x}, p_i\}$ is a unit with constant term equal to 1.

Proof. Let $I = (\alpha_\ell)\mathbb{C}\{\underline{x}\}$. We claim that $\beta_j \in I$ for all $j \geq 0$. By means of Lemma 6.2.1., one reaches the conclusion that $\beta_0 = \beta_1 = \ldots = \beta_\ell = 0$ and $\beta_{\ell+1} = \frac{\ell}{\ell+1}\alpha_\ell$. Let $j > \ell + 1$ and assume $\beta_k \in I$ for all $k < j$. Then

$$\begin{aligned}
\beta_j &= \frac{1}{j}\left((j-1)\alpha_{j-1} + \sum_{k=0}^{j-1}(k+1)\left(\alpha_{k+1}\frac{\partial \beta_{j-k-1}}{\partial x_i} - \beta_{k+1}\frac{\partial \alpha_{j-k-1}}{\partial x_i}\right)\right) \\
&= \frac{\ell}{j}\left(\alpha_\ell\frac{\partial \beta_{j-\ell}}{\partial x_i} - \beta_\ell\frac{\partial \alpha_{j-\ell}}{\partial x_i}\right) + \frac{j-\ell}{j}\left(\alpha_{j-\ell}\frac{\partial \beta_\ell}{\partial x_i} - \beta_{j-\ell}\frac{\partial \alpha_\ell}{\partial x_i}\right) \\
&= \frac{\ell}{j}\alpha_\ell\frac{\partial \beta_{j-\ell}}{\partial x_i} - \frac{j-\ell}{j}\beta_{j-\ell}\frac{\partial \alpha_\ell}{\partial x_i} \in I,
\end{aligned}$$

whenever $j \neq 2\ell$. On the other hand, if $j = 2\ell$, then

$$\beta_{2\ell} = \frac{1}{2}\left(\alpha_\ell\frac{\partial \beta_\ell}{\partial x_i} - \beta_\ell\frac{\partial \alpha_\ell}{\partial x_i}\right) = 0 \in I.$$

The claim then follows. Therefore, for each $k > \ell + 1$, $\beta_k = \alpha_\ell\varepsilon_k$, where $\varepsilon_k \in \mathbb{C}\left\{\underline{x}\right\}$. Thus,

$$\beta(\underline{x}, p_i) = \sum_{k\geq\ell+1}\beta_k p_i^k = \frac{\ell}{\ell+1}\alpha_\ell p_i^{\ell+1} + \sum_{k>\ell+1}\beta_k p_i^k = \frac{\ell}{\ell+1}\alpha_\ell p_i^{\ell+1}\left(1 + \sum_{k>\ell+1}\frac{\ell+1}{\ell}\varepsilon_k p_i^{k-\ell-1}\right),$$

which concludes the proof. □

Lemma 6.2.3. *Let $\alpha_\ell, \beta_0 \in \mathbb{C}\{\underline{x}, y\}$, $\ell = 1, \ldots, n$, be such that*

$$\frac{\partial \alpha_\ell}{\partial x_j}, \frac{\partial \beta_0}{\partial y} \in (\underline{x}, y)\mathbb{C}\{\underline{x}, y\}, \quad j, \ell \in \{1, \ldots, n\}.$$

There are $\gamma_1, \ldots, \gamma_n \in \mathbb{C}\{\underline{x}, y, \underline{p}\}$ such that

$$(\underline{x}, y, \underline{p}) \mapsto (\underline{x} + \underline{\alpha}, y + \beta_0, \underline{p} + \underline{\gamma})$$

is an infinitesimal contact transformation, where $\underline{\alpha} = (\alpha_1, \ldots, \alpha_n)$.

Proof. Given β_0, the existence and uniqueness of $\beta, \gamma_1, \ldots, \gamma_n \in \mathbb{C}\{\underline{x}, y, \underline{p}\}$ such that

$$(\underline{x}, y, \underline{p}) \mapsto (\underline{x} + \underline{\alpha}, y + \beta, \underline{p} + \underline{\gamma}) \tag{6.13}$$

is an infinitesimal contact transformation are assured by Theorem 4.4.3. Similar to the reasoning used in the Lemma 6.2.1, (6.13) is a contact transformation if and only if there exists $\varphi \in \mathbb{C}\{\underline{x}, y, \underline{p}\}$ such that $\varphi(0) \neq 0$ and the equations

$$\varphi = 1 + \frac{\partial \beta}{\partial y} - \sum_{k=1}^{n} (p_k + \gamma_k) \frac{\partial \alpha_k}{\partial y},$$

$$\gamma_j + \sum_{k=1}^{n} \gamma_k \frac{\partial \alpha_k}{\partial x_j} = p_j \frac{\partial \beta}{\partial y} - p_j \sum_{k=1}^{n} (p_k + \gamma_k) \frac{\partial \alpha_k}{\partial y} + \frac{\partial \beta}{\partial x_j} - \sum_{k=1}^{n} p_k \frac{\partial \alpha_k}{\partial x_j}, \quad j = 1, \ldots, n, \tag{6.14}$$

$$\frac{\partial \beta}{\partial p_j} = 0, \quad j = 1, \ldots, n, \tag{6.15}$$

hold. In particular, by equation (6.15) and $\beta - \beta_0 \in (\underline{p})\mathbb{C}\{\underline{x}, y, \underline{p}\}$, we conclude that $\beta = \beta_0$. □

Theorem 6.2.4. *Let Y be a q.o. hypersurface with characteristic exponent $\lambda = (\lambda_1, \ldots, \lambda_n)$ and let m be the ramification number of λ. Let $G = \Gamma(\lambda)_{\mathcal{G}}$, υ be a Puiseux parametrization of Y, and $a = (a_1, \ldots, a_n), b \in \mathbb{N}_0^n$ be such that*

$$b + a_1(\lambda - \mathbb{1}_1) + \ldots + a_n(\lambda - \mathbb{1}_n) > \lambda.$$

Suppose that $\Phi_{\underline{\alpha},\beta}$ is an infinitesimal contact transformation where, for each $i \in \{1, \ldots, n\}$, either $\alpha_i = 0$ or $Supp(\upsilon^*\alpha_i)$ is dominated by $(G, \sigma^{(i)})$, where $\sigma^{(i)} = (\sigma_1^{(i)}, \ldots, \sigma_n^{(i)}) \in Supp(\upsilon^*\alpha_i)$ is such that $\sigma_i^{(i)} > m$. Moreover, assume that $\beta \in (\underline{x}^b \underline{p}^a, x_r y)\mathbb{C}\{\underline{x}, y, \underline{p}\}$, for some $r \in \{1, \ldots, n\}$. Then $(\pi \circ \Phi_{\underline{\alpha},\beta})(\mathbb{P}_Y^* \mathbb{C}^{n+1})$ is a q.o. hypersurface with characteristic exponent λ.

Proof. The goal of this proof is to obtain a Puiseux parametrization for $(\pi \circ \Phi_{\underline{\alpha},\beta})(\mathbb{P}_Y^* \mathbb{C}^{n+1})$. To achieve this, one writes the Puiseux parametrization υ for Y as

$$x_k = t_k^m, \ k = 1, \ldots, n, \quad y = \sum_{\tau \in \Gamma(\lambda)_{\mathcal{G}}} c_\tau \underline{t}^\tau.$$

Then $(\pi\circ\Phi_{\underline{\alpha},\beta})(\mathbb{P}^*_Y\mathbb{C}^{n+1})$ may be parametrized as follows

$$x_k = t_k^m + (v^*\alpha_k)(\underline{t}),\ \ k=1,\dots,n, \quad y = \sum_{\tau\in\Gamma(\lambda)_{\mathcal{G}}} c_\tau \underline{t}^\tau + (v^*\beta)(\underline{t}).$$

By Lemma 4.2.3, for each $i\in\{1,\dots,n\}$, the branches of $\pi_i(\mathbb{P}^*_Y\mathbb{C}^{n+1})$ are strongly normalized and therefore $\operatorname{Supp}(v^*p_i)$ is dominated by $(G, m\lambda - m\mathbb{1}_i)$. Given that $\operatorname{Supp}(\underline{t}^{mb})$ is dominated by (G, mb), then, by part $ii)$ of Proposition 6.1.4, $\operatorname{Supp}(\underline{t}^{mb}(v^*\underline{p}^a))$ is dominated by

$$(G, mb + a_1(m\lambda - m\mathbb{1}_1) + \ldots + a_n(m\lambda - m\mathbb{1}_n)).$$

By Lemma 4.2.3, $\operatorname{Supp}(v^*p_i)\subseteq G$, for each $i\in\{1,\dots,n\}$. For every $\tau,\rho\in\mathbb{N}_0^n\setminus\{0\}$ and $s\in\mathbb{N}_0$, we then have that $\operatorname{Supp}(v^*(\underline{x}^\tau y^s \underline{p}^\rho)) = \operatorname{Supp}(\underline{t}^{m\tau}v^*(y^s)v^*(\underline{p}^\rho))\subseteq G$, by $i)$ of Proposition 6.1.4. Thus, for any $\chi_1\in\mathbb{C}\{\underline{x},y,\underline{p}\}$, it follows that

$$\operatorname{Supp}(v^*\chi_1)\subseteq \bigcup_{\substack{\tau,\rho\in\mathbb{N}_0^n\setminus\{0\}\\ s\in\mathbb{N}_0}} \operatorname{Supp}(\underline{t}^{m\tau}v^*(y^s)v^*(\underline{p}^\rho))\subseteq G.$$

Again by $i)$ of Proposition 6.1.4,

$$\operatorname{Supp}(\underline{t}^{mb}(v^*\underline{p}^a)(v^*\chi_1))\subseteq G\cap(mb + a_1(m\lambda - m\mathbb{1}_1) + \ldots + a_n(m\lambda - m\mathbb{1}_n)).$$

By our initial hypothesis,

$$mb + a_1(m\lambda - m\mathbb{1}_1) + \ldots + a_n(m\lambda - m\mathbb{1}_n) > m\lambda.$$

Therefore, $\operatorname{Supp}(\underline{t}^{mb}(v^*\underline{p}^a)(v^*\chi_1))\subseteq G\cap \llcorner(m\lambda)\setminus\{m\lambda\}$. Following a similar reasoning, from the fact that $\operatorname{Supp}(v^*y)$ is dominated by $(G, m\lambda)$, then

$$\operatorname{Supp}(t_r^m(v^*y)(v^*\chi_2))\subseteq G\cap\llcorner(m\lambda)\setminus\{m\lambda\},$$

for each $r\in\{1,\dots,n\}$ and each $\chi_2\in\mathbb{C}\{\underline{x},y,\underline{p}\}$. Given that $\beta\in(\underline{x}^b\underline{p}^a, x_r y)\mathbb{C}\{\underline{x},y,\underline{p}\}$, for some $r\in\{1,\dots,n\}$, then

$$\beta = \chi_1(\underline{x},y,\underline{p})\underline{x}^b\underline{p}^a + \chi_2(\underline{x},y,\underline{p})x_r y,$$

for some $\chi_1,\chi_2\in\mathbb{C}\{\underline{x},y,\underline{p}\}$. Therefore,

$$\operatorname{Supp}(v^*\beta)\subseteq \operatorname{Supp}(\underline{t}^{mb}(v^*\chi_1)(v^*\underline{p}^a))\cup\operatorname{Supp}(t_r^m(v^*\chi_2)(v^*y))\subseteq G\cap\llcorner(m\lambda)\setminus\{m\lambda\}.$$

Hence,

$$\operatorname{Supp}(v^*y + v^*\beta)\subseteq\operatorname{Supp}(v^*y)\cup\operatorname{Supp}(v^*\beta)\subseteq G\cap\llcorner(m\lambda).$$

Given that $m\lambda\in\operatorname{Supp}(v^*y)\setminus\operatorname{Supp}(v^*\beta)$, then $m\lambda\in\operatorname{Supp}(v^*y+v^*\beta)$, so that $\operatorname{Supp}(v^*y+v^*\beta)$ is dominated by $(G, m\lambda)$.

In what follows, we make the substitution

$$u_1^m = t_1^m + (v^*\alpha_1)(\underline{t}).$$

If $\alpha_1 = 0$, $u_1^m = t_1^m$ and there is nothing left to do. Suppose otherwise. Since $\mathrm{Supp}(v^*\alpha_1)$ is dominated by $(G, \sigma^{(1)})$ and $\sigma_1^{(1)} > m$, it follows that we can write the above equality as

$$u_1^m = t_1^m(1 + \delta_1(\underline{t})),$$

where $\mathrm{Supp}(\delta_1)$ is dominated by $(G, \sigma^{(1)} - m\mathbb{1}_1)$ and $\delta_1 \in (t_1)\mathbb{C}\{\underline{t}\}$. By $ii)$ of Lemma 6.1.7,

$$t_1 = u_1(1 + \varepsilon_1(u_1, t_2, \dots, t_n)),$$

where $\mathrm{Supp}(\varepsilon_1)$ is dominated by $(G, \sigma^{(1)} - m\mathbb{1}_1)$ and $\varepsilon_1 \in (u_1)\mathbb{C}\{u_1, t_2, \dots, t_n\}$. In particular, $\mathrm{Supp}(\varepsilon_1) \subseteq G$. We then make this substitution in each $(v^*\alpha_k)(\underline{t})$, with $k \in \{2, \dots, n\}$, and in $(v^*y + v^*\beta)(\underline{t})$. By Proposition 2.6 of [26], the convergence of $(v^*\alpha_k)(\underline{t})$, with $k \in \{2, \dots, n\}$, of $(v^*y + v^*\beta)(\underline{t})$, and of ε_1 imply that

$$(v^*\alpha_k)(u_1(1 + \varepsilon_1(u_1, t_2, \dots, t_n)), t_2, \dots, t_n), \quad k \in \{2, \dots, n\}$$

and

$$(v^*y + v^*\beta)(u_1(1 + \varepsilon_1(u_1, t_2, \dots, t_n)), t_2, \dots, t_n)$$

are also convergent power series. On the other hand, since $\varepsilon_1 \in (u_1)\mathbb{C}\{u_1, t_2, \dots, t_n\}$ and $\mathrm{Supp}(\varepsilon_1) \subseteq G$, then by part $i)$ of Proposition 6.1.6,

$$\mathrm{Supp}((v^*y + v^*\beta)(u_1(1 + \varepsilon_1(u_1, t_2, \dots, t_n)), t_2, \dots, t_n))$$

is dominated by $(G, m\lambda)$, and

$$\mathrm{Supp}(v^*\alpha_k)(u_1(1 + \varepsilon_1(u_1, t_2, \dots, t_n)), t_2, \dots, t_n),$$

is dominated by $(G, \sigma^{(k)})$, for each $k \in \{2, \dots, n\}$, whenever $\alpha_k \neq 0$.

By successively repeating this process for each substitution

$$u_k^m = t_k^m + (v^*\alpha_k)(\underline{t}),$$

for each $k \in \{2, \dots, n\}$, we obtain a Puiseux parametrization, say $\tilde{v}$, for $(\pi \circ \Phi_{\underline{\alpha},\beta})(\mathbb{P}^*_Y\mathbb{C}^{n+1})$ of the type

$$x_k = u_k^m, \;\; k = 1, \dots, n, \quad y = \sum_{\tau \in \Gamma(\lambda)_{\mathcal{G}}} d_\tau \underline{u}^\tau,$$

where $\underline{u} = (u_1, \dots, u_n)$ and $\mathrm{Supp}(\tilde{v}^*y)$ is dominated by $(G, m\lambda)$. Thus, $(\pi \circ \Phi_{\underline{\alpha},\beta})(\mathbb{P}^*_Y\mathbb{C}^{n+1})$ is a q.o. hypersurface with characteristic exponent λ. □

6.3 Plain Q.O. Surfaces

In what follows, we will restrict ourselves to the case $n = 2$. To motivate this restriction, recall that part $iii)$ of Lemma 5.0.6 only guarantees the finiteness of $(\Gamma(\lambda)_{\mathcal{G}} \setminus \Lambda(\lambda)) \cap \llcorner(m\lambda)$ if $n = 2$. In particular, if the conditions of this result are verified, then

$$(\Gamma(\lambda)_{\mathcal{G}} \setminus \Lambda(\lambda)) \cap \llcorner(m\lambda) \subseteq [m\lambda_1, (m-1)m\lambda_1 - m] \times [m\lambda_2, (m-1)m\lambda_2 - m].$$

For simplicity, we write $G = \Gamma(\lambda)_{\mathcal{G}}$ and $R = [m\lambda_1, (m-1)m\lambda_1 - m] \times [m\lambda_2, (m-1)m\lambda_2 - m]$. Let $S \subseteq \mathbb{Z}^2$. Whenever we write ε_S, $\tilde{\varepsilon}_S$ or δ_S, it is understood that $\varepsilon_S, \tilde{\varepsilon}_S, \delta_S \in \mathbb{C}\{\underline{t}\}$ and

$$\mathrm{Supp}(\varepsilon_S), \mathrm{Supp}(\tilde{\varepsilon}_S), \mathrm{Supp}(\delta_S) \subseteq S.$$

Let Y be a plain q.o. surface with characteristic exponent λ (see Definition 4.3.7). Associated to this q.o. surface, we write a Puiseux parametrization, say v, for Y as

$$x_1 = t_1^m, \quad x_2 = t_2^m, \quad y = \underline{t}^{m\lambda} + \sum_{\tau \in G} c_\tau \underline{t}^\tau.$$

Let $\underline{u} = (u_1, u_2)$. As in the proof of Theorem 6.2.4, if we make reparametrizations

$$t_1 = u_1^m + \varphi_1(\underline{u}), \quad t_2 = u_2^m + \varphi_2(\underline{u}),$$

where $\varphi_1, \varphi_2 \in \mathbb{C}\{\underline{u}\}$, we denote by $\tilde{v}^* y$ the parametrization $(v^* y)(u_1^m + \varphi_1(\underline{u}), u_2^m + \varphi_2(\underline{u}))$.

Recall that every branch of Y is strongly normalized (see Definition 3.1.2). Therefore, $\mathrm{Supp}(v^* y)$ is dominated by $(G, m\lambda)$. Moreover, by the definition of a plain q.o. surface, $m\lambda = (m+1, m+1)$.

Let $\sigma = (\sigma_1, \sigma_2) \in \Lambda(\lambda)_{\mathcal{G}} \cap \mathrm{Supp}(v^* y)$. By Proposition 5.0.4, one can write σ as

$$\sigma = b_{\sigma,1} m \mathbb{1}_1 + b_{\sigma,2} m \mathbb{1}_2 + a_{\sigma,1}(m\lambda - m\mathbb{1}_1) + a_{\sigma,2}(m\lambda - m\mathbb{1}_2),$$

where $b_\sigma = (b_{\sigma,1}, b_{\sigma,2}) \in \mathbb{Z}^2$ and $a_{\sigma,1}, a_{\sigma,2} \in \{0, 1, \ldots, m-1\}$. If, in particular, $\sigma \in \Lambda_i(\lambda)$ for some $i \in \{1, 2\}$, we can write uniquely

$$\sigma = m b_\sigma + a_\sigma (m\lambda - m\mathbb{1}_i),$$

with $b_\sigma \in \mathbb{N}_0^2$ and $a_\sigma \in \{1, \ldots, m-1\}$, by $ii)$ of Remark 5.0.5. Similarly, if $\sigma \in \Gamma(\lambda)$, we also write uniquely

$$\sigma = m b_\sigma + a_\sigma m\lambda,$$

with $b_\sigma \in \mathbb{N}_0^2$ and $a_\sigma \in \{1, \ldots, m-1\}$. Given that these cases will be treated separately, there is no risk of ambiguity here. Furthermore, let $\tau = (\tau_1, \tau_2) \in \mathbb{Z}^2$ and $i \in \{1, 2\}$. Recall that

$$\mathcal{R}_i(\tau) = \llcorner(\tau) \cap \left\{\rho = (\rho_1, \rho_2) \in \mathbb{Z}^2 : \rho_i = \tau_i\right\}.$$

We define the sets

$$L_{\tau,i} := \left\{\tau + \ell m \mathbb{1}_j : \ell \in \mathbb{N}_0, j \in \{1, 2\} \setminus \{i\}\right\}, \qquad M_{\tau,i} := \llcorner(\tau) \setminus \mathcal{R}_i(\tau),$$

and

$$N_{\tau,i} := \mathrm{Supp}(v^* y) \setminus (\{m\lambda\} \cup L_{\tau,i} \cup M_{\tau,i}),$$

for $i = 1, 2$.

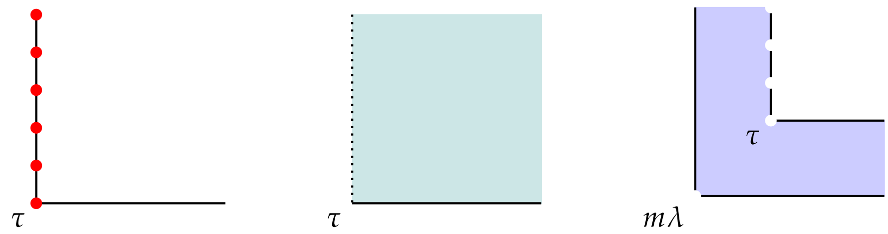

Figure 6.2: Illustration, respectively from left to right, of $L_{\tau,1}$, $M_{\tau,1}$, and $N_{\tau,1}$.

Remark 6.3.1. *Let $\sigma = (\sigma_1, \sigma_2) \in Supp(v^*y)$ such that $\sigma \in \Lambda_i(\lambda)$, for some $i \in \{1, 2\}$.*

i) If $b_{\sigma,i} = 0$, then

$$\sigma_i = a_\sigma(m\lambda_i - m) \leq (m-1)(m+1-m) = m-1 < m+1 = m\lambda_i,$$

which is a contradiction, since $\sigma \in \llcorner(m\lambda)$. Hence $b_{\sigma,i} \geq 1$.

ii) Assume $a_\sigma = 1$. Then

$$\sigma = mb_\sigma + a_\sigma(m\lambda - m\mathbb{1}_i) = m(b_\sigma - \mathbb{1}_i) + m\lambda.$$

Since $b_{\sigma,i} \geq 1$, then $\sigma \in \Gamma(\lambda)$.

Remark 6.3.2. *Let $\tau = (\tau_1, \tau_2) \in \Gamma(\lambda)_{\mathcal{G}} \cap \llcorner(m\lambda)$. Recall that, by ii) of Remark 5.0.5, $a_\tau \in \{0, 1, \ldots, m-1\}$. If $\tau_i = m\lambda_i$, for some $i \in \{1, 2\}$, then*

$$b_{\tau,i}m + a_\tau m\lambda_i = m\lambda_i,$$

so that $b_{\tau,i}m = (1-a_\tau)m\lambda_i$. Hence, there exists $k \in \mathbb{Z}$ such that $a_\tau - 1 = km$, since $\gcd(m, m\lambda_i) = 1$. By the restriction on a_τ, we conclude that $a_\tau = 1$. Therefore, $b_{\tau,i} = 0$ and $b_{\tau,j} \geq 0$, for $j \neq i$. Thus, $\tau \in \Gamma(\lambda)$. This shows that

$$\Gamma(\lambda)_{\mathcal{G}} \cap \mathcal{R}_i(m\lambda) \cap \llcorner(m\lambda) = L_{m\lambda,i}.$$

Lemma 6.3.3. *Let Y be a plain q.o. surface and let $\sigma \in \Gamma(\lambda) \cap \llcorner(m\lambda) \setminus \{m\lambda\}$ be such that $a_\sigma = 1$. Assume Y admits the Puiseux parametrization defined by*

$$x_1 = t_1^m, x_2 = t_2^m, y = \underline{t}^{m\lambda} + \varepsilon_{L_{\sigma,i}} + \varepsilon_{M_{\sigma,i}} + \varepsilon_{N_{\sigma,i}},$$

where $i \in \{1, 2\}$. Let $j \in \{1, 2\} \setminus \{i\}$. Let $\underline{\alpha} = 0$ and $\beta = -\underline{x}^{b_\sigma} y \sum_{\ell \geq 0} \kappa_\ell x_j^\ell \in \mathbb{C}\{\underline{x}, y\}$, where $\kappa_\ell \in \mathbb{C}$ for each $\ell \geq 0$. Then $(\pi \circ \Phi_{\underline{\alpha},\beta})(\mathbb{P}^*_Y\mathbb{C}^3)$ is a plain q.o. surface that admits the following Puiseux parametrization

$$x_1 = t_1^m, x_2 = t_2^m,\ y = \underline{t}^{m\lambda} + \tilde{\varepsilon}_{L_{\sigma,i}} + \tilde{\varepsilon}_{M_{\sigma,i}} + \varepsilon_{N_{\sigma,i}}.$$

In particular,

$$\tilde{\varepsilon}_{L_{\sigma,i}} = \underline{t}^\sigma \sum_{\ell \geq 0} \left(c_{\tau_\ell} - \kappa_\ell - \sum_{s=0}^{\ell-1} \kappa_s c_{\tilde{\tau}_{\ell-s}} \right) t_j^{\ell m}$$

where $\tau_\ell = \sigma + \ell m \mathbb{1}_j$ and $\tilde{\tau}_\ell = m\lambda + m\ell\mathbb{1}_j$, for each $\ell \geq 0$.

Proof. Given that $\underline{\alpha} = 0$ and $\frac{\partial\beta}{\partial y} \in (\underline{x}, y)\mathbb{C}\{\underline{x}, y\}$, by Lemma 6.2.3, $\Phi_{\underline{\alpha},\beta}$ is a contact transformation. Note that $\sigma \in \Gamma(\lambda) \cap \llcorner(m\lambda) \setminus \{m\lambda\}$ implies

$$m\lambda < \sigma = mb_\sigma + a_\sigma m\lambda = m\lambda + mb_\sigma,$$

so that $b_\sigma \neq 0$. Therefore, $\beta \in (x_1 y)\mathbb{C}\{\underline{x}, y, \underline{p}\}$ or $\beta \in (x_2 y)\mathbb{C}\{\underline{x}, y, \underline{p}\}$. Then, since $\underline{\alpha} = 0$, by Theorem 6.2.4, $(\pi \circ \Phi_{\underline{\alpha},\beta})(\mathbb{P}^*_Y\mathbb{C}^3)$ is a plain q.o. surface.

Let $\delta_{N_{\sigma,i}} = \varepsilon_{N_{\sigma,i}} - \underline{t}^{m\lambda} \sum_{\ell\geq 1} c_{\tilde{\tau}_\ell} t_j^{m\ell}$. Note that

$$\mathrm{Supp}(\delta_{N_{\sigma,i}}) \cap \mathcal{R}_i(m\lambda) = \mathrm{Supp}(\delta_{N_{\sigma,i}}) \cap L_{m\lambda,i} = \emptyset.$$

Therefore,

$$\begin{aligned}
v^*y + v^*\beta &= \underline{t}^{m\lambda} + \varepsilon_{L_{\sigma,i}} + \varepsilon_{M_{\sigma,i}} + \varepsilon_{N_{\sigma,i}} \\
&\quad - \underline{t}^{mb_\sigma} \sum_{\ell\geq 0} \kappa_\ell t_j^{\ell m} \left(\underline{t}^{m\lambda} + \underline{t}^{m\lambda} \sum_{s\geq 1} c_{\tilde{\tau}_s} t_j^{sm} + \varepsilon_{L_{\sigma,i}} + \varepsilon_{M_{\sigma,i}} + \delta_{N_{\sigma,i}} \right) \\
&= \underline{t}^{m\lambda} + \underline{t}^{\sigma} \sum_{\ell\geq 0} \left(c_{\tau_\ell} - \kappa_\ell - \sum_{s=0}^{\ell-1} \kappa_s c_{\tilde{\tau}_{\ell-s}} \right) t_j^{\ell m} + \tilde{\varepsilon}_{M_{\sigma,i}} + \varepsilon_{N_{\sigma,i}},
\end{aligned}$$

where

$$\tilde{\varepsilon}_{M_{\sigma,i}} = -\underline{t}^{mb_\sigma} \sum_{\ell\geq 0} \kappa_\ell t_j^{\ell m} \left(\varepsilon_{L_{\sigma,i}} + \varepsilon_{M_{\sigma,i}} + \delta_{N_{\sigma,i}} \right).$$

Indeed $\mathrm{supp}(\tilde{\varepsilon}_{M_{\sigma,i}}) \subseteq M_{\sigma,i}$, given that if $\tau \in \mathrm{Supp}(\varepsilon_{L_{\sigma,i}} + \varepsilon_{M_{\sigma,i}} + \delta_{N_{\sigma,i}})$, then $\tau > m\lambda$. If $\tau_i = m\lambda_i$, then, by Remark 6.3.2, $\tau = m\lambda + m\ell\mathbb{1}_j$ for some $\ell > 0$. Since $\tau \in N_{\sigma,i} \cap L_{m\lambda,i}$, it follows that $\tau \in \mathrm{Supp}(\delta_{N_{\sigma,i}}) \cap L_{m\lambda,i}$. However, by the construction of $\delta_{N_{\sigma,i}}$ this cannot happen. Hence, $\tau_i > m\lambda_i$. Thus, $\tau + mb_\sigma > m\lambda + mb_\sigma = \sigma$ and $\tau_i + mb_{\sigma,i} > \sigma_i$, so that $\tau + mb_\sigma \in M_{\sigma,i}$, and the same follows for $\tau + mb_\sigma + \ell m\mathbb{1}_j$, with $\ell \geq 0$. □

Corollary 6.3.4. *Let Y be a plain q.o. surface and let $\sigma \in \Gamma(\lambda) \cap \llcorner(m\lambda) \setminus \{m\lambda\}$ be such that $a_\sigma = 1$. Assume Y admits the Puiseux parametrization defined by*

$$x_1 = t_1^m, x_2 = t_2^m, y = \underline{t}^{m\lambda} + \varepsilon_{L_{\sigma,i}} + \varepsilon_{M_{\sigma,i}} + \varepsilon_{N_{\sigma,i}},$$

where $i \in \{1, 2\}$. Let $j \in \{1, 2\} \setminus \{i\}$. Also, assume that $\mathrm{Supp}(v^*y) \cap \mathcal{R}_i(m\lambda) = \{m\lambda\}$. Let $\underline{\alpha} = 0$ and $\beta = -\underline{x}^{b_\sigma} y \sum_{\ell\geq 0} \kappa_\ell x_j^\ell$, where $\kappa_\ell \in \mathbb{C}$ for each $\ell \geq 0$. Then $(\pi \circ \Phi_{\underline{\alpha},\beta})(\mathbb{P}^*_Y\mathbb{C}^3)$ is a plain q.o. surface that admits the following Puiseux parametrization

$$x_1 = t_1^m, x_2 = t_2^m,\ y = \underline{t}^{m\lambda} + \tilde{\varepsilon}_{L_{\sigma,i}} + \tilde{\varepsilon}_{M_{\sigma,i}} + \varepsilon_{N_{\sigma,i}},$$

where

$$\tilde{\varepsilon}_{L_{\sigma,i}} = \underline{t}^{\sigma} \sum_{\ell\geq 0} \left(c_{\tau_\ell} - \kappa_\ell \right) t_j^{\ell m}$$

and $\tau_\ell = \sigma + \ell m\mathbb{1}_j$, for each $\ell \geq 0$.

Proposition 6.3.5. *Let Y be a plain q.o. surface. Assume Y admits a Puiseux parametrization defined by*

$$x_1 = t_1^m, \;\; x_2 = t_2^m, \;\; y = \underline{t}^{m\lambda} + \sum_{\tau \in G} c_\tau \underline{t}^\tau,$$

such that $Supp(\upsilon^* y) \cap \mathcal{R}_i(m\lambda) = \{m\lambda\}$. Then, for each $i \in \{1, 2\}$, $\pi_i(\mathbb{P}^*_Y \mathbb{C}^3)$ admits a Puiseux parametrization defined by

$$x_1 = t_1^m, \;\; x_2 = t_2^m, \;\; p_i = \lambda_i \underline{t}^{m\lambda - m\mathbb{1}_i} + \sum_{\tau \in G} d_\tau \underline{t}^\tau,$$

such that $Supp(\upsilon^* p_i^s) \cap \mathcal{R}_i(s(m\lambda - m\mathbb{1}_i)) = \{s(m\lambda - m\mathbb{1}_i)\}$, for all $s \in \mathbb{N}$.

Proof. By Lemma 4.2.3, $\pi_i(\mathbb{P}^*_Y \mathbb{C}^3)$ is a q.o. surface with characteristic exponent $\lambda - \mathbb{1}_i$, with strongly normalized branches, up to a change of coordinates. Therefore, $\mathrm{Supp}(\upsilon^* p_i)$ is dominated by $(G, m\lambda - m\mathbb{1}_i)$. Moreover, by (4.2), we see that

$$\mathrm{Supp}(\upsilon^* p_i) = \left\{\tau - m\mathbb{1}_i : \tau \in \mathrm{Supp}(\upsilon^* y)\right\}.$$

Therefore, if

$$\rho = (\rho_1, \rho_2) \in \mathrm{Supp}(\upsilon^* p_i) \cap \mathcal{R}_i(m\lambda - m\mathbb{1}_i),$$

then $\rho = \tau - m\mathbb{1}_i$, for some $\tau = (\tau_1, \tau_2) \in \mathrm{Supp}(\upsilon^* y)$. On the other hand, since $\rho \in \mathcal{R}_i(m\lambda - m\mathbb{1}_i)$, then $m\lambda_i - m = \rho_i = \tau_i - m$, so that $\tau_i = m\lambda_i$ and, consequently, $\tau \in \mathcal{R}_i(m\lambda) \cap \mathrm{Supp}(\upsilon^* y)$. Hence, $\tau = m\lambda$ and $\rho = m\lambda - m\mathbb{1}_i$. Therefore,

$$\mathrm{Supp}(\upsilon^* p_i) \cap \mathcal{R}_i(m\lambda - m\mathbb{1}_i) \subseteq \{m\lambda - m\mathbb{1}_i\}.$$

Conversely, $m\lambda - m\mathbb{1}_i \in \mathrm{Supp}(\upsilon^* p_i) \cap \mathcal{R}_i(m\lambda - m\mathbb{1}_i)$ is immediately verified. Thus,

$$\mathrm{Supp}(\upsilon^* p_i) \cap \mathcal{R}_i(m\lambda - m\mathbb{1}_i) = \{m\lambda - m\mathbb{1}_i\}.$$

We now claim that $\mathrm{Supp}(\upsilon^* p_i^s) \cap \mathcal{R}_i(s(m\lambda - m\mathbb{1}_i)) = \{s(m\lambda - m\mathbb{1}_i)\}$. Let

$$\rho = (\rho_1, \rho_2) \in \mathrm{Supp}(\upsilon^* p_i^s) \cap \mathcal{R}_i(s(m\lambda - m\mathbb{1}_i)),$$

then there exist $\rho^{(1)}, \ldots, \rho^{(s)} \in \mathrm{Supp}(\upsilon^* p_i)$ such that

$$\rho = \rho^{(1)} + \ldots + \rho^{(s)}.$$

For each $k \in \{1, \ldots, s\}$, write $\rho^{(k)} = (\rho_1^{(k)}, \rho_2^{(k)})$. Since $\rho \in \mathcal{R}_i(s(m\lambda - m\mathbb{1}_i))$, then

$$s(m\lambda_i - m) = \rho_i = \rho_i^{(1)} + \ldots + \rho_i^{(s)}. \tag{6.16}$$

However, from the fact that $\mathrm{Supp}(\upsilon^* p_i) \subseteq \llcorner(m\lambda - m\mathbb{1}_i)$, then $\rho_i^{(k)} \geq m\lambda_i - m$, for each $k \in \{1, \ldots, s\}$. Then, (6.16) forces $\rho_i^{(k)} = m\lambda_i - m$, for each $k \in \{1, \ldots, s\}$. Hence,

$$\rho^{(1)}, \ldots, \rho^{(s)} \in \mathrm{Supp}(\upsilon^* p_i) \cap \mathcal{R}_i(m\lambda - m\mathbb{1}_i).$$

Therefore,

$$\rho^{(1)} = \ldots = \rho^{(s)} = m\lambda - m\mathbb{1}_i$$

and, consequently, $\rho = s(m\lambda - m\mathbb{1}_i)$ so that

$$\mathrm{Supp}(v^* p_i^s) \cap \mathcal{R}_i(m\lambda - m\mathbb{1}_i) \subseteq \{s(m\lambda - m\mathbb{1}_i)\}\,.$$

On the other hand, given that $\mathrm{Supp}(v^* p_i)$ is dominated by $(G, m\lambda - m\mathbb{1}_i)$, by $ii)$ of Proposition 6.1.4, $\mathrm{Supp}(v^* p_i^s)$ is dominated by $(G, s(m\lambda - m\mathbb{1}_i))$. Since $s(m\lambda - m\mathbb{1}_i) \in \mathcal{R}_i(s(m\lambda - m\mathbb{1}_i))$, we conclude that

$$\mathrm{Supp}(v^* p_i^s) \cap \mathcal{R}_i(s(m\lambda - m\mathbb{1}_i)) = \{s(m\lambda - m\mathbb{1}_i)\}\,.$$

□

Lemma 6.3.6. *Let Y be a plain q.o. surface and $\sigma = (\sigma_1, \sigma_2) \in \Lambda_i(\lambda) \cap \llcorner(m\lambda)$, for some $i \in \{1, 2\}$, such that $a_\sigma \geq 2$. Assume Y admits the Puiseux parametrization defined by*

$$x_1 = t_1^m,\;\; x_2 = t_2^m,\;\; y = \underline{t}^{m\lambda} + \varepsilon_{L_{\sigma,i}} + \varepsilon_{M_{\sigma,i}} + \varepsilon_{N_{\sigma,i}},$$

such that $\mathrm{Supp}(v^* y) \cap \mathcal{R}_i(m\lambda) = \{m\lambda\}$. Let $j \in \{1, 2\} \setminus \{i\}$, $\kappa_\ell \in \mathbb{C}$, for each $\ell \in \mathbb{N}_0$, and suppose that

$$\alpha = \underline{x}^{b_\sigma} p_i^{a_\sigma - 1} \sum_{\ell \geq 0} \kappa_\ell x_j^\ell \in \mathbb{C}\{\underline{x}, p_i\}.$$

Then there exists a unique β such that $\Phi_{\alpha\mathbb{1}_i, \beta}$ is an infinitesimal contact transformation. Moreover, $(\pi \circ \Phi_{\alpha\mathbb{1}_i, \beta})(\mathbb{P}^*_Y \mathbb{C}^3)$ is a plain q.o. surface that admits a Puiseux parametrization of the type

$$x_1 = t_1^m,\;\; x_2 = t_2^m,\;\; y = \underline{t}^{m\lambda} + \tilde{\varepsilon}_{L_{\sigma,i}} + \tilde{\varepsilon}_{M_{\sigma,i}} + \varepsilon_{N_{\sigma,i}}. \tag{6.17}$$

In particular,

$$\tilde{\varepsilon}_{L_{\sigma,i}} = \sum_{\ell \geq 0} \left(c_{\tau_\ell} - \frac{\kappa_\ell \lambda_i^{a_\sigma}}{a_\sigma} \right) \underline{t}^{\tau_\ell},$$

where $\tau_\ell = \sigma + \ell m \mathbb{1}_j$, for each $\ell \geq 0$.

Proof. First of all, since $a_\sigma \geq 2$, then $a_\sigma - 1 \geq 1$. This implies that $\frac{\partial \alpha}{\partial x_s} \in (p_i)\mathbb{C}\{\underline{x}, p_i\}$, for each $s \in \{1, 2\}$. Therefore, Lemma 6.2.1 guarantees the existence and uniqueness of such β. Moreover, by Corollary 6.2.2,

$$\beta = \left(\frac{a_\sigma - 1}{a_\sigma} p_i^{a_\sigma} \underline{x}^{b_\sigma} \sum_{\ell \geq 0} \kappa_\ell x_j^\ell \right) \chi(\underline{x}, p_i) \in (\underline{x}^{b_\sigma} p_i^{a_\sigma}) \mathbb{C}\{\underline{x}, y, \underline{p}\}.$$

Since $\sigma \geq m\lambda$ and $a_\sigma \geq 2$, then $\sigma > m\lambda$. Therefore,

$$\lambda < \frac{1}{m}\sigma = \frac{1}{m}(m b_\sigma + a_\sigma(m\lambda - m\mathbb{1}_i)) = b_\sigma + a_\sigma(\lambda - \mathbb{1}_i).$$

Additionally, by Lemma 4.2.3, $\pi_i(\mathbb{P}^*_Y \mathbb{C}^3)$ is a q.o. surface with characteristic exponent $\lambda - \mathbb{1}_i$, with strongly normalized branches, up to a change of coordinates, so that $\mathrm{Supp}(v^* p_i)$ is dominated by $(G, m\lambda - m\mathbb{1}_i)$. Therefore, since $\mathrm{Supp}(\underline{t}^{mb_\sigma})$ is dominated by (G, mb_σ), by $ii)$ of Proposition 6.1.4, we have that $\mathrm{Supp}(\underline{t}^{mb_\sigma}(v^* p_i^{a_\sigma - 1}))$ is dominated by

$$(G, mb_\sigma + (a_\sigma - 1)(m\lambda - m\mathbb{1}_i)) = (G, \sigma - m\lambda + m\mathbb{1}_i).$$

Thus, $\mathrm{Supp}(v^*\alpha)$ is dominated by $(G, \sigma - m\lambda + m\mathbb{1}_i)$. By Remark 6.3.2, $\sigma_i - m\lambda_i + m > m$, since $\sigma \notin \Gamma(\lambda)$. Hence, by Theorem 6.2.4, $(\pi \circ \Phi_{\alpha\mathbb{1}_i,\beta})(\mathbb{P}^*_Y\mathbb{C}^3)$ is a plain q.o. surface.

In what follows, we need to compute the substitution

$$u^m = t_i^m + (v^*\alpha)(\underline{t})$$

or, equivalently,

$$u^m = t_i^m(1 + \delta_1(\underline{t})),$$

where

$$\delta_1(\underline{t}) = \frac{1}{t_i^m}\underline{t}^{mb_\sigma}(v^* p_i^{a_\sigma - 1}) \sum_{\ell \geq 0} \kappa_\ell t_j^{m\ell} \in (t_i)\mathbb{C}\{\underline{t}\}$$

and $\mathrm{Supp}(\delta_1)$ is dominated by $(G, \sigma - m\lambda)$.

We claim that

$$\delta_1(\underline{t}) = \mathrm{DC}(v^* p_i^{a_\sigma - 1})\underline{t}^{\sigma - m\lambda} \sum_{\ell \geq 0} \kappa_\ell t_j^{m\ell} + \chi_1(\underline{t}),$$

where $\mathrm{Supp}(\chi_1) \cap \mathcal{R}_i(\sigma - m\lambda) = \emptyset$.

By hypothesis, $\mathrm{Supp}(v^*y) \cap \mathcal{R}_i(m\lambda) = \{m\lambda\}$, so that, by Proposition 6.3.5,

$$\mathrm{Supp}(v^* p_i^{a_\sigma - 1}) \cap \mathcal{R}_i((a_\sigma - 1)(m\lambda - m\mathbb{1}_i)) = \{(a_\sigma - 1)(m\lambda - m\mathbb{1}_i)\}\,.$$

This, in turn, implies that

$$\begin{aligned}
&\mathrm{Supp}\left(\frac{1}{t_i^m}\underline{t}^{mb_\sigma}(v^* p_i^{a_\sigma - 1})\right) \cap \mathcal{R}_i(\sigma - m\lambda) \\
&= \mathrm{Supp}\left(\frac{1}{t_i^m}\underline{t}^{mb_\sigma}(v^* p_i^{a_\sigma - 1})\right) \cap \mathcal{R}_i(mb_\sigma - m\mathbb{1}_i + (a_\sigma - 1)(m\lambda - m\mathbb{1}_i)) \\
&= \{mb_\sigma - m\mathbb{1}_i + (a_\sigma - 1)(m\lambda - m\mathbb{1}_i)\} = \{\sigma - m\lambda\}\,.
\end{aligned}$$

Therefore,

$$\frac{1}{t_i^m}\underline{t}^{mb_\sigma}(v^* p_i^{a_\sigma - 1}) = \mathrm{DC}(v^* p_i^{a_\sigma - 1})\underline{t}^{\sigma - m\lambda} + \chi_2(\underline{t}),$$

where $\mathrm{Supp}(\chi_2) \cap \mathcal{R}_i(\sigma - m\lambda) = \emptyset$. Hence,

$$\delta_1(\underline{t}) = \mathrm{DC}(v^* p_i^{a_\sigma - 1})\underline{t}^{\sigma - m\lambda} \sum_{\ell \geq 0} \kappa_\ell t_j^{m\ell} + \chi_1(\underline{t}),$$

where $\mathrm{Supp}(\chi_1) \cap \mathcal{R}_i(\sigma - m\lambda) = \emptyset$, which proves the claim. Let $\underline{w} = (w_1, w_2)$, where $w_i = u$ and $w_j = t_j$. Consequently, by part $ii)$ of Lemma 6.1.7,

$$t_i = u(1 + \delta_2(\underline{w})), \tag{6.18}$$

where
$$\delta_2(\underline{w}) = -\frac{\mathrm{DC}(v^* p_i^{a_\sigma - 1})}{m} \underline{w}^{\sigma - m\lambda} \sum_{\ell \geq 0} \kappa_\ell t_j^{m\ell} + \chi_3(\underline{w}) \in (u)\mathbb{C}\{\underline{w}\}, \tag{6.19}$$

$\mathrm{Supp}(\delta_2)$ is dominated by $(G, \sigma - m\lambda)$, and $\mathrm{Supp}(\chi_3) \cap \mathcal{R}_i(\sigma - m\lambda) = \emptyset$.

We then have that, for each $\tau = (\tau_1, \tau_2) \in \llcorner(m\lambda) \cap G$,

$$c_\tau \underline{t}^\tau = c_\tau \underline{w}^\tau (1 + \delta_2(\underline{w}))^{\tau_i} = c_\tau \underline{w}^\tau + c_\tau \tau_i \underline{w}^\tau \delta_2(\underline{w}) + \chi_4(\underline{w}), \tag{6.20}$$

where $\mathrm{Supp}(\chi_4) \subseteq \llcorner(\tau + 2(\sigma - m\lambda))$. Now note that

$$\tau + 2(\sigma - m\lambda) = \sigma + (\tau - m\lambda) + (\sigma - m\lambda) > \sigma,$$

for $\tau \geq m\lambda$ and $\sigma > m\lambda$. On the other hand,

$$\tau_i + 2(\sigma_i - m\lambda_i) = \sigma_i + (\tau_i - m\lambda_i) + (\sigma_i - m\lambda_i) > \sigma_i,$$

since $\sigma_i > m\lambda_i$ (see Remark 6.3.2) and $\tau_i \geq m\lambda_i$. Thus, $\mathrm{Supp}(\chi_4) \cap \mathcal{R}_i(\sigma) = \emptyset$.

Moreover, from (6.19) and (6.20), we write, for each $\tau = (\tau_1, \tau_2) \in \llcorner(m\lambda) \cap G$,

$$c_\tau \underline{t}^\tau = c_\tau \underline{w}^\tau + c_\tau \tau_i \,\mathrm{DC}(\delta_2) \underline{w}^{\tau + \sigma - m\lambda} \sum_{\ell \geq 0} \kappa_\ell t_j^{m\ell} + \chi_5(\underline{w}), \tag{6.21}$$

where $\mathrm{Supp}(\chi_5) \cap \mathcal{R}_i(\sigma) = \emptyset$, since $\mathrm{Supp}(\underline{w}^\tau \chi_3) \cap \mathcal{R}_i(\tau + \sigma - m\lambda) = \emptyset$, $\mathrm{Supp}(\underline{w}^\tau \chi_3) \subseteq \llcorner(\tau + \sigma - m\lambda)$, and $\tau + \sigma - m\lambda \geq \sigma$.

We want to determine when

$$\mathrm{Supp}\left(c_\tau \underline{w}^\tau + c_\tau \tau_i \,\mathrm{DC}(\delta_2) \underline{w}^{\tau + \sigma - m\lambda} \sum_{\ell \geq 0} \kappa_\ell t_j^{m\ell}\right) \cap L_{\sigma,i} \neq \emptyset.$$

One immediate case is when $\tau = (\tau_1, \tau_2) \in \mathrm{Supp}(v^* y)$ and $\tau \in L_{\sigma,i}$. In this case,

$$\tau_i + \sigma_i - m\lambda_i > \tau_i = \sigma_i$$

and, therefore,

$$\mathrm{Supp}\left(c_\tau \underline{w}^\tau + c_\tau \tau_i \,\mathrm{DC}(\delta_2) \underline{w}^{\tau + \sigma - m\lambda} \sum_{\ell \geq 0} \kappa_\ell t_j^{m\ell}\right) \cap L_{\sigma,i} = \{\tau\}.$$

Given that $\tau < \tau + \sigma - m\lambda$, then

$$\mathrm{Supp}\left(c_\tau \underline{w}^\tau\right) \cap \mathrm{Supp}\left(c_\tau \tau_i \,\mathrm{DC}(\delta_2) \underline{w}^{\tau + \sigma - m\lambda} \sum_{\ell \geq 0} \kappa_\ell t_j^{m\ell}\right) = \emptyset$$

and, consequently, the only remaining case is when $\tau \in \mathrm{Supp}(v^* y)$ and $\tau + \sigma - m\lambda \in L_{\sigma,i}$. The latter implies that $\tau_i + \sigma_i - m\lambda_i = \sigma_i$, that is, $\tau_i = m\lambda_i$. Therefore, $\tau \in \mathcal{R}_i(m\lambda)$. Our hypothesis implies that $\tau = m\lambda$.

Since, by Corollary 6.2.2,

$$v^* \beta = \left(\frac{a_\sigma - 1}{a_\sigma} \underline{t}^{m b_\sigma} (v^* p_i)^{a_\sigma} \sum_{\ell \geq 0} \kappa_\ell t_j^{m\ell}\right) \chi_6(\underline{t}),$$

where χ_6 is a unit with constant term equal to 1, then, by a similar reasoning to the one developed for $v^*\alpha$, it follows that

$$\begin{aligned} v^*\beta &= \frac{a_\sigma - 1}{a_\sigma} \mathrm{DC}(v^* p_i^{a_\sigma}) \underline{t}^{m b_\sigma + a_\sigma (m\lambda - m\mathbb{1}_i)} \sum_{\ell \geq 0} \kappa_\ell t_j^{m\ell} + \chi_7(\underline{t}) \\ &= \frac{a_\sigma - 1}{a_\sigma} \mathrm{DC}(v^* p_i^{a_\sigma}) \underline{t}^{\sigma} \sum_{\ell \geq 0} \kappa_\ell t_j^{m\ell} + \chi_7(\underline{t}), \end{aligned}$$

where $\mathrm{Supp}(\chi_7) \cap \mathcal{R}_i(\sigma) = \emptyset$ and $\mathrm{Supp}(\chi_7) \subseteq \llcorner(\sigma)$. By $ii)$ of Proposition 6.1.6, applying the substitution in (6.18) yields

$$\tilde{v}^*\beta = \frac{a_\sigma - 1}{a_\sigma} \mathrm{DC}(v^* p_i^{a_\sigma}) \underline{w}^{\sigma} \sum_{\ell \geq 0} \kappa_\ell t_j^{m\ell} + \chi_8(\underline{w}), \tag{6.22}$$

where $\mathrm{Supp}(\chi_8) \cap \mathcal{R}_i(\sigma) = \emptyset$ and $\mathrm{Supp}(\chi_8) \subseteq \llcorner(\sigma)$. Therefore, collecting the coefficients of $\underline{w}^{\tau_\ell} = \underline{w}^{\sigma} t_j^{m\ell}$ in (6.21) and (6.22), for each $\tau_\ell = \sigma + m\ell\mathbb{1}_j$, $\ell \geq 0$, yields

$$\begin{aligned} \tilde{c}_{\tau_\ell} \underline{w}^{\tau_\ell} &= \left(c_{\tau_\ell} - c_{m\lambda} m \lambda_i \kappa_\ell \frac{\mathrm{DC}(v^* p_i^{a_\sigma - 1})}{m} + \kappa_\ell \frac{a_\sigma - 1}{a_\sigma} \mathrm{DC}(v^* p_i^{a_\sigma}) \right) \underline{w}^{\tau_\ell} \\ &= \left(c_{\tau_\ell} - \lambda_i^{a_\sigma} \kappa_\ell + \frac{a_\sigma - 1}{a_\sigma} \lambda_i^{a_\sigma} \kappa_\ell \right) \underline{w}^{\tau_\ell} = \left(c_{\tau_\ell} - \frac{\lambda_i^{a_\sigma} \kappa_\ell}{a_\sigma} \right) \underline{w}^{\tau_\ell}, \end{aligned}$$

where $\tilde{c}_{\tau_\ell}$ denotes the coefficient of $\underline{w}^{\tau_\ell}$ in the obtained Puiseux parametrization for $(\pi \circ \Phi_{\alpha\mathbb{1}_i,\beta})(\mathbb{P}^*_Y\mathbb{C}^3)$.

Lastly, since $\mathrm{Supp}(\delta_2)$ is dominated by $(G, \sigma - m\lambda)$, by part $iii)$ of Proposition 6.1.6, for every $\tau = (\tau_1, \tau_2) \in \llcorner(m\lambda)$,

$$c_\tau \underline{t}^{\tau} = c_\tau \underline{w}^{\tau} + \chi_9(\underline{w}),$$

where $\mathrm{Supp}(\chi_9) \subseteq \llcorner(\tau + \sigma - m\lambda) \subseteq \llcorner(\sigma)$, we have shown that $\varepsilon_{N_{\sigma,i}}$ remains unaltered after applying the constructed infinitesimal contact transformation and further reparametrization. □

Remark 6.3.7. *Let $\sigma = (\sigma_1, \sigma_2) \in \llcorner(m\lambda) \cap \Lambda(\lambda)$. As usual, we write*

$$\sigma = b_\sigma m + a_{\sigma,1}(m\lambda - m\mathbb{1}_1) + a_{\sigma,2}(m\lambda - m\mathbb{1}_2),$$

with $a_{\sigma,1}, a_{\sigma,2} \in \{0, \dots, m-1\}$, by Proposition 5.0.4. If

$$a_{\sigma,1} + a_{\sigma,2} < 2,$$

then $\sigma \in \Lambda_i(\lambda)$, for some $i \in \{1, 2\}$. In particular, $a_{\sigma,i} = 1$ and $\sigma \in \Gamma(\lambda)$, by Remark 6.3.2. Therefore, if

$$\sigma \in \llcorner(m\lambda) \cap \Lambda(\lambda) \setminus (\Lambda_1(\lambda) \cup \Lambda_2(\lambda)),$$

then $a_{\sigma,1} + a_{\sigma,2} \geq 2$.

Lemma 6.3.8. *Let Y be a plain q.o. surface. Assume Y admits the Puiseux parametrization υ defined by*

$$x_1 = t_1^m, \;\; x_2 = t_2^m, \;\; y = \underline{t}^{m\lambda} + \varepsilon_G$$

such that $Supp(\upsilon^* y) \cap (\mathcal{R}_1(m\lambda) \cup \mathcal{R}_2(m\lambda)) = \{m\lambda\}$. Let

$$\sigma = (\sigma_1, \sigma_2) \in Supp(\upsilon^* y) \cap \Lambda(\lambda) \setminus (\Lambda_1(\lambda) \cup \Lambda_2(\lambda))$$

such that $b_\sigma = 0$. Let $a = (a_{\sigma,1}, a_{\sigma,2})$, $\kappa \in \mathbb{C}$, $\beta = \kappa \underline{p}^a$, and $\underline{\alpha} = (\alpha_1, \alpha_2)$, where

$$\alpha_j = \frac{\kappa a_{\sigma,j}}{|a|-1} \underline{p}^{a-\mathbb{1}_j},$$

for each $j \in \{1,2\}$. Then $(\pi \circ \Phi_{\underline{\alpha},\beta})(\mathbb{P}^*_Y \mathbb{C}^3)$ is a plain q.o. surface that admits a Puiseux parametrization of the type

$$x_1 = t_1^m, \;\; x_2 = t_2^m, \;\; y' = \underline{t}^{m\lambda} + \varepsilon_{R \setminus \llcorner(\sigma)} + \varepsilon_{\llcorner(\sigma)},$$

where $\varepsilon_{R\setminus \llcorner(\sigma)}$ is a subseries of ε_G. In particular,

$$\tilde{c}_\sigma = c_\sigma - \frac{\kappa \lambda^a}{|a|-1},$$

where c_σ and $\tilde{c}_\sigma$ are, respectively, the coefficient of $\underline{t}^\sigma$ in $\upsilon^* y$ and $\tilde{\upsilon}^* y'$.

Proof. If $\kappa = 0$, there is nothing to prove. Suppose otherwise. From Remark 6.3.7, we see that $|a| \geq 2$. Therefore, by Corollary 4.2 of [21], $\Phi_{\underline{\alpha},\beta}$ is an infinitesimal contact transformation.

Given that $\Gamma(\lambda) \subseteq \Lambda_j(\lambda)$, $\sigma \notin \Lambda_j(\lambda)$, for each $j \in \{1,2\}$, $\mathrm{Supp}(\upsilon^* y) \subseteq \llcorner(m\lambda)$, and $m\lambda \in \Gamma(\lambda)$, then $\sigma > m\lambda$. Therefore,

$$\lambda < \frac{1}{m}\sigma = \frac{1}{m}(a_{\sigma,1}(m\lambda - m\mathbb{1}_1) + a_{\sigma,2}(m\lambda - m\mathbb{1}_2)) = a_{\sigma,1}(\lambda - \mathbb{1}_1) + a_{\sigma,2}(\lambda - \mathbb{1}_2),$$

and $\beta \in (\underline{p}^a)\mathbb{C}\{\underline{x}, y, \underline{p}\}$. On the other hand, since

$$\sigma = a_{\sigma,1}(m\lambda - m\mathbb{1}_1) + a_{\sigma,2}(m\lambda - m\mathbb{1}_2) \notin \Lambda_1(\lambda) \cup \Lambda_2(\lambda),$$

then $a_{\sigma,1}, a_{\sigma,2} \geq 1$. Moreover, by (4.2), $\mathrm{Supp}(\upsilon^* p_j)$ is dominated by $(G, m\lambda - m\mathbb{1}_j)$, for each $j \in \{1,2\}$. The latter implies, by part ii) of Proposition 6.1.4, that for each $j \in \{1,2\}$, $\mathrm{Supp}(\upsilon^* \alpha_j)$ is dominated by

$$(G, a_{\sigma,i}(m\lambda - m\mathbb{1}_i) + (a_{\sigma,j} - 1)(m\lambda - m\mathbb{1}_j)) = (G, \sigma - m\lambda + m\mathbb{1}_j),$$

where $i \in \{1,2\} \setminus \{j\}$. In particular, since $\mathrm{Supp}(\upsilon^* y) \cap (\mathcal{R}_1(m\lambda) \cup \mathcal{R}_2(m\lambda)) = \{m\lambda\}$ and $\sigma \in \mathrm{Supp}(\upsilon^* y) \setminus \{m\lambda\}$, it follows that $\sigma_j - m\lambda_j + m > m$, for $\sigma_j > m\lambda_j$. Hence, we are in position to apply Theorem 6.2.4 in order to conclude that $(\pi \circ \Phi_{\underline{\alpha},\beta})(\mathbb{P}^*_Y \mathbb{C}^3)$ is a plain q.o. surface.

We now need to compute the reparametrization

$$u_1^m = t_1^m + v^*\alpha_1(\underline{t}),$$

or equivalently,

$$u_1^m = t_1^m(1 + \delta_1(\underline{t})),$$

where

$$\delta_1(\underline{t}) = \frac{\kappa a_{\sigma,1}}{|a|-1}\frac{1}{t_1^m}(v^*p_1^{a_{\sigma,1}-1})(v^*p_2^{a_{\sigma,2}}) \in (t_1)\mathbb{C}\{\underline{t}\}$$

and $\mathrm{Supp}(\delta_1)$ is dominated by

$$(G, \sigma - m\lambda + m\mathbb{1}_1 - m\mathbb{1}_1) = (G, \sigma - m\lambda).$$

By part ii) of Lemma 6.1.7,

$$t_1 = u_1(1 + \varepsilon_1(u_1, t_2)), \tag{6.23}$$

where $\varepsilon_1 \in (u_1)\mathbb{C}\{u_1, t_2\}$, $\mathrm{Supp}(\varepsilon_1)$ is dominated by $(G, \sigma - m\lambda)$, and

$$\mathrm{DC}(\varepsilon_1) = -\frac{\mathrm{DC}(\delta_1)}{m} = -\frac{\kappa a_{\sigma,1}\lambda^{a-\mathbb{1}_1}}{m(|a|-1)}.$$

The right-most equality follows from part ii) of Proposition 6.1.4 and the fact that, by (4.2), $\mathrm{DC}(v^*p_j) = \lambda_j$, for each $j \in \{1, 2\}$.

Consequently, by part i) of Proposition 6.1.6, applying (6.23) in $v^*\alpha_2$ and $v^*y + v^*\beta$ yields that $\mathrm{Supp}(v^*\alpha_2)$ is dominated by $(G, \sigma - m\lambda + m\mathbb{1}_2)$ and $\mathrm{Supp}(v^*y + v^*\beta)$ is dominated by $(G, m\lambda)$. In particular, part ii) of Proposition 6.1.6 guarantees that $\mathrm{DC}(v^*\alpha_2)$ and $\mathrm{DC}(v^*y + v^*\beta)$ remain unaltered after this substitution.

Let $\underline{w} = (u_1, t_2)$. We now want to determine the coefficient of $\underline{w}^\sigma$ in

$$(v^*y + v^*\beta)(u_1(1 + \varepsilon_1(u_1, t_2)), t_2).$$

Let $\tau = (\tau_1, \tau_2) \in \llcorner(m\lambda)$. Then, by part iii) of Proposition 6.1.6,

$$\begin{aligned} c_\tau \underline{t}^\tau &= c_\tau \underline{w}^\tau (1 + \varepsilon_1(\underline{w}))^{\tau_1} \\ &= c_\tau \underline{w}^\tau + \tau_1 \,\mathrm{DC}(\varepsilon_1)\underline{w}^{\tau+\sigma-m\lambda} + \chi(\underline{w}), \end{aligned} \tag{6.24}$$

where $\mathrm{Supp}(\chi) \subseteq \llcorner(\tau + \sigma - m\lambda) \setminus \{\tau + \sigma - m\lambda\}$. Since $\tau \geq m\lambda$, then

$$\tau + \sigma - m\lambda \geq \sigma$$

and equality only holds when $\tau = m\lambda$. Hence, the coefficient of $\underline{w}^\sigma$ in

$$(v^*y + v^*\beta)(u_1(1 + \varepsilon_1(u_1, t_2)), t_2).$$

is given by

$$c_\sigma + m\lambda_1 \,\mathrm{DC}(\varepsilon_1) + \kappa\lambda^a = c_\sigma - \frac{\kappa a_{\sigma,1}\lambda^a}{|a|-1} + \kappa\lambda^a.$$

Moreover, given that $\llcorner(\tau + \sigma - m\lambda) \subseteq \llcorner(\sigma)$, we see that (6.24) implies that $\varepsilon_{R\setminus\llcorner(\sigma)}$ remains unaltered after reparametrization. Following this very reasoning for the substitution

$$u_2^m = t_2^m(1 + v^*\alpha_2(u_1, t_2))$$

enables us to conclude that

$$\tilde{c}_\sigma = c_\sigma - \frac{\kappa a_{\sigma,1}\lambda^a}{|a|-1} - \frac{\kappa a_{\sigma,2}\lambda^a}{|a|-1} + \kappa\lambda^a = c_\sigma - \frac{\kappa\lambda^a}{|a|-1},$$

and that $\varepsilon_{R\setminus\llcorner(\sigma)}$ remains unaltered. □

6.4 Main Theorem

For the following results, we will consider several Puiseux parametrizations. Without loss of generality, and for the sake of simplicity, we let c_τ denote the coefficient of $\underline{t}^\tau$ for different Puiseux parametrizations, when no ambiguity is given. Moreover, we write a Puiseux parametrization of a q.o. surface Y_d as

$$x_1 = t_1^m, \;\; x_2 = t_2^m, \;\; y_d = \underline{t}^{m\lambda} + \sum_{\tau\in\llcorner(m\lambda)} c_\tau \underline{t}^\tau.$$

Lemma 6.4.1. *Let $f(x) = \sum_{\ell\geq 0} c_\ell x^\ell$, $g(x) = \sum_{\ell>0} d_\ell x^\ell \in \mathbb{C}\{x\}$. Consider $h(x) = \sum_{\ell\geq 0}\kappa_\ell x^\ell \in \mathbb{C}[[x]]$, where the κ_ℓ are defined recursively as follows.*

$$\kappa_0 = c_0, \;\; \kappa_\ell = c_\ell - \sum_{j=0}^{\ell-1}\kappa_j d_{\ell-j}, \; \ell \geq 1.$$

Then $h(x) \in \mathbb{C}\{x\}$.

Proof. Using the Cauchy product between h and g, one observes that

$$h(x) + h(x)g(x) = \sum_{\ell\geq 0}\left(\kappa_\ell + \sum_{j=0}^{\ell-1}\kappa_j d_{\ell-j}\right)x^\ell = \sum_{\ell\geq 0} c_\ell x^\ell = f(x).$$

Then, $h(x)(1 + g(x)) = f(x)$. Given that $1 + g(x) \in \mathbb{C}\{x\}$ is a unit, then

$$h(x) = (1 + g(x))^{-1} f(x) \in \mathbb{C}\{x\}.$$

□

Lemma 6.4.2. *Let Y be a plain q.o. surface. Then Y is microlocally equivalent to a plain q.o. surface Y', which admits a Puiseux parametrization of the type*

$$x_1 = t_1^m, \;\; x_2 = t_2^m, \;\; y' = \underline{t}^{m\lambda} + \varepsilon_{\Gamma(\lambda)_{\mathcal{G}}},$$

where Supp$(v^*y') \cap (\mathcal{R}_1(m\lambda) \cup \mathcal{R}_2(m\lambda)) = \{m\lambda\}$.

Proof. We consider $\beta_1 = -y\sum_{\ell\geq 0}\kappa_\ell x_2^\ell$, where the κ_ℓ are recursively defined by

$$\kappa_0 = 0,\ \kappa_\ell = c_{\tau_\ell} - \sum_{s=0}^{\ell-1}\kappa_s c_{\tau_{\ell-s}},$$

where $\tau_\ell = m\lambda + m\ell\mathbb{1}_2$, for each $\ell \geq 0$. We claim that $\beta_1 \in \mathbb{C}\{\underline{x}, y\}$. Indeed, let

$$f(\underline{t}) = \sum_{\ell>0} c_{\tau_\ell}\underline{t}^{\tau_\ell} = \underline{t}^{m\lambda}\sum_{\ell>0} c_{\tau_\ell} t_2^{m\ell}.$$

Note that $f(\underline{t})$ is a subseries of v^*y. Hence, $f(\underline{t})$ converges in an open disk of radius r. Therefore, $\underline{x}^\lambda\sum_{\ell>0} c_{\tau_\ell}x_2^\ell$ converges in an open disk of radius $r' \leq r^m$. By Lemma 6.4.1, $\sum_{\ell>0}\kappa_\ell x_2^\ell \in \mathbb{C}\{\underline{x}\}$ and, consequently, $\beta_1 \in \mathbb{C}\{\underline{x}, y\}$. Since $\kappa_0 = 0$, we are safe to apply the contact transformation of Lemma 6.3.3 with $\sigma = m\lambda + m\mathbb{1}_2$, obtaining a plain q.o. surface Y_1, microlocally equivalent to Y, such that

$$\mathrm{Supp}(v^*y_1) \cap L_{m\lambda,1} = \{m\lambda\}.$$

Therefore, by Remark 6.3.2,

$$\mathrm{Supp}(v^*y_1) \cap \mathcal{R}_1(m\lambda) = \{m\lambda\}.$$

By applying this very argument to Y_1, with $\sigma = m\lambda + m\mathbb{1}_1$ and $\beta_2 = -y\sum_{\ell\geq 0}\kappa_\ell x_1^\ell$, where

$$\kappa_0 = 0, \quad \kappa_\ell = c_{\tau_\ell} - \sum_{s=0}^{\ell-1}\kappa_s c_{\tau_{\ell-s}}, \quad \ell \geq 1,$$

and $\tau_\ell = m\lambda + m\ell\mathbb{1}_1$, for $\ell \geq 1$, we obtain a plain q.o. surface Y_2, microlocally equivalent to Y_1, such that, by Remark 6.3.2,

$$\mathrm{Supp}(v^*y_2) \cap \mathcal{R}_2(m\lambda) = \mathrm{Supp}(v^*y_2) \cap L_{m\lambda,2} = \{m\lambda\}.$$

Since $\mathcal{R}_1(m\lambda) \subseteq N_{\sigma,2}$, then Lemma 6.3.3, in particular, guarantees that

$$\mathrm{Supp}(v^*y_2) \cap (\mathcal{R}_1(m\lambda) \cup \mathcal{R}_2(m\lambda)) = \{m\lambda\}.$$

Given that the set of all contact transformations forms a group under composition and we applied finitely many contact transformations, the plain q.o. surface Y_2 is microlocally equivalent to Y and verifies the desired conditions. □

Remark 6.4.3. *For a plain q.o. surface Y, let $\sigma \in \Gamma(\lambda) \cap \mathrm{Supp}(v^*y)$. One then may write*

$$\sigma = mb_\sigma + a_\sigma m\lambda,$$

with $a_\sigma \in \{1, \ldots, m-1\}$ and $b_\sigma \in \mathbb{N}_0^2$. Thus,

$$\sigma = mb_\sigma + a_\sigma(m\lambda - m\mathbb{1}_i) + a_\sigma m\mathbb{1}_i \in \Lambda_i(\lambda),$$

for all $i \in \{1, 2\}$.

Remark 6.4.4. *Let $\tau = (\tau_1, \tau_2), \sigma = (\sigma_1, \sigma_2) \in \llcorner(m\lambda)$. If $\tau_1 < \sigma_1$ or $\tau_2 < \sigma_2$, then $\tau \notin \llcorner(\sigma)$. Therefore, if $\tau <_{lex} \sigma$, then $\tau \in \llcorner(m\lambda) \setminus \llcorner(\sigma)$.*

We let R^* be the subset of R defined as the elements $\sigma \in R \setminus (\Lambda_1(\lambda) \cup \Lambda_2(\lambda))$ such that if

$$\sigma = mb_\sigma + a_{\sigma,1}(m\lambda - m\mathbb{1}_1) + a_{\sigma,2}(m\lambda - m\mathbb{1}_2) \in \Lambda(\lambda),$$

then $b_\sigma \neq 0$.

Lemma 6.4.5. *Let Y be a plain q.o. surface that admits a Puiseux parametrization of the type*

$$x_1 = t_1^m, \ \ x_2 = t_2^m, \ \ y = \underline{t}^{m\lambda} + \varepsilon_{\Gamma(\lambda)_{\mathcal{G}}},$$

such that $\mathrm{Supp}(v^* y) \cap (\mathcal{R}_1(m\lambda) \cup \mathcal{R}_2(m\lambda)) = \{m\lambda\}$. Then Y is microlocally equivalent to a plain q.o. surface Y', which admits a Puiseux parametrization of the type

$$x_1 = t_1^m, \ \ x_2 = t_2^m, \ \ y' = \underline{t}^{m\lambda} + \varepsilon_{R^*} + \varepsilon_{\llcorner(m\lambda)\setminus R},$$

such that $\mathrm{Supp}(\tilde{v}^* y') \cap (\mathcal{R}_1(m\lambda) \cup \mathcal{R}_2(m\lambda)) = \{m\lambda\}$.

Proof. Let $\sigma = (\sigma_1, \sigma_2) \in R \cap \mathrm{Supp}(v^* y) \setminus \{m\lambda\}$. If $\sigma \in \Lambda_i(\lambda) \setminus \Gamma(\lambda)$ for some $i \in \{1, 2\}$, let

$$\alpha = \frac{c_\sigma a_\sigma}{\lambda_i^{a_\sigma}} \underline{x}^{b_\sigma} p_i^{a_\sigma - 1} \in \mathbb{C}\{\underline{x}, p_i\}.$$

Since $a_\sigma \geq 2$ by $ii)$ of Remark 6.3.1, then, by Lemma 6.3.6, we conclude that Y is microlocally equivalent to a plain q.o. surface that admits a Puiseux parametrization such that $\sigma \notin \mathrm{Supp}(\tilde{v}^* y')$ and that

$$\sum_{\tau \in \llcorner(m\lambda) \setminus \llcorner(\sigma)} c_\tau \underline{t}^\tau$$

remains unaltered. By Remark 6.4.3, if $\sigma \in \Gamma(\lambda)$ and $a_\sigma \geq 2$, then $\sigma \in \Lambda_i(\lambda)$, for some $i \in \{1, 2\}$ and we may apply this very reasoning. On the other hand, if $\sigma \in \Gamma(\lambda)$ and $a_\sigma = 1$, one considers

$$\beta = -c_\sigma \underline{x}^{b_\sigma} y \in \mathbb{C}\{\underline{x}, y\}.$$

Therefore, by Corollary 6.3.4, we conclude that Y is microlocally equivalent to a plain q.o. surface that admits a Puiseux parametrization such that $\sigma \notin \mathrm{Supp}(\tilde{v}^* y')$ and that

$$\sum_{\tau \in \llcorner(m\lambda) \setminus \llcorner(\sigma)} c_\tau \underline{t}^\tau$$

remains unaltered. Lastly, suppose that $\sigma \in \Lambda(\lambda) \setminus (\Lambda_1(\lambda) \cup \Lambda_2(\lambda))$ and $b_\sigma = 0$. Let $a = (a_{\sigma,1}, a_{\sigma,2})$,

$$\alpha_i = \frac{c_\sigma a_{\sigma,i}}{\lambda^a} \underline{p}^{a - \mathbb{1}_i},$$

for each $i \in \{1, 2\}$, and $\beta = \frac{c_\sigma(|a|-1)}{\lambda^a} \underline{p}^a$. Via Lemma 6.3.8, one concludes that Y is microlocally equivalent to a plain q.o. surface that admits a Puiseux parametrization such that $\sigma \notin \mathrm{Supp}(\tilde{v}^* y')$ and that

$$\sum_{\tau \in \llcorner(m\lambda) \setminus \llcorner(\sigma)} c_\tau \underline{t}^\tau$$

remains unaltered.

Recall the set of all contact transformations forms a group under composition. Endowing $R \cap \mathrm{Supp}(v^* y)$ with the lexicographic order, we obtain the desired result by applying the finitely many contact transformations described above for each $\sigma \in (R \cap \mathrm{Supp}(v^* y)) \setminus R^*$, according to this ordering (see Remark 6.4.4). □

Finally, we end with the main result.

Theorem 6.4.6. *Let Y be a plain q.o. surface. Then Y is microlocally equivalent to a plain q.o. surface Y' that admits a Puiseux parametrization of the type*

$$x_1 = t_1^m, \quad x_2 = t_2^m, \quad y' = \underline{t}^{m\lambda} + \varepsilon_R + \varepsilon_{\Gamma(\lambda)},$$

where $Supp(\varepsilon_R) \subseteq R^*$ and $Supp(\varepsilon_{\Gamma(\lambda)}) \subseteq int(\llcorner(Fr(\lambda)))$.

Proof. By Lemmas 6.4.2 and 6.4.5, Y is microlocally equivalent to a plain q.o. surface Y_1 that admits a Puiseux parametrization of the type

$$x_1 = t_1^m, \quad x_2 = t_2^m, \quad y_1 = \underline{t}^{m\lambda} + \varepsilon_R + \varepsilon_{\llcorner(m\lambda)\setminus R},$$

such that

$$\mathrm{Supp}(v^* y_1) \cap (\mathcal{R}_1(m\lambda) \cup \mathcal{R}_2(m\lambda)) = \{m\lambda\},$$

and $\mathrm{Supp}(\varepsilon_R) \subseteq R^*$.

Let $i \in \{1,2\}$ and $j \in \{1,2\} \setminus \{i\}$. For each $e \in \{1, \ldots, m-1\}$ and each

$$g_i \in \{m\lambda_i + 1, \ldots, (m-1)m\lambda_i - m\},$$

let $\sigma^{(g_i,e)} = (\sigma_1, \sigma_2) \in \Gamma(\lambda)_{\mathcal{G}}$ be such that $\sigma_i = g_i$, $\sigma_j \in \mathbb{N}$ is the least positive integer such that $\sigma_j > (m-1)m\lambda_j - m$, and $a_{\sigma^{(g_i,e)}} = e$. This choice is possible given that $\sigma^{(g_i,e)} \in \Lambda_i(\lambda)$, when g_i and e are admissible. In particular, this choice is unique. For simplicity, we write

$$\sigma^{(g_i,e)} = mb_\sigma + a_\sigma(m\lambda - m\mathbb{1}_i).$$

Let $g_i \in \{m\lambda_i + 1, \ldots, (m-1)m\lambda_i - m\}$. If $e = 1$, then $\sigma^{(g_i,e)} \in \Gamma(\lambda)$ by ii) of Remark 6.3.1. Let

$$\beta = -\underline{x}^{b_\sigma} y \sum_{\ell \geq 0} c_{\tau_\ell} x_j^\ell,$$

where $\tau_\ell = \sigma + m\ell\mathbb{1}_j$, for each $\ell \geq 0$. We claim that $\beta \in \mathbb{C}\{\underline{x}, y\}$. Indeed, let

$$f(\underline{t}) = \sum_{\ell \geq 0} c_{\tau_\ell} \underline{t}^{\tau_\ell} = \underline{t}^{m\lambda} \sum_{\ell \geq 0} c_{\tau_\ell} t_j^{m\ell}.$$

Note that $f(\underline{t})$ is a subseries of $v^* y$. Hence, $f(\underline{t})$ converges in an open disk of radius r. Therefore, $\underline{x}^\lambda \sum_{\ell \geq 0} c_{\tau_\ell} t_j^\ell$ converges in an open disk of radius $r' \leq r^m$. Thus, $\beta \in \mathbb{C}\{\underline{x}, y\}$. By Corollary 6.3.4, we obtain a plain q.o. surface Y_2 microlocally equivalent to Y_1 such that

$$\mathrm{Supp}(\tilde{v}^* y_2) \cap L_{\sigma^{(g_i,1)},i} = \{m\lambda\}.$$

Note that, since $\sigma_i \geq m\lambda_i+1 > m\lambda_i$ and $\sigma_j > (m-1)m\lambda_j - m > m\lambda_j$, then $\mathcal{R}_1(m\lambda) \cup \mathcal{R}_2(m\lambda) \subseteq N_{\sigma,i}$. Therefore, Corollary 6.3.4 guarantees, in particular, that

$$\mathrm{Supp}(\tilde{v}^* y_2) \cap (\mathcal{R}_1(m\lambda) \cup \mathcal{R}_2(m\lambda) \cup L_{\sigma^{(g_i,1)},i}) = \{m\lambda\}\,.$$

Moreover, $\sigma_j > (m-1)m\lambda_j - m$ implies that ε_R remains unaltered.

Assume that we have a plain q.o. surface Y_{e-1} microlocally equivalent to Y_2 such that

$$\mathrm{Supp}(\tilde{v}^* y_{e-1}) \cap (\mathcal{R}_1(m\lambda) \cup \mathcal{R}_2(m\lambda) \cup L_{\sigma^{(g_i,1)},i} \cup \ldots \cup L_{\sigma^{(g_i,e-1)},i}) = \{m\lambda\}\,,$$

and that ε_R remains unaltered. Let $\kappa_\ell = \frac{c_{\tau_\ell} e}{\lambda_i^e}$, where $\tau_\ell = \sigma^{(g_i,e)} + m\ell\mathbb{1}_j$, for each $\ell \geq 0$. Let

$$\alpha = \underline{x}^{b_\sigma} p_i^{a_\sigma - 1} \sum_{\ell \geq 0} \kappa_\ell x_j^\ell = \frac{e}{\lambda_i^e} \underline{x}^{b_\sigma} p_i^{a_\sigma - 1} \sum_{\ell \geq 0} c_{\tau_\ell} x_j^\ell,$$

then $\alpha \in \mathbb{C}\{\underline{x}, p_i\}$ given that $\sum_{\ell \geq 0} c_{\tau_\ell} x_j^\ell \in \mathbb{C}\{\underline{x}\}$, as in the reasoning for $e = 1$. Applying Lemma 6.3.6 to Y_{e-1} then gives a plain q.o. surface Y_e microlocally equivalent to Y_{e-1} such that

$$\mathrm{Supp}(\tilde{v}^* y_e) \cap L_{\sigma^{(g_i,e)},i} = \{m\lambda\}\,.$$

For each $s \in \{1, \ldots, e-1\}$, $a_{\sigma^{(g_i,s)}} \neq a_{\sigma^{(g_i,e)}}$ implies $\sigma^{(g_i,s)} \notin L_{\sigma^{(g_i,e)},i}$, so that $\sigma^{(g_i,s)} \in N_{\sigma^{(g_i,e)},i}$. Additionally, $\mathcal{R}_1(m\lambda) \cup \mathcal{R}_2(m\lambda) \subseteq N_{\sigma^{(g_i,e)},i}$. Therefore,

$$\mathrm{Supp}(\tilde{v}^* y_e) \cap (\mathcal{R}_1(m\lambda) \cup \mathcal{R}_2(m\lambda) \cup L_{\sigma^{(g_i,1)},i} \cup \ldots \cup L_{\sigma^{(g_i,e)},i}) = \{m\lambda\}\,,$$

and ε_R remains unaltered for $\sigma_j > (m-1)m\lambda_j - m$, so that $R \subseteq N_{\sigma^{(g_i,e)},i}$. Notice that we are able to do this since i) of Remark 6.3.1 guarantees that $b_{\sigma^{(g_i,e)},i} \geq 1$. Therefore, by repeated application of Lemma 6.3.6, we find a plain q.o. surface Y_d microlocally equivalent to Y_e, for all $e \in \{1, \ldots, m-1\}$, such that

$$\mathrm{Supp}(\tilde{v}^* y_d) \cap (\mathcal{R}_1(m\lambda) \cup \mathcal{R}_2(m\lambda) \cup L_{\sigma^{(g_i,1)},i} \cup \ldots \cup L_{\sigma^{(g_i,m-1)},i}) = \{m\lambda\}\,, \tag{6.25}$$

and that ε_R remains unaltered. Notice that (6.25) can be written as

$$\mathrm{Supp}(\tilde{v}^* y_r) \cap (\mathcal{R}_1(m\lambda) \cup \mathcal{R}_2(m\lambda) \cup \mathcal{R}_i(\tilde{\sigma}^{(g_i)})) = \emptyset,$$

where $\tilde{\sigma}^{(g_i)}$ has the i-th coordinate equal to g_i and the j-th coordinate equal to $(m-1)m\lambda_j - m + 1$. Since $\tilde{\sigma}^{(\tilde{g}_i)} \in N_{\tilde{\sigma}^{(g_i)},i}$, whenever $\tilde{g}_i > g_i$, by applying this very reasoning successively to each

$$g_i \in \{m\lambda_i + 1, \ldots, (m-1)m\lambda_i - m\}$$

yields a plain q.o. surface Y_{d_i} microlocally equivalent to Y_d such that

$$\mathrm{Supp}(\tilde{v}^* y_{d_i}) \cap \left(\mathcal{R}_1(m\lambda) \cup \mathcal{R}_2(m\lambda) \cup F^i\right) = \{m\lambda\}\,,$$

where

$$F^i = \bigcup_{g_i = m\lambda_i + 1}^{(m-1)m\lambda_i - m} \mathcal{R}_i(\tilde{\sigma}^{(g_i)}),$$

and ε_R remains unaltered. We claim that by applying this argument successively for each $i \in \{1,2\}$ we obtain a plain q.o. surface Y_{d_2} such that

$$\mathrm{Supp}(\tilde{v}^* y_{d_2}) \cap \left(L_{m\lambda,1} \cup L_{m\lambda,2} \cup F^1 \cup F^2\right) = \{m\lambda\}$$

and that ε_R remains unaltered. Indeed, if $\tau = (\tau_1, \tau_2) \in F^1$, then $\tau_1 \leq (m-1)m\lambda_1 - m$, so that $\tau \in N_{\sigma^{(g_2,e)},2}$, for each $g_2 \in \{m\lambda_2 + 1, \ldots, (m-1)m\lambda_2 - m\}$ and each $e \in \{1, \ldots, m-1\}$. Hence the subseries ε_{F^1} in $\tilde{v}^* y_{d_1}$ remains unaltered. The claim then follows.

Recall that $\mathrm{Fr}(\lambda) = (m-1)m\lambda - m\mathbb{1}$, by Lemma 5.0.1. Let $Y' = Y_{d_2}$. We claim that

$$\mathrm{Supp}(\tilde{v}^* y') \subseteq R^* \cup \mathrm{int}(\llcorner(\mathrm{Fr}(\lambda))).$$

In fact, let $\tau = (\tau_1, \tau_2) \in \llcorner(m\lambda) \setminus (R^* \cup \mathrm{int}(\mathrm{Fr}(\lambda)))$. If $\tau \in R \setminus R^*$, then $\tau \notin \mathrm{Supp}(v^* y')$, for ε_R remains unaltered from $v^* y$, i.e.,

$$R \cap \mathrm{Supp}(\tilde{v}^* y') = R \cap \mathrm{Supp}(v^* y) \subseteq R^*.$$

Hence, assume that $\tau \notin R$. Then

$$m\lambda_i \leq \tau_i \leq (m-1)m\lambda_i - m$$

and $\tau_j > (m-1)m\lambda_j - m$, for some $i, j \in \{1,2\}$. Hence, $\tau \in \mathcal{R}_i(\tilde{\sigma}^{(g_i)})$, with $g_i = \tau_i$. Consequently, $\tau \notin \mathrm{Supp}(v^* y')$.

Finally, since the set of all contact transformations forms a group under composition and one applied finitely many contact transformations, it follows by transitivity that Y is microlocally equivalent to Y'. $\square$

6.5 Examples

In this section, we showcase applications of the algorithm developed in Theorem 6.4.6 and its limitations, particularly when the conditions for the characteristic exponent λ are relaxed.

Example 6.5.1. *Let Y be a plain q.o. surface with characteristic exponent $\lambda = (\frac{5}{4}, \frac{5}{4})$. We consider a branch ζ of Y defined by*

$$y = x_1^{5/4} x_2^{5/4} + x_1^{5/4} x_2^{9/4} + x_1^{7/4} x_2^{7/4}.$$

Then $\mathbb{P}_Y^* \mathbb{C}^3$ admits the parametrization v defined by

$$\begin{aligned}
x_1 &= t_1^4, \qquad x_2 = t_2^4, \\
y &= t_1^5 t_2^5 + t_1^5 t_2^9 + t_1^7 t_2^7, \\
p_1 &= \tfrac{5}{4} t_1 t_2^5 + \tfrac{5}{4} t_1 t_2^9 + \tfrac{7}{4} t_1^3 t_2^7, \\
p_2 &= \tfrac{5}{4} t_1^5 t_2 + \tfrac{9}{4} t_1^5 t_2^5 + \tfrac{7}{4} t_1^7 t_2^3.
\end{aligned}$$

Initially, $Supp(v^*y) = \{(5,5),(5,9),(7,7)\}$. We proceed by successive infinitesimal contact transformations $\Phi_{\underline{\alpha},\beta}$ to eliminate terms. We aim to eliminate $t_1^5t_2^9 \in L_{m\lambda,1}$. We set $\underline{\alpha} = 0$ and $\beta = -y\sum_{\ell\geq 1}\kappa_\ell x_2^\ell$, where $\kappa_\ell = (-1)^{\ell+1}$, for each $\ell \geq 1$. This yields a plain q.o. surface $Y_1 = (\pi \circ \Phi_{\underline{\alpha},\beta})(\mathbb{P}_Y^*\mathbb{C}^3)$ with

$$y_1 = t_1^5t_2^5 + t_1^7t_2^7 - t_1^7t_2^{11} + t_1^7t_2^{15} - t_1^7t_2^{19} + t_1^7t_2^{23} - t_1^7t_2^{27} - t_1^7t_2^{31} \pmod{(t_1,t_2)^{40}\mathbb{C}\{t_1,t_2\}}.^\dagger$$

Notice that $(7,7),(7,11) \in \Gamma(\lambda)_{\mathcal{G}} \setminus (\Lambda_1(\lambda)\cup\Lambda_2(\lambda))$. To eliminate the higher-order terms $t_1^7t_2^{15}, t_1^7t_2^{19}, t_1^7t_2^{23}$, and $t_1^7t_2^{31}$, we apply $\pi\circ\Phi_{\alpha\mathbb{1}_1,\beta}$ to $\mathbb{P}_{Y_1}^*\mathbb{C}^3$ with

$$\alpha = \tfrac{192}{125}x_1(1-x_2+x_2^2-x_2^3+x_2^4)p_1^2,$$

and β determined by Lemma 6.3.6, given that

$$(7,15) = 1\cdot(4,0) + 0\cdot(0,4) + 3\cdot(1,5) \in \Lambda_1(\lambda).$$

This results in a microlocally equivalent plain q.o. surface Y_2 that admits a Puiseux parametrization with $x_1 = t_1^4$, $x_2 = t_2^4$, and

$$y_2 = t_1^5t_2^5 + t_1^7t_2^7 - t_1^7t_2^{11} - \tfrac{21}{5}t_1^9t_2^{17} + \tfrac{42}{5}t_1^9t_2^{21} - \tfrac{147}{25}t_1^{11}t_2^{19} \pmod{(t_1,t_2)^{40}\mathbb{C}\{t_1,t_2\}}.$$

To eliminate the t_1^9 terms, we apply $(\pi\circ\Phi_{\underline{\alpha},\beta})$ to $\mathbb{P}_{Y_2}^*\mathbb{C}^3$, where $\underline{\alpha} = 0, \beta = x_1y(\frac{21}{5}x_2^3 - \frac{42}{5}x_2^4)$, resulting in a microlocally equivalent q.o. surface Y_3 that admits a Puiseux parametrization such that $x_1 = t_1^4$, $x_2 = t_2^4$, and

$$y_3 = t_1^5t_2^5 + t_1^7t_2^7 - t_1^7t_2^{11} - \frac{42}{25}t_1^{11}t_2^{19} \pmod{(t_1,t_2)^{40}\mathbb{C}\{t_1,t_2\}}.$$

We emphasize the phenomenon of "translation" that occurs. As in the proof of Lemma 6.3.4, one would expect for each term $(t_1t_2)^\tau$ in y_2 to generate a term $(t_1t_2)^{\sigma+\tau-m\lambda}$ in y_3. This is precisely what is verified here if we let $\sigma = (9,17)$ and $\tau = (7,7)$.

A final transformation with $\alpha = -\frac{8064}{3125}x_1^2x_2p_1^2$ yields the plain q.o. surface Y_4 with a Puiseux parametrization where $x_1 = t_1^4$, $x_2 = t_2^4$, and

$$y_4 = t_1^5t_2^5 + t_1^7t_2^7 - t_1^7t_2^{11} \pmod{(t_1,t_2)^{40}\mathbb{C}\{t_1,t_2\}}.$$

Example 6.5.2. *Let Y be a plain q.o. surface with characteristic exponent $\lambda = (\frac{5}{4},\frac{5}{4})$. We consider a branch ζ of Y parametrized by*

$$x_1 = t_1^4, \qquad x_2 = t_2^4,$$
$$y = t_1^5t_2^5 + \sum_{\tau\in R\setminus\{m\lambda\}} c_\tau \underline{t}^\tau + \sum_{\tau\in \llcorner(m\lambda)\setminus R} c_\tau \underline{t}^\tau,$$

where $R = ([5,11]\times[5,11])$. By Theorem 6.4.6, Y is microlocally equivalent to a plain q.o. surface Y_1 that admits the following Puiseux parametrization

$$x_1 = t_1^4, \qquad x_2 = t_2^4,$$
$$y = t_1^5t_2^5 + \sum_{\tau\in R^*} d_\tau \underline{t}^\tau + \sum_{\tau\in\Gamma(\lambda)} d_\tau \underline{t}^\tau,$$

† Terms of order ≥ 40 are omitted to preserve the illustrative flow of the example.

where

$$\operatorname{Supp}\left(\sum_{\tau\in\Gamma(\lambda)} d_\tau \underline{t}^\tau\right) \subseteq \operatorname{int}(\llcorner((11,11)))$$

and $R^* = \{(7,7),(11,7),(7,11),(11,11)\}$.

Example 6.5.3. *Let* Y *be the q.o. surface defined by the branch* $y = x_1^2 x_2^{\frac{7}{4}} + x_1^2 x_2^{\frac{9}{2}}$. *The associated semigroups for* Y *are*

$$\Gamma(\lambda) = \langle(4,0),(0,4),(8,7)\rangle,$$

$$\Lambda_1(\lambda) = \langle(4,0),(0,4),(4,7)\rangle, \quad \Lambda_2(\lambda) = \langle(4,0),(0,4),(8,3)\rangle.$$

A Puiseux parametrization for this branch yields

$$x_1 = t_1^4, x_2 = t_2^4, \quad y = t_1^8 t_2^7 + t_1^8 t_2^{18}.$$

The induced parametrization of $\mathbb{P}_Y^*\mathbb{C}^3$ *then yields*

$$\begin{aligned} &x_1 = t_1^4, x_2 = t_2^4, \qquad y = t_1^8 t_2^7 + t_1^8 t_2^{18}, \\ &p_1 = 2t_1^4 t_2^7 + 2t_1^4 t_2^{18}, \quad p_2 = \frac{7}{4} t_1^8 t_2^3 + \frac{9}{2} t_1^8 t_2^{14}. \end{aligned}$$

In order to eliminate the term $t_1^8 t_2^{18}$, *one writes*

$$(8,18) = 0\cdot(4,0) + 1\cdot(0,4) + 2\cdot(4,7) \in \Lambda_1(\lambda).$$

Therefore, if one applies the infinitesimal contact transformation $\Phi_{\alpha\mathbb{1}_1,\beta}$ *constructed in 6.2.1, where* $\alpha = \frac{1}{2}x_2 p_1$, *to* $\mathbb{P}_Y^*\mathbb{C}^3$ *then* Y *is microlocally equivalent to a q.o. surface* Y_1 *that admits the following Puiseux parametrization*

$$x_1 = t_1^4, x_2 = t_2^4, \qquad y_1 = t_1^8 t_2^7 - \frac{1}{4} t_1^8 t_2^{29} \pmod{(t_1,t_2)^{40}\mathbb{C}\{t_1,t_2\}}.$$

Now note that $(8,29) \in Supp(v^* y_1) \setminus (R \cup \Gamma(\lambda) \cup \Lambda_1(\lambda) \cup \Lambda_2(\lambda))$ *so that we are not able to eliminate the term* $-\frac{1}{4}t_1^8 t_2^{29}$ *via the constructed infinitesimal contact transformations. This example illustrates the difficulties that arise when one relaxes the condition* iii) *of Lemma 5.0.6 for some* $i \in \{1,\ldots,n\}$.

Example 6.5.4. *Let* Y *be the q.o. surface defined by the branch* $y = x_1^{\frac{7}{4}} x_2^{\frac{3}{2}} + x_1^{\frac{25}{4}} x_2^{\frac{3}{2}}$. *The associated semigroups for* Y *are*

$$\Gamma(\lambda) = \langle(4,0),(0,4),(7,6)\rangle,$$

$$\Lambda_1(\lambda) = \langle(4,0),(0,4),(3,6)\rangle, \quad \Lambda_2(\lambda) = \langle(4,0),(0,4),(7,2)\rangle.$$

A Puiseux parametrization for this branch yields

$$x_1 = t_1^4, x_2 = t_2^4, y = t_1^7 t_2^6 + t_1^{25} t_2^6.$$

The induced parametrization of $\mathbb{P}^*_Y\mathbb{C}^3$ *is then given by*

$$x_1 = t_1^4, x_2 = t_2^4, \quad y = t_1^7 t_2^6 + t_1^{25} t_2^6,$$
$$p_1 = \frac{7}{4} t_1^3 t_2^6 + \frac{25}{4} t_1^{21} t_2^6, p_2 = \frac{3}{2} t_1^7 t_2^2 + \frac{3}{2} t_1^{25} t_2^2.$$

In order to eliminate the term $t_1^{25} t_2^6$, *one writes*

$$(25, 6) = 1 \cdot (4, 0) + 0 \cdot (0, 4) + 3 \cdot (7, 2)$$

in order to then apply the infinitesimal contact transformation $\Phi_{\alpha\mathbb{1}_2,\beta}$ *to* $\mathbb{P}^*_Y\mathbb{C}^3$, *where* $\alpha = \frac{8}{9} x_1 p_2^2$. *Consequently, one obtains for* $(\pi \circ \Phi_{\underline{\alpha},\beta})(\mathbb{P}^*_Y\mathbb{C}^3)$ *the Puiseux parametrization*

$$x_1 = t_1^4, x_2 = t_2^4, y = t_1^7 t_2^6 + \frac{3}{8} t_1^{79} t_2^6 + \frac{15}{256} t_1^{151} t_2^6 + \frac{5}{1024} t_1^{223} t_2^6 + \frac{15}{65636} t_1^{295} t_2^6 \pmod{(t_1)^{300}\mathbb{C}\{t_1, t_2\}}.$$

Further analysis verifies the extremely delicate nature of the coefficients and the respective generated terms with exponents $m\lambda + \ell\mathbb{1}_2$, *with* $\ell \in \mathbb{N}_0$. *This example illustrates the difficulties that arise when one tries to work with the case* $b_{\sigma,i} = 0$ *for some* $i \in \{1, \ldots, n\}$.

7

Future Work

In the words of the celebrated Scottish writer Robert Louis Stevenson, "To travel hopefully is a better thing than to arrive, and the true success is to labour", it would be somewhat disappointing if one arrived at the end of this dissertation without unanswered questions. Fortunately, this is not the case.

As stated in the abstract, we are left in Theorem 6.4.6 with an analytic problem. If one could prove the convergence of the composition of infinitely many analytic transformations, then Theorem 3.2.3 would be fully generalized. An alternative approach to solving this problem would be to follow O. Zariski's steps and attempt to find a ring with some "good properties" (see Section 3.2) and attempt to follow his strategy. One of the main problems here is the phenomenon of "translation" that happens with the terms inside R^*, when eliminating terms in $\mathrm{int}(\llcorner(\mathrm{Fr}(\lambda)))$.

One question that naturally arose in the course of this research was whether one can generalize Theorems 4.3.3 and 6.4.6 to the case of several characteristic exponents or even to higher dimensions. One would need to rebuild the concepts of Chapter 5 from scratch and explore the properties of $\Gamma(\tilde{\lambda}_1, \ldots, \tilde{\lambda}_s)$.

For the case of one characteristic exponent $\lambda = (\frac{m+1}{m}, \frac{m+1}{m})$, we know that $\mathbb{P}^*_Y\mathbb{C}^3$ has an isolated singularity at the origin. Is there some notion of "minimality" of the singular locus at the origin of $\mathbb{P}^*_Y\mathbb{C}^{n+1}$ when we consider the case of one characteristic exponent $\lambda = \frac{m+1}{m}\mathbb{1}$? Moreover, further attention needs to be allocated to the behaviour of the applied contact transformations in the generalized setting.

Future research could apply the framework established in this dissertation to generalized monomials and semi-roots (see [7]), with the aim of relating these concepts to the case of irreducible plane curves.

Finally, there have been recent studies by F. Aroca and J.M. Tornero in [4] regarding ν-q.o. surfaces, a concept proposed by H. Hironaka in order to generalize q.o. surfaces. One possible route of research is to study their limits of tangents and attempt to apply the ideas developed in this dissertation to this framework.

Bibliography

[1] S. Abhyankar. "Local Uniformization on Algebraic Surfaces Over Ground Fields of Characteristic $p \neq 0$". In: *Annals of Mathematics, Vol. 63, No. 3* (1956) (cit. on pp. vi, vii).

[2] S. Abhyankar. "On the Ramification of Algebraic Functions". In: *American Journal of Mathematics* 77.3 (1955), pp. 575–592. ISSN: 00029327, 10806377. URL: http://www.jstor.org/stable/2372643 (visited on 2026-01-22) (cit. on p. 1).

[3] A. Araújo and O. Neto. "Limits of tangents of quasi-ordinary hypersurfaces". In: *Proceedings of the American Mathematical Society* 141.1 (2012-09), pp. 1–11. DOI: 10.1090/s0002-9939-2012-11126-8 (cit. on p. 15).

[4] F. Aroca and J. M. Tornero. "On ν-Quasiordinary Surface Singularities and Their Resolution". In: *Mediterranean Journal of Mathematics* (2024) (cit. on p. 67).

[5] A. Assi. "The Frobenius Vector of a Free Affine Semigroup". In: *Journal of Algebra and Its Applications* 11 (2012), p. 1250065. DOI: 10.1142/S021949881250065X (cit. on pp. 12, 23, 24, 27).

[6] E. Brieskorn and H. Knörrer. *Plane algebraic curves*. 1st ed. Birkhäuser Basel, 1986. ISBN: 9783764317690 (cit. on pp. 1, 7).

[7] J. Cabral and A. Casimiro. "Computational Decomposition of Minimal Polynomial for Quasi-Ordinary Hypersurface Branch Truncation". In: *Experimental Mathematics* (2026) (cit. on p. 67).

[8] T. De Jong and G. Pfister. *Local analytic geometry: Basic theory and applications*. Springer Science & Business Media, 2013 (cit. on p. 16).

[9] M. González Villa. "Newton process and semigroups of irreducible quasi-ordinary power series". In: *Revista de la Real Academia de Ciencias Exactas, Fisicas y Naturales. Serie A. Matematicas* 108.1 (2013-08), pp. 259–279. DOI: 10.1007/s13398-013-0139-1 (cit. on pp. vi, vii, 8, 9).

[10] G.-M. Greuel, C. Lossen, and E. Shustin. *Introduction to singularities and deformations*. Springer, 2007 (cit. on p. 16).

[11] G.-M. Greuel and G. Pfister. *A Singular Introduction to Commutative Algebra*. Springer Berlin, Heidelberg, 2007 (cit. on pp. 5, 35).

[12] G.-M. Greuel, G. Pfister, O. Bachmann, C. Lossen, and H. Schönemann. *A Singular introduction to commutative algebra*. Springer, 2008 (cit. on p. 10).

[13] M. Hernandes and N. M. P. Panek. "On the $\mathcal{A}$-equivalence of quasi-ordinary parameterizations". In: *Revista Matemática Complutense, Vol. 32* (2018) (cit. on pp. vi, vii).

[14] H. Hironaka. "Resolution of Singularities of an Algebraic Variety Over a Field of Characteristic Zero: I". In: *Annals of Mathematics, Vol. 79, No. 1* (1964) (cit. on pp. vi, vii).

[15] J. M. Lee. *Introduction to Smooth Manifolds*. 2nd. Springer International Publishing, 2012 (cit. on p. 5).

[16] M. S. Lie, E. Trell, and R. M. Santilli. "Marius Sophus Lie's Doctoral Thesis: Over en Classe Geometriske Transformationer". In: *Algebras, Groups and Geometries* 15.4 (1998). English translation by Erik Trell; Commentary by Erik Trell and Ruggero Maria Santilli, pp. 395–445 (cit. on p. 13).

[17] J. Lipman. "Desingularization of Two-Dimensional Schemes". In: *Annals of Mathematics, Vol. 107, No. 2* (1978) (cit. on pp. vi, vii).

[18] J. Lipman. "Quasi-Ordinary Singularities of Embedded Surfaces". PhD thesis. Harvard University, 1965 (cit. on p. 9).

[19] J. Lipman. "Topological invariants of quasi-ordinary singularities". In: *Memoirs of the American Mathematical Society* (1988) (cit. on pp. 1, 8, 9).

[20] J. M. Lourenço. *The `aidisclose` package: Generative AI disclosure checklist and statements*. Version 1.11.0. 2025. URL: https://ctan.org/pkg/aidisclose/aidisclose-doc.pdf (cit. on p. viii).

[21] A. R. Martins and O. Neto. "On the group of germs of contact transformations". In: *Portugaliae Mathematica* 72.4 (2015-10), pp. 393–406. DOI: 10.4171/pm/1972 (cit. on pp. 21, 56).

[22] A. McInerney. "First Steps in Differential Geometry - Riemannian, Contact, Symplectic". In: Springer New York, 2013, pp. 271–337 (cit. on p. 13).

[23] O. Neto and A. Araújo. "Limits of Tangents of a Quasi-Ordinary Hypersurface". In: *Proceedings of the American Mathematical Society, Vol. 141, No. 1* (2013) (cit. on pp. vi, vii).

[24] I. Newton and H. Turnball. "188. Newton to Oldenburg, 24 October 1676". In: Cambridge University Press, 2020-11, pp. 110–161. ISBN: 9781108627375. DOI: 10.1017/9781108627375.032 (cit. on pp. vi, vii).

[25] J. C. Rosales and P. A. García-Sánchez. "On free affine semigroups." In: *Semigroup Forum*. Vol. 58. 1999 (cit. on p. 26).

[26] J. M. Ruiz. *The Basic Theory of Power Series*. Vieweg+Teubner Verlag Wiesbaden, 1993 (cit. on p. 47).

[27] B. Sturmfels. *Gröbner Bases and Convex Polytopes*. 8th ed. American Mathematical Society, 1996 (cit. on p. 33).

[28] Y. Suchikova, N. Tsybuliak, J. A. T. da Silva, and S. Nazarovets. "GAIDeT (Generative AI Delegation Taxonomy): A Taxonomy for Humans to Delegate Tasks to Generative Artificial Intelligence in Scientific Research and Publishing". In: *Accountability in Research* (2025), pp. 1–27. DOI: 10.1080/08989621.2025.2544331 (cit. on p. viii).

[29] The LaTeX Project team. *LaTeX - A document preparation system*. 1985. URL: https://www.latex-project.org (visited on 2026-01-14) (cit. on p. viii).

[30] O. Zariski. *The moduli problem for plane branches*. University Lecture Series 039. American Mathematical Society, 2006. ISBN: 0-8218-2983-1 (cit. on pp. 1, 10, 11).

[31] O. Zariski. "The Reduction of the Singularities of an Algebraic Surface". In: *Annals of Mathematics, Vol. 40, No. 3* (1939) (cit. on pp. vi, vii).